

Spinal Nullity at Singular Infinity: A Consolidated Package of Laws, Equations, Proof Sketches, Sensor Theory, and Finite Experiments

Consolidated research package

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Status note. This report is a research consolidation, not a publication-ready proof. Statements marked as theorems/laws include proof sketches from the stated hypotheses; they should still be checked before formal use. Statements marked conjectural or speculative are proposed targets. The finite experiments are sanity checks of finite shadows; they are not proofs of the transfinite results.

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1 Executive Summary

The theory studies a nonnumeric null ideal on a countable product

$$X = \prod_{n < \omega} \kappa_n,$$

where $\kappa_0 < \kappa_1 < \dots$ are regular measurable or stronger cardinals with $\lambda = \sup_n \kappa_n$, and each coordinate has a κ_n -complete nonprincipal ultrafilter U_n . The dual small ideal is

$$I_n = U_n^* = \{S \subseteq \kappa_n : S \notin U_n\}.$$

The singular-spine growth condition is

$$\left| \prod_{i < n} \kappa_i \right| < \kappa_n. \quad (\text{Growth})$$

The central object is the J -spinal null ideal. For an ideal $J \supseteq \text{Fin}$ on ω ,

$$A \in Z_U^J \iff \exists S_n \in I_n \forall x \in A, \{n : x(n) \in S_n\} \notin J.$$

Thus a bad event is null if every branch through it is trapped on a recurrent set of coordinates, where “recurrent” is defined by J -positivity. For $J = \text{Fin}$, this means infinitely often.

The core slogan is

spinal nullity = recurring local rarity.

The theory expanded from slaloms and Prikry category into a robust calculus of recurrence

ideals, sensors, rough observation, and information thresholds. The largest new blocks of results are:

- **Causal collapse:** adaptive causal observation collapses to blind slaloms whenever local additivity dominates past histories.
- **Tomography and incidence:** product-cylinder nullity is exactly the J -positivity of local bottleneck coordinates.
- **Closed finite-rank correction:** in the countably complete countable-spine setting, closed Fin -nullity stabilizes at stage ω and uses finite countdown ranks, not arbitrary transfinite closed ranks.
- **Blur/join calculus:** coordinate noise ideals compose with recurrence ideals by ideal join.
- **Sensor rough-nullity:** every coordinate sensor creates two observed ideals, universal and existential, producing a three-valued observation logic: definitely null, ambiguous, definitely positive.
- **Faithful sensor and Nyquist laws:** finite/subcritical sensors can certify nullity but cannot certify positivity; faithful observation requires resolving recurrent aliasing.
- **Normal sensor phase diagram:** for normal local ultrafilters, each coordinate sensor has cost 0, 1 bit, or full width $\log_2 \kappa_n$.
- **Portfolio cover theorem:** sensor fusion design reduces to set cover modulo the effective ideal $H \vee J$.

Most compact combined law

With a coordinate-noise ideal H , monitoring ideal J , and effective ideal

$$L = H \vee J,$$

robust observation has this form:

$$\boxed{\text{noise} + \text{monitoring tolerance} + \text{sensor ambiguity are measured by } H \vee J.}$$

For normal ultrafilter sensors, robust faithful observation with bit budgets μ_n is possible exactly when

$$\boxed{\{n \in N : \mu_n = 0\} \cup \{n \in P : 2^{\mu_n} < \kappa_n\} \in H \vee J,}$$

where N is the set of nonprincipal but not already faithful coordinates, and P is the set of principal/default-collapse coordinates.

2 Status Map of Principal Laws

Name	Statement	Status
Spinal nullity	$A \in Z_U^J$ iff one coordinate-small slalom catches every branch on a J -positive coordinate set.	Definition/core
Orthogonal filter	$Z_U^J = (F_U^J)^\perp$, where F_U^J is generated by J -fault-tolerant large protocols.	Proved/sketched
Causal collapse	Causal adaptive slaloms flatten to blind slaloms if $ \prod_{i < n} X_i < \text{add}(I_n)$.	Proved/sketched
Trace law	$P_E \in Z_U^J$ iff $E \notin J$.	Proved/sketched
Tomography	$\prod_n A_n \in Z_U^J$ iff $\{n : A_n \in I_n\} \notin J$.	Proved/sketched
Incidence	$Q_A \subseteq C_S^J$ iff $\{n : A_n \subseteq S_n\} \notin J$.	Proved/sketched
Slalom order	$C_S^J \subseteq C_T^J$ iff $\{n : S_n \not\subseteq T_n\} \in J$.	Proved/sketched
Observer compression	$\bigcup_{S \in \Pi} C_S^J \in Z_U^J$ iff Π is bounded in $(\prod I_n, \leq_J)$.	Proved/sketched
Staggered completeness	λ -many null hulls compress by delayed diagonal union.	Proved/sketched
Fault-control law	$A \in Z_U^J$ iff a large-set controller avoids A under J -small faults.	Proved/sketched
Causality barrier	Future-looking observers can strictly exceed spinal nullity.	Counterexample
Category identity	For $J = \text{Fin}$, $Z_U = \text{Rare}(\tau_U) = \text{Meager}(\tau_U)$.	Proved/sketched; ordinary topology refinements require care
Closed stabilization	omega- For countably complete U_n , closed derivative stabilizes at ω .	Proved/sketched
Closed finite countdown	Closed Fin-nullity iff every node dies at a finite derivative stage.	Proved/sketched
Closed J -alarm law	Closed J -nullity = finite countdown ranks plus J -positive small-alarm recurrence.	Proved/sketched
Product hazard	For product cylinders, rank is time to next local bottleneck.	Proved/sketched
Blur/join	$[A]_J \in Z_U^K$ iff $A \in Z_U^{J \vee K}$.	Proved/sketched
Modal robustness	$\Box_J C_S^K = C_S^{J \vee K}$, $\Diamond_J D_B^K = D_B^{J \vee K}$.	Proved/sketched
Asynchronous recurrence	$C_S^J \cap C_T^J = C_{S \cap T}^J$ for all slaloms iff J is prime.	Proved/sketched
Cross-ideal implication	For full support, $C_S^K \subseteq C_T^L$ iff $L \subseteq K$ and $S \leq_K T$.	Proved/sketched
Schedule factorization	For nondegenerate S , $h_S^{-1}(B) \in Z_U^L$ iff $B \cap L = \emptyset$.	Proved/sketched
Sensor pullback	$\sigma^{-1}(B) \in Z_I^J$ iff $B \in Z_{K^\vee}^J$.	Proved/sketched
Sensor image	$B = \sigma[A]$ for some $A \in Z_I^J$ iff $B \in Z_{K^\exists}^J$.	Proved/sketched
Rough trichotomy	Observed events are definitely null, ambiguous, or definitely positive depending on $Z_{K^\vee}^J \subseteq Z_{K^\exists}^J$.	Proved/sketched
Faithful sensor	Images of all null sets are null iff the sensor is almost injective modulo the ultrafilter.	Proved/sketched
Subcritical collapse	$ Y_n < \kappa_n$ makes the pushforward ultrafilter principal at a default symbol.	Proved/sketched
Conditional Nyquist	Refinement capacity must exceed unresolved fiber-resolution modulo $H \vee J$.	Proved/sketched
Normal phase diagram	Under normality, sensor repair cost is 0 bits, 1 bit, or full width.	Proved/sketched

Name	Statement	Status
Portfolio cover	Fusion plus one repair bit is faithful iff nonprincipal supports cover coordinates modulo $H \vee J$.	Proved/sketched
Analytic perfect box	Analytic positive sets contain full U -boxes.	Conjectural target
PCF/Tukey formulas	Cardinal invariants of Z_U^J equal reduced-product invariants.	Conjectural target
Quantum/projection analogue	Replace local sets by projections in commuting tensor products.	Speculative target

3 Core Setup and Slalom Geometry

Definition 3.1 (Spinal nullity). *Let $J \supseteq \text{Fin}$ be an ideal on ω . A slalom is a sequence $S = \langle S_n : n < \omega \rangle$ with $S_n \in I_n$. Its J -recurrence hull is*

$$C_S^J = \{x \in X : \{n : x(n) \in S_n\} \notin J\}.$$

Then

$$Z_U^J = \{A \subseteq X : \exists S \ A \subseteq C_S^J\}.$$

For $J = \text{Fin}$, $x \in C_S$ means $x(n) \in S_n$ infinitely often. This is a nonnumeric Borel-Cantelli principle: no probability values are assigned, but recurrent entry into coordinatewise null/rare traps makes the event globally negligible.

Law 3.2 (Orthogonal filter form). *For $B_n \in U_n$, define*

$$D_B^J = \{x : \{n : x(n) \notin B_n\} \in J\}.$$

Let F_U^J be the filter generated by the regions D_B^J . Then

$$\boxed{Z_U^J = (F_U^J)^\perp.}$$

Proof sketch. If $S_n \in I_n$, set $B_n = \kappa_n \setminus S_n \in U_n$. Then $X \setminus C_S^J = D_B^J$. Thus $A \subseteq C_S^J$ iff $A \cap D_B^J = \emptyset$. This is exactly orthogonality to the large-protocol filter. $\square$

3.1 Trace, tomography, and incidence

Theorem 3.3 (Trace law). *Fix points $a_n \in \kappa_n$, and for $E \subseteq \omega$ define*

$$P_E = \{x : \forall n \in E, x(n) = a_n\}.$$

Then

$$\boxed{P_E \in Z_U^J \iff E \notin J.}$$

Consequently, Z_U^J remembers J :

$$J = \{E \subseteq \omega : P_E \notin Z_U^J\}.$$

Proof sketch. If $E \notin J$, the singleton slalom $S_n = \{a_n\}$ for $n \in E$ catches every point of P_E on a J -positive set. Conversely, if $E \in J$, then given any slalom S , choose a point satisfying the trace constraints on E and avoiding S_n off E . Its hit set is contained in E , hence J -small. $\square$

Theorem 3.4 (Spinal tomography). *For nonempty $A_n \subseteq \kappa_n$, let $Q_A = \prod_n A_n$. Then*

$$\boxed{Q_A \in Z_U^J \iff E_A := \{n : A_n \in I_n\} \notin J.}$$

Proof sketch. If $E_A \notin J$, the slalom $S_n = A_n$ on E_A catches every branch. If $E_A \in J$, then outside E_A , $A_n \in U_n$. For any slalom S , choose $x(n) \in A_n \setminus S_n$ outside E_A ; this is possible because $A_n \in U_n$ and $S_n \in I_n$. The branch only hits S on E_A , so it is not caught. $\square$

Theorem 3.5 (Null incidence equation). *For a product cylinder $Q_A = \prod_n A_n$ and slalom S ,*

$$\boxed{Q_A \subseteq C_S^J \iff \{n : A_n \subseteq S_n\} \notin J.}$$

Proof sketch. If $A_n \subseteq S_n$ on a J -positive set, every branch through Q_A hits the slalom there. Conversely, if that set is J -small, choose a branch in $A_n \setminus S_n$ at all other coordinates. It avoids recurrent capture. $\square$

3.2 J-spectrum

The map $J \mapsto Z_U^J$ is an anti-order embedding:

$$J \subseteq K \iff Z_U^K \subseteq Z_U^J.$$

Larger J means a stronger recurrence demand: J -positive is harder than K -positive when $J \subseteq K$. The trace law separates ideals exactly.

3.3 Slalom order and observer compression

Define

$$S \leq_J T \iff \{n : S_n \not\subseteq T_n\} \in J.$$

Theorem 3.6 (Exact hull order).

$$C_S^J \subseteq C_T^J \iff S \leq_J T.$$

Proof sketch. The forward direction follows by deleting the J -small coordinate set on which $S_n \not\subseteq T_n$. Conversely, if $E = \{n : S_n \not\subseteq T_n\} \notin J$, choose $x(n) \in S_n \setminus T_n$ on E , and choose $x(n) \notin T_n$ elsewhere. Then $x \in C_S^J \setminus C_T^J$. $\square$

Theorem 3.7 (Observer compression). *For a family $\Pi \subseteq \prod_n I_n$,*

$$\bigcup_{S \in \Pi} C_S^J \in Z_U^J \iff \Pi \text{ is bounded in } (\prod_n I_n, \leq_J).$$

Proof sketch. If Π is bounded by T , then each $C_S^J \subseteq C_T^J$. Conversely, if the union is contained in C_T^J , then each $C_S^J \subseteq C_T^J$, so $S \leq_J T$ by the exact hull order. $\square$

This produces the cardinal invariant dictionary:

$$\text{add}(Z_U^J) = \mathfrak{b}(\prod I_n, \leq_J), \quad \text{cof}(Z_U^J) = \mathfrak{d}(\prod I_n, \leq_J),$$

with covering/uniformity similarly expressed by branch-vs-slalom evasion relations.

3.4 Staggered completeness

A raw coordinatewise union of λ slaloms may be too large at early coordinates. The correct construction is delayed.

Lemma 3.8 (Staggered diagonal compression). *Let $\theta_n = \text{add}(I_n)$. Suppose $\{S^\xi : \xi < \mu\}$ is a family of slaloms and there is a delay map $d : \mu \rightarrow \omega$ with*

$$|\{\xi : d(\xi) < n\}| < \theta_n \text{ for every } n.$$

Set

$$T_n = \bigcup_{\xi : d(\xi) < n} S_n^\xi.$$

Then $T_n \in I_n$, and

$$\bigcup_{\xi < \mu} C_{S^\xi}^J \subseteq C_T^J.$$

Proof sketch. Smallness follows from the delay bound. If x is caught by S^ξ on a J -positive set, deleting the finite set $\{n : n \leq d(\xi)\}$ preserves J -positivity; on the remaining hit coordinates, $S_n^\xi \subseteq T_n$. $\square$

In the singular spine, λ -sized families can be partitioned into countably many pieces of sizes below later κ_n 's, so Z_U^J is λ^+ -complete. The proof is not brute coordinatewise union; it is staggered absorption.

4 Causal Collapse, Category, and Control

4.1 Minimal causal collapse

The ultrafilter assumptions are stronger than needed for causal memory collapse.

Theorem 4.1 (Minimal causal collapse). *Let $X = \prod_n X_n$ and let I_n be proper ideals. Assume*

$$\left| \prod_{i < n} X_i \right| < \text{add}(I_n) \quad \text{for each } n.$$

A causal adaptive slalom is a sequence of maps

$$\pi_n : \prod_{i < n} X_i \rightarrow I_n.$$

It catches x if

$$\{n : x(n) \in \pi_n(x \upharpoonright n)\} \notin J.$$

Then causal adaptive J -nullity equals blind J -slalom nullity.

Proof. Flatten by

$$S_n = \bigcup_{s \in \prod_{i < n} X_i} \pi_n(s).$$

The union is indexed by fewer than $\text{add}(I_n)$ many small sets, so $S_n \in I_n$. Any adaptive hit is a hit by S_n . The converse is trivial because blind slaloms are adaptive slaloms with no history dependence. $\square$

4.2 Direct category collapse

For $J = \text{Fin}$, define a direct basic region

$$N_{s,B} = \{x : s \subseteq x, \forall n \geq |s|, x(n) \in B_n\}, \quad B_n \in I_n^+.$$

Call A directly rare if every $N_{s,B}$ has a pure tail shrink disjoint from A . Under the same additivity condition, direct rarity is equivalent to blind slalom nullity. In the ultrafilter spine, rectangularization upgrades this to the stronger category identity

$$\boxed{Z_U = \text{Rare}(\tau_U) = \text{Meager}(\tau_U)}.$$

The important correction is that minimal additivity gives causal/direct collapse; the full topological meager identity also uses ultrafilter-style rectangularization.

4.3 Fault-tolerant control

A controller chooses $B_n(h) \in I_n^*$ after a finite history h . A play x is J -legal if

$$\{n : x(n) \notin B_n(x \upharpoonright n)\} \in J.$$

The controller wins against A if no J -legal play lies in A .

Theorem 4.2 (Fault-tolerant safety). *Under causal collapse,*

$$A \in Z_I^J \iff A \text{ is avoidable by a causal large-set controller under } J\text{-small faults.}$$

Moreover, every winning adaptive strategy compresses to one blind memoryless protocol.

Proof sketch. Write $S_n(h) = X_n \setminus B_n(h) \in I_n$. A bad branch in A must hit the adaptive small sets on a J -positive set. By causal collapse, the adaptive slalom flattens to one blind slalom. Conversely, a blind catching slalom gives the protocol $B_n = X_n \setminus S_n$. $\square$

4.4 Causality barrier

Warning 4.3 (Future-looking observers are stronger). *Causal dependence is essential. Let $X_0 = X_1 = \kappa$ and $I_0 = [\kappa]^{<\kappa}$. The diagonal*

$$D = \{(\alpha, \alpha) : \alpha < \kappa\}$$

cannot be caught at coordinate 0 by a blind small set. But a future-looking rule $\pi_0(x(1)) = \{x(1)\}$ catches every diagonal point. Thus future-adaptive rarity can strictly exceed spinal nullity.

This explains why any J -category or topology using infinite/future stems must prove a pure-extension theorem eliminating future dependence; otherwise it builds in clairvoyance.

4.5 Causal morphisms

Let (X_n, I_n) and (Y_n, K_n) be two spines. A map

$$F : \prod_n X_n \rightarrow \prod_n Y_n$$

is causal coordinatewise if

$$F(x)(n) = \varphi_{n,x|n}(x(n)).$$

If K_n -small sets have I_n -small preimages under all local maps, then

$$B \in Z_K^J \Rightarrow F^{-1}(B) \in Z_I^J.$$

If I_n -small sets have K_n -small images and history unions are absorbed by K_n , then

$$A \in Z_I^J \Rightarrow F[A] \in Z_K^J.$$

These laws are the functorial backbone of causal spinal systems.

5 Closed Sets, Ranks, and Alarm Laws

5.1 Derivative and the finite-rank correction

For a closed set $[T] \subseteq X$, where T is a tree of finite prefixes, define the spinal derivative

$$T^0 = T, \quad T^{m+1} = D(T^m),$$

where

$$D(A) = \left\{ s \in A : \{ \alpha : s \frown \alpha \in A \} \in U_{|s|} \right\}.$$

At limits, take intersections.

An important correction emerged during development. In the current countably complete, countable-spine, $J = \text{Fin}$ setting, the closed derivative stabilizes at ω :

$$\boxed{D(T^\omega) = T^\omega.}$$

Proof. If $s \in T^\omega$, then for each finite m , since $s \in T^{m+1}$, the successor set

$$A_m(s) = \{ \alpha : s \frown \alpha \in T^m \}$$

belongs to $U_{|s|}$. Countable completeness gives

$$\bigcap_{m < \omega} A_m(s) \in U_{|s|}.$$

Every α in this intersection has $s \frown \alpha \in T^m$ for all m , hence in T^ω . Therefore $s \in D(T^\omega)$. The reverse inclusion is immediate. $\square$

Consequently,

$$\boxed{[T] \in Z_U \iff T^\omega = \emptyset.}$$

If $T^\omega \neq \emptyset$, it contains a U -perfect kernel and hence a full U -box by rectangularization. If $T^\omega = \emptyset$, each node dies at a finite derivative stage.

Define

$$\rho(s) = \min\{m \geq 1 : s \notin T^m\}.$$

Then closed Fin -nullity has a finite countdown rank:

$$\boxed{\rho : T \rightarrow \omega \setminus \{0\}.}$$

This corrects the earlier broad transfinite-rank target for closed Fin -null sets in the present countable-spine setting. Transfinite ranks may still reappear for non-countably-complete filters, longer spines, projective/coanalytic ranks, or richer J -clock structures.

5.2 Finite Bellman equation

For nonexistent successors, set $\rho = 0$. Then

$$\boxed{r(s) = 1 + \text{E} \max_{U_{|s|}} (\text{alpha} - r(s \frown \text{row} \text{alpha})).}$$

Since the rank is finite-valued, esssup is an essential maximum. Moreover,

$$\boxed{\{ \alpha : \rho(s \frown \alpha) = \rho(s) - 1 \} \in U_{|s|}.}$$

Thus typical large moves decrease the finite counter by exactly one.

5.3 Closed alarm law

Define the mandatory alarm set at a node:

$$A_s = \{\alpha : s \frown \alpha \in T \text{ and } \rho(s \frown \alpha) \geq \rho(s)\}.$$

Then $A_s \in I_{|s|}$. If a branch hit these alarms only finitely often, eventually the finite rank would strictly decrease at every step forever, impossible. Therefore every branch in $[T]$ hits local alarms infinitely often.

Flattening the alarms by

$$S_n = \bigcup_{s \in T \cap \prod_{i < n} \kappa_i} A_s$$

gives $S_n \in I_n$ by the growth condition, and $[T] \subseteq C_S$. Hence

$$[T] \in Z_U \iff T \text{ admits a finite countdown Lyapunov rank.}$$

For larger J , finite ranks alone force infinitely many alarms but not necessarily J -positive alarms. The corrected closed J -law is:

$$[T] \in Z_U^J \iff \text{finite countdown rank plus } J\text{-positive small-alarm recurrence.}$$

5.4 Product-cylinder hazard equation

Let

$$Q_A = \prod_n A_n, \quad E_A = \{n : A_n \in I_n\}.$$

For the product tree, at a compatible node of length m , let

$$e(m) = \min(E_A \cap [m, \infty))$$

if it exists. Then

$$\rho(s) = e(m) - m + 1.$$

If no such $e(m)$ exists, the node survives into the perfect kernel. Thus product-cylinder rank is exactly

$$1 + \text{time to the next local bottleneck.}$$

Tomography becomes a dynamical law:

$$Q_A \in Z_U^J \iff E_A \notin J.$$

5.5 Projection catastrophe and safe projection

Let $M \subseteq \omega$, and let $E = \omega \setminus M$. If $E \notin J$, the trace set

$$P_E = \{x : \forall n \in E, x(n) = a_n\}$$

is null, but its projection to M is the whole target space. Thus projections can send closed null sets onto everything.

If $E \in J$, projection is exact:

$$A \in Z_U^J \iff \pi_M(A) \in Z_{U \upharpoonright M}^{J \upharpoonright M}$$

for coordinate projections that discard only J -small mass. Projection is safe exactly when it discards only J -small coordinates.

6 Recurrence Logic, Blur, and Modal Robustness

6.1 Blur and ideal join

Define $x \equiv_J y$ iff $\{n : x(n) \neq y(n)\} \in J$, and define the J -blur

$$[A]_J = \{y : \exists x \in A, y \equiv_J x\}.$$

For ideals J, K , their join is

$$J \vee K = \{E : \exists A \in J, \exists B \in K, E \subseteq A \cup B\}.$$

Theorem 6.1 (Blur-join law).

$$[A]_J \in Z_U^K \iff A \in Z_U^{J \vee K}.$$

Proof sketch. If $A \subseteq C_S^{J \vee K}$ and $y \equiv_J x \in A$, deleting the J -small difference set from the hit set leaves a $(J \vee K)$ -positive, hence K -positive, hit set. Conversely, if $[A]_J \subseteq C_S^K$ but some $x \in A$ hits S on a set in $J \vee K$, modify x on the J -part of the hit set to avoid S . The modified point remains in $[A]_J$ but has only K -small hits, contradiction. $\square$

If A is J -saturated, this gives the saturation collapse:

$$A = [A]_J \Rightarrow (A \in Z_U^K \iff A \in Z_U^{J \vee K}).$$

In particular, on J -saturated events,

$$A \in Z_U^{\text{Fin}} \iff A \in Z_U^J.$$

6.2 Modal operators

Define

$$\Diamond_J A = [A]_J, \quad \Box_J A = X \setminus \Diamond_J (X \setminus A).$$

Then

$$\Diamond_J \Diamond_K = \Diamond_{J \vee K}, \quad \Box_J \Box_K = \Box_{J \vee K}.$$

For alarm hulls and safety regions

$$D_B^K = \{x : \{n : x(n) \notin B_n\} \in K\}, \quad B_n \in U_n,$$

we have

$$\Box_J C_S^K = C_S^{J \vee K}$$

and

$$\Diamond_J D_B^K = D_B^{J \vee K}.$$

Therefore

$$A \in Z_U^{J \vee K} \iff \exists B_n \in U_n \text{ such that } \Diamond_J D_B^K \cap A = \emptyset.$$

This is the robust shield equation: external J -noise plus internal K -fault tolerance equals $J \vee K$ -safety.

6.3 Alarm and safety phase transitions

For a slalom S , let $\text{supp}(S) = \{n : S_n \neq \emptyset\}$. Then

$$\diamond_J C_S^K = \begin{cases} C_S^K, & J \upharpoonright \text{supp}(S) \subseteq K, \\ X, & \text{otherwise.} \end{cases}$$

Dually, for a protocol B , let $\text{vul}(B) = \{n : B_n \neq \kappa_n\}$. Then

$$\square_J D_B^K = \begin{cases} D_B^K, & J \upharpoonright \text{vul}(B) \subseteq K, \\ \emptyset, & \text{otherwise.} \end{cases}$$

A basic alarm or safety protocol is either stable under the allowed noise or collapses completely.

6.4 Asynchronous recurrence

For two slaloms,

$$C_{S \cap T}^J \subseteq C_S^J \cap C_T^J.$$

Equality for all S, T holds iff J^+ is closed under finite intersections, equivalently iff J is prime. For $J = \text{Fin}$, equality fails: two alarms can each recur infinitely often while never firing together.

Countable OR also fails:

$$\bigcup_k C_{S^k}^J \subsetneq C_{\bigvee_k S^k}^J$$

for every proper $J \supseteq \text{Fin}$. A master observer may catch a branch by switching tests over time even when no fixed observer catches recurrently.

6.5 Cross-ideal implication

Let $M_S = \{n : S_n \neq \emptyset\}$ and $R_{S,T} = \{n : S_n \subseteq T_n\}$. Then

$$C_S^K \subseteq C_T^L \iff \forall H \subseteq M_S, H \notin K \Rightarrow H \cap R_{S,T} \notin L.$$

For full-support S , this simplifies to

$$C_S^K \subseteq C_T^L \iff L \subseteq K \text{ and } S \leq_K T.$$

Hence for full-support nonempty S ,

$$C_S^K \in Z_U^L \iff L \subseteq K.$$

No slalom can amplify weak recurrence into a stronger recurrence standard.

7 Schedule Factorization and Temporal Transport

7.1 One-monitor schedule semantics

Let S be a full-support nondegenerate slalom, meaning $S_n \in I_n$ and $\kappa_n \setminus S_n \in U_n$. Define the hit-schedule map

$$h_S : X \rightarrow \mathcal{P}(\omega), \quad h_S(x) = \{n : x(n) \in S_n\}.$$

For $B \subseteq \mathcal{P}(\omega)$, set

$$A_B^S = h_S^{-1}(B).$$

Theorem 7.1 (Hit-schedule factorization). *For every ideal $L \supseteq \text{Fin}$,*

$$A_B^S \in Z_U^L \iff B \cap L = \emptyset.$$

Proof sketch. If $B \cap L = \emptyset$, then every allowed schedule in B is L -positive, so the original slalom S catches A_B^S . Conversely, if $H \in B \cap L$, construct a branch with schedule exactly H , and outside H also avoid any proposed witnessing slalom. This uses that $\kappa_n \setminus S_n$ is large, so it cannot be swallowed by a small trap. $\square$

Modulo Z_U^L , only the L -small schedules matter:

$$A_B^S =_{Z_U^L} A_C^S \iff B \cap L = C \cap L.$$

The monitor skeleton quotient is isomorphic to $\mathcal{P}(L)$.

7.2 Schedule blur

Define schedule blur

$$\nabla_J B = \{H : \exists G \in B, H \triangle G \in J\}.$$

Then

$$\Diamond_J A_B^S = A_{\nabla_J B}^S.$$

The blur-join law reduces to the schedule identity

$$\nabla_J B \cap K = \emptyset \iff B \cap (J \vee K) = \emptyset.$$

7.3 Finite-alphabet monitor theorem

Suppose each coordinate is partitioned into cells

$$\kappa_n = G_n \sqcup R_n^1 \sqcup \dots \sqcup R_n^m,$$

where $G_n \in U_n$ is the good cell and each $R_n^i \in I_n$ is a rare signal cell. Let $\sigma(x)(n) \in \{0, 1, \dots, m\}$ be the observed signal.

For $B \subseteq \{0, 1, \dots, m\}^\omega$,

$$A_B^\sigma = \{x : \sigma(x) \in B\}.$$

Then

$$A_B^\sigma \in Z_U^J$$

iff there are finite rare-letter selectors $M_n \subseteq \{1, \dots, m\}$ such that every word $w \in B$ satisfies

$$\boxed{\{n : w(n) \in M_n\} \notin J.}$$

Thus finite symbolic nullity becomes a recurrent set-cover problem over rare letters.

7.4 Temporal/index transport

Let $\phi : \omega_Y \rightarrow \omega_X$ be a coordinate-time map. Define the transported ideal

$$\phi_* K = \{B \subseteq \omega_X : \phi^{-1}(B) \in K\}.$$

If

$$F(x)(m) = g_m(x(\phi(m)))$$

and local small-preimage preservation holds, with each fiber $\phi^{-1}(\{n\})$ small enough for I_n -additivity, then

$$\boxed{B \in Z_Y^K \Rightarrow F^{-1}(B) \in Z_X^{\phi_* K}.$$

If local small-image preservation holds and $\phi_* K \subseteq L$, then

$$\boxed{A \in Z_X^L \Rightarrow F[A] \in Z_Y^K.}$$

For $K = \text{Fin}$, $\phi_* \text{Fin}$ contains all finite sets iff ϕ is finite-to-one. Infinite repetition creates artificial recurrence.

8 Coordinate Sensors and Rough Observation

8.1 Universal pullback semantics

Let $\sigma_n : X_n \rightarrow Y_n$ be onto coordinate sensors, and $\sigma = \prod_n \sigma_n$. For local ideals I_n , define

$$K_n^\forall = \{M \subseteq Y_n : \sigma_n^{-1}(M) \in I_n\}.$$

Theorem 8.1 (Exact coarse-graining). *For every $B \subseteq \prod_n Y_n$,*

$$\sigma^{-1}(B) \in Z_I^J \iff B \in Z_{K^\forall}^J.$$

Proof sketch. A symbolic slalom $M_n \in K_n^\forall$ pulls back to I_n -small traps. Conversely, given source traps S_n , define

$$M_n = \{y : \sigma_n^{-1}(\{y\}) \subseteq S_n\}.$$

Then $M_n \in K_n^\forall$. If an observed word y were hit by M_n only on a J -small set, choose a lift avoiding S_n elsewhere, contradicting source capture. $\square$

Thus pullback embeds the symbolic quotient:

$$\mathcal{P}(Y)/Z_{K^\forall}^J \hookrightarrow \mathcal{P}(X)/Z_I^J.$$

8.2 Existential image semantics

Define the existential representative ideal

$$K_n^\exists = \{M \subseteq Y_n : \exists S \in I_n, M \subseteq \sigma_n[S]\}.$$

Then $K_n^\forall \subseteq K_n^\exists$.

Theorem 8.2 (Null-image theorem).

$$B \in Z_{K^\exists}^J \iff \exists A \in Z_I^J \text{ such that } \sigma[A] = B.$$

Equivalently,

$$\{\sigma[A] : A \in Z_I^J\} = Z_{K^\exists}^J.$$

Proof sketch. If $A \subseteq C_S^J$, then $\sigma[A]$ is caught by $M_n = \sigma_n[S_n] \in K_n^\exists$. Conversely, if B is caught by $M_n \in K_n^\exists$, choose $S_n \in I_n$ with $M_n \subseteq \sigma_n[S_n]$, and choose one lift x_y of each $y \in B$ using representatives inside S_n whenever $y(n) \in M_n$. Then $A = \{x_y : y \in B\}$ is null and $\sigma[A] = B$. $\square$

8.3 Rough nullity trichotomy

An observed event $B \subseteq Y$ has lifts $A \subseteq X$ with $\sigma[A] = B$. Combining pullback and null-image laws:

$$B \in Z_{K^\forall}^J \iff \text{every lift of } B \text{ is null.}$$

$$B \in Z_{K^\exists}^J \iff \text{some lift of } B \text{ is null.}$$

Therefore:

- $B \in Z_{K^\forall}^J$: definitely null.
- $B \in Z_{K^\exists}^J \setminus Z_{K^\forall}^J$: ambiguous, with null and non-null source explanations.
- $B \notin Z_{K^\exists}^J$: definitely non-null; no null lift exists.

The ambiguity band is

$$\text{Amb}_\sigma^J = Z_{K^\exists}^J \setminus Z_{K^\forall}^J.$$

It is the precise mathematical region where the sensor can hallucinate source rarity.

8.4 Robust rough trichotomy

Let H be a coordinate-noise ideal and put $L = H \vee J$. Then:

$$B \in Z_{K^\forall}^L \iff \text{every lift } A \text{ satisfies } \Diamond_H A \in Z_I^J.$$

$$B \in Z_{K^\exists}^L \iff \text{some lift } A \text{ satisfies } \Diamond_H A \in Z_I^J.$$

The same three-valued observation logic holds after replacing J by the effective ideal $H \vee J$.

8.5 Aliasing locality and null sections

Let

$$\Delta_\sigma = \{n : K_n^\exists \neq K_n^\forall\}.$$

Then

$$Z_{K^\exists}^L = Z_{K^\forall}^L \iff \Delta_\sigma \in L.$$

Aliasing is harmless exactly when it lies inside the effective budget L .

A coordinatewise section is a right inverse $r : Y \rightarrow X$ with $\sigma \circ r = \text{id}_Y$. There is a null section iff

$$Y \in Z_{K^\exists}^J.$$

Robustly,

$$Y \in Z_{K^\exists}^{H \vee J} \iff \exists r : Y \rightarrow X \text{ with } \Diamond_{Hr}[Y] \in Z_I^J.$$

Thus a null source set may contain one representative of every observed behavior.

8.6 Subcritical sensor collapse

In the measurable spine, if $|Y_n| < \kappa_n$, then κ_n -completeness forces a unique default symbol d_n with

$$\sigma_n^{-1}(\{d_n\}) \in U_n.$$

The pushforward ultrafilter is principal at d_n , and

$$\sigma^{-1}(B) \in Z_U^J \iff \forall y \in B, \{n : y(n) \neq d_n\} \notin J.$$

Subcritical observed nullity is recurrent deviation from the default trace.

Existentially, subcritical sensors are maximally wild: if this occurs on a J -positive coordinate set, every observed event has a null source lift.

9 Faithful Sensors, Nyquist Laws, and Fusion

9.1 Faithful sensor theorem

For a coordinate sensor $\sigma : \kappa \rightarrow Y$ and ultrafilter U , let $V = \sigma_* U$. Let

$$\text{Sing}(\sigma) = \{y \in Y : |\sigma^{-1}(\{y\})| = 1\}.$$

Theorem 9.1 (Local faithful sensor). *σ preserves small saturations,*

$$S \in U^* \Rightarrow \sigma^{-1}(\sigma[S]) \in U^*,$$

iff

$$\boxed{\text{Sing}(\sigma) \in V.}$$

Equivalently, σ is one-to-one over a V -large set of observed symbols.

Proof sketch. If singleton fibers are V -large and S is small, then $\sigma[S]$ cannot be V -large, since its intersection with singleton fibers would pull back to a large subset of S . Conversely, if singleton fibers are not V -large, choose two disjoint selectors through the V -large set of nonsingleton fibers; one selector is U -small but its saturation is U -large. $\square$

Globally,

$$\boxed{\sigma[Z_U^J] \subseteq Z_{\sigma_* U}^J \iff \{n : \text{Sing}(\sigma_n) \notin \sigma_{n*} U_n\} \in J.}$$

Thus observation is exact backward always, but faithful forward only when almost injective modulo J .

9.2 Universal/existential ultrafilter dichotomy

For ultrafilter coordinates,

$$K_n^\forall = (\sigma_{n*} U_n)^*.$$

Moreover,

$$K_n^\exists = \begin{cases} K_n^\forall, & \text{Sing}(\sigma_n) \in \sigma_{n*} U_n, \\ \mathcal{P}(Y_n), & \text{otherwise.} \end{cases}$$

Thus a coordinate is either faithful at the existential level or completely ambiguous.

If nonfaithful coordinates are J -positive, then

$$\boxed{\{\sigma[A] : A \in Z_U^J\} = \mathcal{P}(Y).}$$

Every observed event is the image of a source-null event.

9.3 Binary anomaly certificate

Every null proof can be represented by a binary monitor:

$$\boxed{A \in Z_U^J \iff \exists \sigma_n : \kappa_n \rightarrow \{0, 1\} \text{ with } \sigma_n^{-1}(\{1\}) \in I_n \text{ and } \sigma[A] \subseteq \text{Rec}_J(1).}$$

Here

$$\text{Rec}_J(1) = \{z \in 2^\omega : \{n : z(n) = 1\} \notin J\}.$$

This reconciles two facts: finite monitors can certify chosen null sets, but a fixed finite observation channel cannot faithfully preserve arbitrary source positivity.

9.4 Conditional Nyquist

For a base sensor $\sigma : \kappa \rightarrow Y$, a refinement $\tau : \kappa \rightarrow \Theta$ fuses to (σ, τ) . Define

$$\text{Iso}(\sigma, \tau) = \{\alpha : (\sigma, \tau)^{-1}(\{(\sigma(\alpha), \tau(\alpha))\}) = \{\alpha\}\}.$$

The fiber-resolution number is

$$\text{res}_U(\sigma) = \min\{|\Theta| : \exists \tau : \kappa \rightarrow \Theta, \text{Iso}(\sigma, \tau) \in U\}.$$

Given bit budgets μ_n and effective ideal $L = H \vee J$, robust faithful refinement exists iff

$$\{n : 2^{\mu_n} < \text{res}_{U_n}(\sigma_n)\} \in L.$$

For a trivial sensor, $\text{res}_U = \kappa$, so faithful observation from scratch needs $\log_2 \kappa$ bits. After a high-arity coarse sensor, the cost may be much smaller: it is the cost of splitting remaining recurrent aliases.

9.5 One-bit or full-width resolution

A partial σ -transversal is a set $T \subseteq \kappa$ meeting each fiber in at most one point. The sensor is U -sectionable if some partial transversal lies in U .

For a uniform κ -complete ultrafilter,

$$\text{res}_U(\sigma) = \begin{cases} 1, & \text{already faithful,} \\ 2, & U\text{-sectionable but not already faithful,} \\ \kappa, & \text{not } U\text{-sectionable.} \end{cases}$$

Thus below κ , unresolved aliasing is either one-bit repairable or full-width.

9.6 Normal selector theorem

If U is normal, then every map $f : \kappa \rightarrow Y$ is either constant on a U -large set or injective on a U -large set.

Proof sketch. Let $M = \{\alpha : f(\alpha) \notin f[\alpha]\}$, the first-occurrence set. If $M \in U$, f is injective on M . If not, on $R = \kappa \setminus M \in U$, define the regressive function $g(\alpha) = \min\{\beta < \alpha : f(\beta) = f(\alpha)\}$. Normality gives a U -large level set of g , contained in one fiber of f . $\square$

Therefore, for normal U ,

$$\text{res}_U(\sigma) = \begin{cases} 1, & \text{SingSrc}(\sigma) \in U, \\ \kappa, & \exists y, \sigma^{-1}(\{y\}) \in U, \\ 2, & \text{otherwise.} \end{cases}$$

The middle case of unresolved nonprincipal aliasing is always one-bit repairable.

9.7 Normal fusion phase diagram

For a normal coordinate sensor, define:

$$F = \text{already faithful}, \quad N = \text{nonprincipal but not faithful}, \quad P = \text{principal/default-collapsed}.$$

The costs are

$$F : 0 \text{ bits}, \quad N : 1 \text{ bit}, \quad P : \log_2 \kappa \text{ bits}.$$

For fewer than κ fused sensors, the fused sensor has a U -large default fiber iff every component has one. Thus principal/default collapse survives exactly while all fused sensors are principal. Nonprincipal information is contagious.

Globally, with $L = H \vee J$, robust faithful observation with bit budgets μ_n is possible iff

$$\boxed{\{n \in N : \mu_n = 0\} \cup \{n \in P : 2^{\mu_n} < \kappa_n\} \in L.}$$

One bit everywhere suffices iff $P \in L$.

9.8 Sensor portfolio cover theorem

Let $\mathcal{S}$ be a portfolio of available sensors. For each sensor σ , let

$$P_\sigma = \{n : \sigma_n \text{ has a } U_n\text{-large fiber}\}$$

and

$$R_\sigma = \omega \setminus P_\sigma.$$

For a subfamily Γ of size $< \lambda$, let Σ^Γ be the fusion.

Theorem 9.2 (Portfolio cover). *Modulo the effective ideal $L = H \vee J$,*

$$\boxed{P_{\Sigma^\Gamma} =_L \bigcap_{\sigma \in \Gamma} P_\sigma.}$$

Hence

$$\boxed{\Sigma^\Gamma + \text{one repair bit is robustly faithful} \iff \bigcap_{\sigma \in \Gamma} P_\sigma \in L.}$$

Equivalently,

$$\boxed{\bigcup_{\sigma \in \Gamma} R_\sigma \text{ covers } \omega \text{ modulo } L.}$$

Thus sensor design reduces to set cover in the quotient Boolean algebra $\mathcal{P}(\omega)/L$. Define

$$\text{npcov}_L(\mathcal{S}) = \min \left\{ |\Gamma| : \omega \setminus \bigcup_{\sigma \in \Gamma} R_\sigma \in L \right\}.$$

This is the exact number of portfolio sensors needed so that, after fusion, one additional repair bit suffices.

10 Corroborative Finite Experiments

Finite nonprincipal ultrafilters do not exist. The experiments below are finite shadows: threshold ideals, principal large-point models, and finite ignored-coordinate ideals. They corroborate algebraic shapes but do not prove transfinite claims.

10.1 Core finite-shadow data

Experiment	Finite model	Instances/checks	Failures	Meaning
Derivative/perfect subtree	threshold survival at each node	16 to 1000 sampled sub-sets across products up to $(3, 3, 3)$	0	root survival equals existence of large branching subtree
Rank recursion	threshold derivative rank	300 random trees per configuration	0	finite rank = 1 + threshold-order statistic of child ranks
Incidence equation	finite budgets, $J = \mathcal{P}(D)$	up to 197,568 containments	0	product containment iff coordinate inclusion on recurrent coordinates
Fubini collapse	two-dimensional Fubini smallness	up to 1,280 finite sets	original budgets failed; prefix-expanded 0 failures	later additivity must absorb previous histories
Support cellularity	principal finite model	exact antichain computations	0	max antichain = product over small support

Experiment	Finite model	Instances/checks	Failures	Meaning
Strategy flattening	finite adaptive forbidden sets	100 random strategies per configuration	0	flattened size matches history blow-up
Slalom order	finite ignored-coordinate ideals	exhaustive/sampled up to 5,000 pairs	0	$C_S \subseteq C_T$ iff $S \leq_J T$
Blur-join	finite products, finite ideals	35,584 instances	0	$[A]_J \in Z^K$ iff $A \in Z^{J \vee K}$
Cross-ideal implication	finite slaloms	32,000 sampled checks	0	positive-lens criterion
Coarse-graining pullback	random coordinate maps	up to 25,600 checks	0	exact sensor pullback theorem
Rough trichotomy	finite sensors	ambiguous cases explicitly constructed	0	lower/upper nullity band works
Modal naturality	blur and sensor pullback/image	500 random tests per configuration	0	coordinate sensors commute with coordinate blur
Portfolio cover	random $F/N/P$ sensor portfolios	2,000 trials per row	0	one-bit design = quotient set cover

10.2 Selected finite tables

Table 3: Adaptive-to-blind flattening: max flattened sizes matched past-history blow-up.

Product sizes	Budgets	Trials	Failures	Max flattened / predicted
(3, 3)	(1, 1)	100	0	(1, 3)/(1, 3)
(3, 3, 3)	(1, 1, 1)	100	0	(1, 3, 3)/(1, 3, 3)
(4, 3, 2)	(1, 1, 1)	100	0	(1, 3, 2)/(1, 3, 2)
(4, 3, 3)	(1, 2, 1)	100	0	(1, 3, 3)/(1, 3, 3)

10.3 Representative code snippets

Table 4: Product-cylinder hazard formula in finite shadow.

Allowed sizes	Small coordinates	Predicted ranks
(2, 5, 5, 1, 5, 5, 5, 1)	{0, 3, 7}	(1, 3, 2, 1, 4, 3, 2, 1)
(5, 5, 1, 5, 1, 5, 5, 5)	{2, 4}	(3, 2, 1, 2, 1, ∞ , ∞ , ∞)
(5, 5, 5, 5, 5)	$\emptyset$	(∞ , ∞ , ∞ , ∞ , ∞)

Table 5: Sensor/fusion finite shadows.

Experiment	Model	Checks	Failures
Derivative commutes with sensor pullback	random sensors + trees	2,000 per row	0
Subcritical sensor law	finite principal models	15,000 languages	0
Image-preservation criterion	random sensors, exhaustive subsets	50–100 sensors	0
Local faithful sensor theorem	finite principal analogue	5,000 sensors per row	0
Null universal section	random sensors	50 sensors per row	0
Aliasing locality	product-cylinder witnesses	100 sensors per row	0

Listing 1: Finite incidence equation check for $J=P(D)$.

```

lhs = all(
    any(x[n] in Svec[n] for n in range(N) if n not in D)
    for x in Q_A
)
rhs = any(Avec[n].issubset(Svec[n]) for n in range(N) if n not in D)
assert lhs == rhs

```

Listing 2: Finite strategy flattening.

```

# Adaptive forbidden sets are indexed by histories t of length n.
# Flatten by unioning over all histories at coordinate n.
flat[n] = union(strategy[(n,t)] for t in histories[n])
# Transfinite theorem: this union remains small when
# |histories[n]| < add(I_n).

```

Listing 3: Finite portfolio cover search.

```

# F/N/P phases per sensor-coordinate.
# One-bit repair is possible iff selected sensors cover every
# nonignored coordinate by F or N, i.e. avoid P everywhere modulo L.
for Gamma in subfamilies(sensors):
    if all(any(phase[s][n] in {'F','N'}) for s in Gamma)
        for n in relevant_coordinates):
            record_cover(Gamma)

```

Table 6: Portfolio-cover simulation. 0-bit requires faithful cover; 1-bit requires nonprincipal cover.

Coords	Ignored	Sensors	$p(F)$	$p(N)$	0-bit possible	1-bit possible
10	2	8	0.15	0.35	8.2%	97.2%
10	2	8	0.05	0.55	0.0%	99.5%
12	3	10	0.10	0.30	2.4%	95.0%
12	3	10	0.20	0.20	35.8%	94.8%
16	4	14	0.08	0.42	0.9%	99.8%

11 Conjectures, Open Targets, and Speculative Directions

11.1 Analytic perfect-box theorem

Conjecture 11.1 (Analytic perfect box). *For suitable λ -analytic sets in the singular product,*

$$A \notin Z_U \implies A \text{ contains a full } U\text{-box } \prod_n B_n, \quad B_n \in U_n.$$

Projection catastrophe shows that this cannot be proved by naive projection from closed witnesses: positive observed sets can be projections of closed null source sets. A proof would need a spinal uniformization principle selecting non-null fibers.

11.2 Spinal Prikry property

For $J = \text{Fin}$, positive-cylinder forcing is densely equivalent to diagonal Prikry forcing. The target is a pure-extension decision theorem:

$$\forall p, \varphi \quad \exists q \leq^* p \quad q \text{ decides } \varphi.$$

The category/control/slalom compression results are strong evidence for this, but a full forcing proof remains a target.

11.3 Rank spectrum beyond closed Fin-nullity

The closed Fin-null rank collapses to finite countdown ranks because the derivative stabilizes at ω . Transfinite rank phenomena may reappear in:

- longer-than-countable coordinate spines;
- filters without countable completeness;
- coanalytic/projective ranks;
- J -structured recurrence clocks for ideals larger than Fin .

A remaining target is to formulate the correct J -clock rank theory for closed Z_U^J -null sets.

11.4 PCF/Tukey formulas

The cardinal invariant dictionary suggests formulas such as

$$\text{cof}(Z_U^J) \approx \text{cf} \left(\prod_n \chi(U_n)/J \right),$$

where $\chi(U_n)$ is the character of U_n . This is a PCF/Tukey problem: compute bounding and dominating numbers of products of local ideals modulo recurrence ideals.

11.5 Sensor-selective ultrafilters

Define an ultrafilter U to be sensor-selective if every partition into U -small fibers admits a U -large transversal. Normal ultrafilters have this property by the Normal Selector Lemma. A likely equivalence is:

$$U \text{ sensor-selective} \iff \text{every nonprincipal pushforward of } U \text{ is Rudin-Keisler equivalent to } U.$$

This should be investigated as a local large-cardinal/ultrafilter property.

11.6 Zero-bit synergy

One repair bit reduces fusion design to set cover of nonprincipal supports. Without the repair bit, one needs the fused equivalence relation to have singleton classes on a U_n -large set. The target is to classify when a family of nonfaithful sensors becomes jointly faithful:

$$\bigcap_{\sigma \in \Gamma} E_{\sigma,n} = \Delta \quad \text{on a } U_n\text{-large set.}$$

This is a joint-identification problem, not merely a support-cover problem.

11.7 High-fusion singular boundary

The portfolio fusion theorem assumes $|\Gamma| < \lambda$, so $|\Gamma| < \kappa_n$ for all sufficiently large n . For $|\Gamma| \geq \lambda$, intersections of default fibers can exceed local completeness. It is plausible that λ -many subcritical sensors can break default collapse even though every $< \lambda$ -fusion cannot. This is a genuine singular-cardinal boundary.

11.8 Commuting projection analogue

A speculative direction is a commuting tensor-product history model where coordinate subsets are replaced by projections and local small ideals by projection ideals. The realistic first target is the commuting case, seeking projection analogues of tomography, incidence, observer compression, and Lyapunov ranks. Noncommuting/contextual variants are much harder and should not be overclaimed.

12 Applications and Interpretation

12.1 Set theory and generalized descriptive set theory

The tame ideal Z_U provides a candidate null side for singular generalized Baire spaces at countable cofinality. It has simultaneous forms:

limsup slalom ideal, Prikry/Ellentuck category ideal, forcing quotient ideal, control avoidance ideal.

This convergence is evidence that the construction is canonical in the $J = \text{Fin}$ setting.

12.2 Verification and control

A closed bad set is negligible iff there is a finite countdown Lyapunov rank and recurrent small alarms on every bad run. In finite threshold shadows, this gives computable robust-safety algorithms:

$$\rho(s) = 1 + \text{threshold-order statistic of child ranks.}$$

Monitoring interpretation:

bad behavior is negligible iff it forces recurrent locally rare alarms.

12.3 Anomaly detection

The sensor theory separates nullity certificates from positivity certificates:

- A set is null iff it admits a binary recurrent-anomaly certificate designed for that set.
- A fixed finite monitor cannot certify source positivity from observed possibility in a measurable spine.
- Positivity certificates require resolving recurrent aliasing: break principal/default collapse and select nonprincipal aliases.

The practical law is:

finite monitors are valid for safety certificates, not for typicality certificates.

12.4 Sensor design

For normal local ultrafilters, each coordinate is in phase F, N, P . Robust design is:

cover P modulo $H \vee J$; one bit selects the remaining N .

Thus minimum-cost sensor design becomes weighted set cover in $\mathcal{P}(\omega)/(H \vee J)$.

13 Compact Equation Index

Name	Equation
Spinal nullity	$A \in Z_U^J \iff A \subseteq C_S^J \text{ for one } I_n\text{-small slalom.}$
Orthogonal filter	$Z_U^J = (F_U^J)^\perp.$
Trace	$P_E \in Z_U^J \iff E \notin J.$
Tomography	$\prod_n A_n \in Z_U^J \iff \{n : A_n \in I_n\} \notin J.$
Incidence	$Q_A \subseteq C_S^J \iff \{n : A_n \subseteq S_n\} \notin J.$
Slalom order	$C_S^J \subseteq C_T^J \iff S \leq_J T.$
Observer compression	$\bigcup_{S \in \Pi} C_S^J \in Z_U^J \iff \Pi \text{ is } \leq_J\text{-bounded.}$
Causal collapse	causal adaptive J -slaloms = blind J -slaloms under $ \text{past}_n < \text{add}(I_n).$
Closed finite rank	$[T] \in Z_U \iff T^\omega = \emptyset \iff T \text{ has finite countdown rank.}$
Product hazard	$\rho(s) = 1 + \text{distance to next local bottleneck.}$
Blur-join	$[A]_J \in Z_U^K \iff A \in Z_U^{J \vee K}.$
Modal alarm	$\Box_J C_S^K = C_S^{J \vee K}.$
Modal safety	$\Diamond_J D_B^K = D_B^{J \vee K}.$
Asynchronous meet	$C_S^J \cap C_T^J = C_{S \cap T}^J \text{ for all } S, T \text{ iff } J \text{ is prime.}$
Cross-ideal implication	Full support: $C_S^K \subseteq C_T^L \iff L \subseteq K \text{ and } S \leq_K T.$
Schedule factor	$h_S^{-1}(B) \in Z_U^L \iff B \cap L = \emptyset.$
Sensor pullback	$\sigma^{-1}(B) \in Z_I^J \iff B \in Z_{K^\vee}^J.$
Sensor image	$B = \sigma[A] \text{ for some } A \in Z_I^J \iff B \in Z_{K^\exists}^J.$
Rough trichotomy	$Z_{K^\vee}^J \subseteq Z_{K^\exists}^J$: definitely null / ambiguous / definitely positive.
Robust roughness	replace J by $H \vee J.$
Faithful sensor	$\sigma[Z_U^J] \subseteq Z_{\sigma_* U}^J \iff \text{Sing}(\sigma_n) \in \sigma_n * U_n \text{ for all but } J\text{-small many } n.$
Subcritical collapse	$ Y_n < \kappa_n \Rightarrow \sigma_n * U_n \text{ is principal at a default symbol.}$
Conditional Nyquist	$\{n : 2^{\mu_n} < \text{res}_{U_n}(\sigma_n)\} \in H \vee J.$
Normal resolution	$\text{res}_U(\sigma) = 1, 2, \kappa \text{ according to faithful / nonprincipal / principal phases.}$
Portfolio cover	$\Sigma^\Gamma + 1 \text{ bit faithful} \iff \bigcap_{\sigma \in \Gamma} P_\sigma \in H \vee J.$

14 Final Synthesis

The theory began with one definition:

$$A \in Z_U^J \iff \exists S_n \in I_n \forall x \in A, \{n : x(n) \in S_n\} \notin J.$$

It now has several mutually reinforcing interpretations:

recurring local rarity

large-protocol avoidability

Prikry/Ellentuck category for $J = \text{Fin}$

closed finite countdown safety

modal robustness by ideal join

rough observation under sensors

information-theoretic sensor design modulo $H \vee J$.

The most important corrections and discoveries were:

1. Closed Fin-nullity in the countably complete countable-spine setting is finite-rank, because the derivative stabilizes at ω .
2. Future-looking observation is strictly stronger than spinal nullity; causality is essential.
3. Projection can send closed null sets onto everything; analytic regularity must use uniformization, not naive projection.
4. Sensor observation is exact backward but rough forward: universal and existential observed smallness differ.
5. Finite/subcritical sensors can certify nullity but cannot certify source positivity.
6. Under normality, nonprincipal aliasing is one-bit repairable; principal/default collapse needs full critical-width information.
7. Sensor portfolio design reduces to set cover modulo the effective ideal $H \vee J$.

The compact final slogan is:

spinal nullity is recurrence logic of local rarity,

and observation is faithful exactly when recurrent aliasing is resolved.

Round 2 Addendum: Critical Default-Breaking and Zero-Bit Synergy

Continuing the spinal nullity program at singular infinity

April 29, 2026 - exploratory draft, appended to the consolidated report

Status note. This addendum continues the same exploratory program as the consolidated report. The main result below is a theorem under the standing spine hypotheses: a family of individually default-collapsed binary sensors cannot break default collapse below the local critical cardinal, but a critical-width family can become fully faithful. The finite computations are finite shadows and sanity checks; they are not proofs of the transfinite statements.

1. Why this round looked here

The previous package ended with two unresolved sensor questions: zero-bit synergy and the high-fusion singular boundary. The one-bit portfolio theorem said that, after enough nonprincipal support has been fused, one extra repair bit per coordinate can make observation faithful. It did not answer whether many sensors with individually negligible or default behavior can become faithful with no repair bit.

The blind spot is this: a single default-collapsed sensor has essentially no positive-observation power, so it is tempting to discard it. But a critical number of such sensors can code identity by rare exceptions. This produces a sharp singular threshold.

Exceptional finding of this round. Individually default-collapsed binary sensors are powerless under any subcritical fusion, but at exactly the local critical width they can become fully faithful. Globally along the singular spine, the critical width is the singular cardinal

$$\mathfrak{db}_{\text{spine}}^0(\mathcal{U}, J) = \lambda = \sup_{n < \omega} \kappa_n.$$

This turns the prior high-fusion boundary from a speculative target into a precise theorem for default binary portfolios.

2. Local default-breaking number

Let U be a nonprincipal κ -complete ultrafilter on κ . A coordinate sensor $\sigma : \kappa \rightarrow Y$ is called *principal/default-collapsed* if there is a symbol $d \in Y$ such that

$$D_\sigma = \sigma^{-1}(\{d\}) \in U.$$

A family Γ of sensors has fusion

$$\Sigma_\Gamma(\alpha) = \langle \sigma(\alpha) : \sigma \in \Gamma \rangle.$$

The fused sensor is locally faithful exactly when its singleton-fiber source set is U -large:

$$\text{Sing}(\Sigma_\Gamma) = \{\alpha < \kappa : \Sigma_\Gamma^{-1}(\{\Sigma_\Gamma(\alpha)\}) = \{\alpha\}\} \in U.$$

Definition 1 (Default-breaking number). Let $\mathfrak{db}^0(U)$ be the least size of a family Γ of binary principal/default-collapsed sensors such that Σ_Γ is faithful.

Theorem 2 (Local critical default-breaking law). *For every nonprincipal κ -complete ultrafilter U on κ ,*

$$\mathfrak{db}^0(U) = \kappa.$$

More precisely:

- (a) *any fusion of $< \kappa$ principal/default-collapsed sensors remains principal/default-collapsed;*
- (b) *there is a family of κ binary principal/default-collapsed sensors whose fusion is injective.*

Proof sketch. For (a), choose for each $\gamma \in \Gamma$ a default fiber $D_\gamma \in U$. If $|\Gamma| < \kappa$, then κ -completeness gives

$$D = \bigcap_{\gamma \in \Gamma} D_\gamma \in U.$$

Every point of D has the same fused output, so the fusion still has a U -large default fiber. Since U is nonprincipal, this large fiber prevents singleton-fiber faithfulness.

For (b), for each $\xi < \kappa$ define the binary singleton detector

$$\sigma_\xi(\alpha) = \begin{cases} 1, & \alpha = \xi, \\ 0, & \alpha \neq \xi. \end{cases}$$

The default fiber $\kappa \setminus \{\xi\}$ is in U , so each detector is principal/default-collapsed. But the fused vector $\langle \sigma_\xi(\alpha) : \xi < \kappa \rangle$ is the one-hot code of α , hence the fusion is injective. Therefore $\mathfrak{db}^0(U) = \kappa$.

3. Singular spine equation

Return to the standing spine

$$X = \prod_{n < \omega} \kappa_n, \quad \lambda = \sup_{n < \omega} \kappa_n,$$

with U_n a nonprincipal κ_n -complete ultrafilter and J a proper ideal on ω with $\text{Fin} \subseteq J$.

A *global default binary sensor* is a product sensor whose coordinate map is binary and principal/default-collapsed at every coordinate. A portfolio Γ of such sensors is globally faithful modulo J when the fused coordinate sensor is locally faithful for all but a J -small set of coordinates.

Theorem 3 (Singular critical fusion equation). *For every proper $J \supseteq \text{Fin}$,*

$$\mathfrak{db}_{\text{spine}}^0(\mathcal{U}, J) = \lambda.$$

That is, no portfolio of size $< \lambda$ made only of default binary sensors can be globally faithful, while a portfolio of size λ can be globally faithful.

Proof sketch. Let $|\Gamma| = \mu < \lambda$. Since the κ_n increase cofinally to λ , there is N such that $\mu < \kappa_n$ for all $n \geq N$. By the local law, at every such coordinate n the fusion of the μ default sensors still has a U_n -large default fiber and is not faithful. The bad coordinate set is cofinite, and no proper ideal containing Fin can contain a cofinite set. Hence Γ is not globally faithful modulo J .

For the upper bound, index sensors by pairs (n, ξ) with $\xi < \kappa_n$. There are

$$\sum_{n < \omega} \kappa_n = \lambda$$

such sensors. At coordinate m , the sensor (n, ξ) is constant 0 if $n \neq m$, and at coordinate n it is the singleton detector for ξ . Every coordinate therefore receives all singleton detectors for its own κ_n , so the fused coordinate map is injective for every n . The portfolio is globally faithful.

Corollary 4 (Critical-width discontinuity). *Default information has a transfinite cliff. Below κ_n fused default sensors, coordinate n still looks default-collapsed. At width κ_n , identity coding becomes possible. Along the singular spine, the global cliff is exactly λ .*

4. Exact zero-bit synergy criterion

The critical default breaker is one example of a more general zero-bit fusion law.

For a local sensor family Γ , define the separated-source set

$$\text{Sep}_\Gamma = \{\alpha < \kappa : \forall \beta \neq \alpha \exists \sigma \in \Gamma [\sigma(\alpha) \neq \sigma(\beta)]\}.$$

This is the set of points whose fused equivalence class is singleton.

Theorem 5 (Zero-bit synergy law). *For a local family Γ ,*

$$\boxed{\Sigma_\Gamma \text{ is faithful} \iff \text{Sep}_\Gamma \in U.}$$

For the full spine and effective recurrence ideal $L = H \vee J$,

$$\boxed{\Sigma_\Gamma[\mathcal{Z}_U^J] \subseteq \mathcal{Z}_{\Sigma_\Gamma * \mathcal{U}}^J \iff \{n : \text{Sep}_{\Gamma, n} \notin U_n\} \in L.}$$

When $H = \text{Fin}$ or no coordinate-noise ideal is being used, replace L by J .

Proof sketch. The fused fiber of α is singleton exactly when $\alpha \in \text{Sep}_\Gamma$. Applying the faithful sensor theorem to the fused map gives the local equivalence. The global equivalence is the coordinatewise local theorem modulo the effective ideal $L = H \vee J$.

Interpretation. Zero-bit synergy is not a support-cover problem. It is a separating-code problem: the chosen sensors must separate a U -large set of source points from all of their aliases. The earlier one-bit portfolio cover theorem is a useful relaxation; this law is the exact zero-bit criterion.

5. Finite experiments and sanity checks

Finite nonprincipal ultrafilters do not exist, so the tests below use finite threshold shadows. Let the finite universe have size k and let “small” mean size at most r . A set is treated as “large” when its complement is small. Singleton detectors have default fibers of size $k - 1$, so they are default-collapsed in this finite shadow.

5.1 Deterministic singleton-detector bank

Take the first m singleton detectors. Their common default class has size $k - m$. The singleton-source set has size m for $m < k$ and size k for $m = k$. Thus the finite phases are:

$$\text{P phase if } m \leq r, \quad \text{F phase if } m \geq k - r,$$

with an intermediate nonprincipal/nonfaithful band between them. In the transfinite κ -complete limit, both finite thresholds collapse to the single cardinal threshold κ .

Universe k	Small budget r	Principal/default range	Faithful range
8	1	$0 \leq m \leq 1$	$7 \leq m \leq 8$
12	2	$0 \leq m \leq 2$	$10 \leq m \leq 12$
16	2	$0 \leq m \leq 2$	$14 \leq m \leq 16$
24	3	$0 \leq m \leq 3$	$21 \leq m \leq 24$
32	4	$0 \leq m \leq 4$	$28 \leq m \leq 32$

5.2 Zero-bit synergy by row-column separation

A simple finite witness shows that two individually nonfaithful sensors can be jointly faithful. On $A \times B$, the row sensor has fibers of size $|B|$ and the column sensor has fibers of size $|A|$; neither has singleton fibers when both factors have size > 1 . Their fusion is the identity on $A \times B$.

Grid	Source size	Row singletons	Column singletons	Fused singletons	Max fused class
3×4	12	0	0	12	1
4×4	16	0	0	16	1
5×6	30	0	0	30	1

5.3 Random separating codes

Balanced random sensors are not default-collapsed, but they test the separating-code side of the zero-bit law. For q -ary random sensors, full finite separation appears after about the expected coding scale. The table reports 500 trials per row.

k	Alphabet q	Tolerance r	Mean sensors	Median	Max
16	2	0	8.09	8	17
16	4	0	4.41	4	9
32	2	0	10.36	10	18
32	4	0	5.44	5	11
64	2	0	12.27	12	18
64	4	0	6.39	6	11
32	2	2	8.83	9	13

Experiment verdict. The deterministic default-bank computations exactly match the finite phase equations. The row-column test confirms zero-bit synergy: no individual singleton fibers, but full separation after fusion. The random-code test confirms that zero-bit synergy behaves like a separating-code problem, not like a coordinate support-cover problem.

6. New applications suggested by the law

Sensor-bank design. A designer now has three mathematically different ways to obtain faithful observation at coordinate n :

1. use one genuinely faithful or high-width sensor;
2. use a nonprincipal sensor and add one repair bit, as in the normal phase diagram;
3. use κ_n many default binary singleton detectors, which is zero-bit after fusion but critical-width in portfolio size.

Any subcritical collection of default detectors is provably unable to certify source positivity. This is a strong negative design rule.

Privacy and re-identification. The theorem gives a transfinite version of a privacy cliff. Each singleton detector looks harmless: it only reports a rare exception and is default almost everywhere. A critical-width bank of such harmless logs becomes an identity code. The warning is: privacy risk is not additive in the apparent information of individual default channels; it can jump at the critical fusion width.

Distributed diagnostics. In a distributed monitoring system, many devices may each record one rare local signature. Below the local critical width the combined record is still default-collapsed; at critical width the record can become faithful. The law tells when a rare-event detector bank can move from safety certification to typicality or positivity certification.

Verification and anomaly detection. The old slogan was: finite monitors certify nullity but not positivity. The refined slogan is:

default monitors certify nullity; positivity needs nonprincipal information, repair bits, or critical-width default fusion.

This prevents overinterpreting sparse anomaly logs as evidence of typical behavior.

Set-theoretic invariant. The number $\mathfrak{db}_{\text{spine}}^0(\mathcal{U}, J) = \lambda$ is a new portfolio invariant of the spinal sensor calculus. It measures how many zero-typical-information binary channels are needed to reconstruct faithful source information. It is independent of the recurrence ideal J as long as J is proper and contains Fin .

7. Open targets after this round

1. **Exact zero-bit optimization.** Given a concrete portfolio, compute the least subfamily Γ such that $\text{Sep}_{\Gamma, n}$ is U_n -large modulo $H \vee J$. Finite shadows identify this as a large-set separating-code problem.
2. **PCF/Tukey version.** Replace singleton detectors by restricted families and compute the resulting default-breaking number. The unrestricted value is λ ; restricted portfolios may produce new reduced-product invariants.
3. **Normal-ultrafilter synergy.** Classify when several normal-phase N sensors become jointly faithful with no repair bit. The exact criterion is $\text{Sep}_{\Gamma} \in U$, but structural normal-form classifications remain open.
4. **Definable sensor banks.** The singleton-detector upper bound uses arbitrary indexed detectors. If the sensor bank must be definable or low-complexity, the critical number may rise or the upper construction may fail. This is a promising descriptive-set-theoretic refinement.

8. Round 2 summary

This round found a useful new law rather than only a new analogy. The law is:

$$\mathfrak{db}^0(U_n) = \kappa_n \quad \text{locally, and} \quad \mathfrak{db}_{\text{spine}}^0(\mathcal{U}, J) = \lambda \quad \text{globally.}$$

It explains exactly how many individually default-collapsed binary sensors are needed before zero-bit fusion can become faithful. Subcritical fusion is provably powerless; critical fusion can code identity by one-hot rare exceptions. The accompanying zero-bit synergy equation

$$\text{faithful zero-bit fusion} \iff \text{Sep}_\Gamma \in U$$

turns the previous zero-bit-synergy target into a separating-code problem.

The most useful application is a sharp design and privacy principle: sparse default logs may be safe below the critical width, but at the critical width they can reconstruct the source. This is the new practical warning carried by the transfinite theory.

Round 3 Addendum: Transversal Repair Spectrum and the Sensor Rank Equation

Continuing the spinal nullity program at singular infinity

April 29, 2026 - exploratory draft, appended to the consolidated report

Status note. This round continues the same draft research program. The main theorem below is a direct structural derivation from the sensor definitions: after a sensor fusion has created alias classes, the amount of additional repair alphabet needed is governed by how many partial transversals are required to cover a large set. I do not claim literature priority; the claim is that this is a useful new law inside the present spinal-nullity calculus. The finite computations are finite shadows and sanity checks, not proofs of the transfinite large-cardinal statements.

1. Why this round looked here

Round 2 found the critical default-breaking law

$$\mathfrak{db}^0(U) = \kappa, \quad \mathfrak{db}_{\text{spine}}^0(\mathcal{U}, J) = \lambda,$$

for default binary sensor banks. The remaining blind spot was what happens *after* an arbitrary fusion that is neither faithful nor fully default-collapsed. The previous theory had three important special cases:

- already faithful sensors need no repair;
- normal nonprincipal aliasing is one-bit repairable;
- principal/default collapse needs full critical-width repair.

The exceptional finding of this round is that these are not separate phenomena. They are the three visible endpoints of a single alias-transversal spectrum.

New principle. The missing repair alphabet is not controlled by entropy or by the number of observed symbols alone. It is controlled by the least number of partial alias transversals whose union is large. In slogan form,

repair capacity = large transversal cover of the alias relation.

This turns the conditional Nyquist law into an exact structural equation for arbitrary fused sensors.

2. Alias relations and repair symbols

Let U be a nonprincipal κ -complete ultrafilter on κ . A fused local sensor is a map

$$\Sigma : \kappa \longrightarrow Y.$$

Its alias relation is

$$\alpha E_{\Sigma} \beta \iff \Sigma(\alpha) = \Sigma(\beta).$$

A repair sensor with alphabet θ is a map $\tau : \kappa \rightarrow \theta$. It makes a point α isolated when no other point in the same alias class receives the same repair symbol:

$$\text{Iso}(E_{\Sigma}, \tau) = \{\alpha < \kappa : \forall \beta \neq \alpha (\beta E_{\Sigma} \alpha \Rightarrow \tau(\beta) \neq \tau(\alpha))\}.$$

The repaired fusion (Σ, τ) is faithful modulo U iff $\text{Iso}(E_{\Sigma}, \tau) \in U$.

Definition 1 (Repair-symbol number). *For an equivalence relation E on κ , define*

$$\text{res}_U(E) = \min\{\theta : \exists \tau : \kappa \rightarrow \theta \text{ with } \text{Iso}(E, \tau) \in U\}.$$

Here $\theta = 1$ means no repair is needed: singleton alias classes already cover a U -large set.

Definition 2 (Large transversal cover number). *A set $T \subseteq \kappa$ is a partial E -transversal when*

$$|T \cap [\alpha]_E| \leq 1 \quad \text{for every } \alpha < \kappa.$$

Define

$$\text{tcov}_U(E) = \min\{\mu : \exists \langle T_i : i < \mu \rangle \text{ partial } E\text{-transversals with } \bigcup_{i < \mu} T_i \in U\}.$$

Theorem 1 (Transversal repair law). *For every equivalence relation E on κ ,*

$$\text{tcov}_U(E) \leq \text{res}_U(E) \leq \text{tcov}_U(E) + 1.$$

Consequently, whenever $\text{tcov}_U(E)$ is infinite,

$$\boxed{\text{res}_U(E) = \text{tcov}_U(E).}$$

For finite $\text{tcov}_U(E) = m$, the exact value is either m or $m + 1$; the extra symbol is a default “dump” color for non-isolated remainders.

Proof sketch. Suppose $\tau : \kappa \rightarrow \theta$ isolates a U -large set. For each color $i < \theta$, the set

$$T_i = \text{Iso}(E, \tau) \cap \tau^{-1}(\{i\})$$

meets each E -class at most once. Thus the isolated large set is covered by at most θ partial transversals, so $\text{tcov}_U(E) \leq \theta$.

Conversely, suppose $\langle T_i : i < \mu \rangle$ are partial transversals with $A = \bigcup_{i < \mu} T_i \in U$. Make them disjoint by replacing T_i with $T_i \setminus \bigcup_{j < i} T_j$. Color points of T_i by i and color everything outside A by one reserved default symbol. Then every point of A is isolated. This uses at most $\mu + 1$ symbols. If μ is infinite, $\mu + 1 = \mu$ cardinally, giving exact equality. $\square$

Interpretation. A repair symbol is useful only when it selects at most one representative from each unresolved alias class. Many repair symbols are therefore many partial transversals. Faithfulness is achieved exactly when these transversals cover a large set.

3. Unified Nyquist equation

Return to the singular spine $X = \prod_{n < \omega} \kappa_n$, with U_n on κ_n , monitoring ideal J , coordinate-noise ideal H , and effective ideal

$$L = H \vee J.$$

Let Σ_n be the fused sensor at coordinate n , and let $E_n = E_{\Sigma_n}$ be its local alias relation. A repair budget of μ_n bits gives

$$\theta_n = 2^{\mu_n}$$

repair symbols.

Theorem 2 (Sensor rank / Nyquist equation). *Robust faithful observation after repair is possible exactly when*

$$\boxed{\{n < \omega : 2^{\mu_n} < \text{res}_{U_n}(E_n)\} \in L.}$$

Equivalently, outside an L -small coordinate set, the repair alphabet must be large enough to cover a U_n -large subset of the alias classes by partial transversals.

Proof sketch. At coordinate n , the repaired sensor is faithful iff there is a 2^{μ_n} -symbol coloring isolating a U_n -large set, which is exactly $2^{\mu_n} \geq \text{res}_{U_n}(E_n)$. The spinal faithful-sensor theorem globalizes local failure modulo the effective ideal $L = H \vee J$. $\square$

This equation subsumes the earlier conditional Nyquist law. The earlier quantity $\text{res}_{U_n}(\sigma_n)$ is now identified as an alias-transversal invariant of E_{σ_n} .

Corollary 1 (Normal-ultrafilter collapse of the spectrum). *If U is normal and E is the kernel of a map $f : \kappa \rightarrow Y$, then*

$$\text{tcov}_U(E) \in \{1, \kappa\}.$$

More precisely:

- *if singleton E -classes cover a U -large set, then $\text{res}_U(E) = 1$;*
- *if there is no U -large alias class but singleton classes are not U -large, then $\text{tcov}_U(E) = 1$ and $\text{res}_U(E) = 2$;*
- *if some alias class is U -large, then $\text{tcov}_U(E) = \text{res}_U(E) = \kappa$.*

Proof sketch. The normal selector lemma says that f is either constant on a U -large set or injective on a U -large set. In the injective case there is a U -large partial transversal, so one repair bit suffices unless the sensor was already faithful. In the constant case, every partial transversal meets the U -large class in at most one point; fewer than κ such transversals cover only a U -small part of that class. $\square$

Corollary 2 (Round 2 recovered). *For a subcritical fusion of $< \kappa$ default-collapsed binary sensors, the common default fiber is one U -large alias class. Hence*

$$\text{tcov}_U(E) = \text{res}_U(E) = \kappa.$$

At the critical one-hot bank, E is equality and $\text{res}_U(E) = 1$.

Thus Round 2's critical cliff is the singular-cardinal limit of a repair-rank step function: below the critical bank, the residual alias class is still U -large and has full repair rank; at the critical bank, the alias relation collapses to equality.

4. Exact finite shadow: alias-class capacity equations

Finite models have no nonprincipal ultrafilters, but they give an exact shadow of the repair law. Let the finite source have k points. Let the alias classes of E have sizes

$$s_1, \dots, s_b, \quad \sum_{i=1}^b s_i = k.$$

Let “large” mean size at least $k - r$, where r is the allowed small remainder.

Transversal cover capacity. With m partial transversals, the most points one can cover is

$$T_E(m) = \sum_{i=1}^b \min(s_i, m).$$

Therefore

$$\boxed{\text{tcov}_r(E) = \min\{m : T_E(m) \geq k - r\}.$$

Repair-symbol isolation capacity. With q repair symbols, the maximum number of isolated points inside one alias class of size s is

$$I_q(s) = \begin{cases} s, & s \leq q, \\ q - 1, & s > q. \end{cases}$$

The reason is simple: if the class has more points than colors, at least one color must be repeated, and all points of that repeated color stop being isolated. Hence

$$I_E(q) = \sum_{i=1}^b I_q(s_i),$$

and the finite repair-symbol number is

$$\boxed{\text{res}_r(E) = \min\{q : I_E(q) \geq k - r\}.$$

These two equations are the finite version of the transfinite transversal repair law. They also show exactly where the finite +1 gap can appear: if a large alias class has leftover non-isolated points, one extra dump color may be needed.

5. Finite experiments and sanity checks

The following computations used the two finite capacity equations above. They are exact finite shadows, not approximations.

5.1 Deterministic alias profiles

Here $r = 4$, so the target large set has size $k - 4$.

Alias profile	k	target	tcov_r	res_r	gap
1^{48}	48	44	1	1	0
2^{24}	48	44	2	2	0
3^{16}	48	44	3	3	0
4^{12}	48	44	4	4	0
12^4	48	44	11	12	1
$40 + 1^8$	48	44	36	37	1
$20 + 10 + 10 + 4 + 4$	48	44	16	17	1
$31 + 16 + 8 + 4 + 2 + 1$	62	58	27	28	1

The profiles with uniformly small classes have $\text{res}_r = \text{tcov}_r$. The profiles with a dominant residual class show the finite dump-color gap $\text{res}_r = \text{tcov}_r + 1$.

5.2 Residual repair after a singleton-detector bank

For a finite default singleton-detector bank on $k = 64$ points with tolerance $r = 4$, after m singleton detectors the alias profile is

$$1^m + (64 - m).$$

The finite residual repair equation is

$$\text{res}_r(m) = 1 \quad (m \geq 60), \quad \text{res}_r(m) = 61 - m \quad (m < 60).$$

m singleton detectors	default class size	tcov_r	res_r
0	64	60	61
8	56	52	53
16	48	44	45
32	32	28	29
48	16	12	13
56	8	4	5
60	4	1	1
64	0	1	1

This finite linear decline becomes the transfinite discontinuity of Round 2: for every $m < \kappa$, cardinal subtraction leaves the residual default class U -large, so the repair rank remains κ ; at $m = \kappa$, it drops to 1.

5.3 Random alias partitions

The next table assigns k points uniformly to b observed aliases and computes tcov_r and res_r from the exact formulas over 1000 trials.

k	aliases b	r	mean tcov_r	mean res_r	$\text{res} = \text{tcov}$	$\text{res} = \text{tcov} + 1$
48	4	4	12.77	13.54	23.3%	76.7%
48	8	4	7.42	8.09	33.4%	66.6%
64	8	4	9.84	10.54	29.6%	70.4%
64	16	4	5.96	6.56	40.0%	60.0%
128	16	8	10.00	10.70	30.2%	69.8%
128	32	8	6.08	6.64	44.1%	55.9%

The computed gap was never larger than 1, matching the theorem. More aliases reduce the repair rank because each partial transversal can cover more of the source.

5.4 Random q -ary sensor fusion

Finally, take $k = 64$, $r = 4$, and fuse t random q -ary sensors. Equivalence classes are points with the same observed code. Each row uses 500 trials.

alphabet q	sensors t	mean tcov_r	mean res_r	already faithful
2	0	60.00	61.00	0.0%
2	2	17.23	18.01	0.0%
2	4	5.93	6.47	0.0%
2	6	2.84	3.17	0.0%
2	8	1.89	2.00	1.4%
4	0	60.00	61.00	0.0%
4	2	5.98	6.56	0.0%
4	4	1.89	2.01	1.0%
4	6	1.00	1.00	99.6%
4	8	1.00	1.00	100.0%

Experiment verdict. The random tests support the new interpretation: sensor fusion improves observation by shrinking the alias-transversal repair rank. The rank can be near full width even when many observed symbols exist, and it can drop to one before the code is literally injective if singleton classes already cover the allowed large set.

6. Applications suggested by the repair spectrum

1. Sensor-bank design. The design objective should not be only “increase alphabet size” or “increase entropy.” After any proposed fusion, compute or estimate the alias-class transversal rank $\text{res}_U(E)$. The exact global design law is

$$2^{\mu_n} \geq \text{res}_{U_n}(E_n)$$

outside an L -small coordinate set. This gives a direct budget equation for how many extra probe bits are actually needed.

2. Privacy and re-identification. Alias classes are anonymity classes. The finite equation

$$\text{res}_r(E) = \min\{q : \sum_i I_q(s_i) \geq k - r\}$$

turns a class-size histogram into an exact side-channel re-identification risk: it says how large a new auxiliary label must be to isolate all but r people. The transfinite warning is sharper: a subcritical collection of default logs may leave full repair rank, while a critical collection can collapse the rank to one.

3. Active diagnostics. In distributed diagnostics, a repair symbol is a targeted probe. The theorem says probes should be allocated as partial transversals through current alias classes. A probe family is sufficient exactly when its transversals cover a large set of possible states.

4. Verification and anomaly detection. The previous slogan was that finite monitors certify nullity but not positivity. The refined slogan is:

positivity certification requires enough repair transversals through alias classes.

A system can therefore report not just “ambiguous” but a numerical or cardinal repair rank measuring how far the observation is from faithful.

5. PCF/Tukey invariant. For a spinal fused sensor Σ , the sequence

$$n \longmapsto \text{res}_{U_n}(E_{\Sigma_n})$$

defines a reduced-product invariant modulo $L = H \vee J$. The next PCF/Tukey target is to compute the cofinal behavior of these repair-rank profiles under available sensor operations. This may be the sensor-side analogue of the earlier cardinal-invariant dictionary for slaloms.

7. Open targets after this round

1. **Exact finite-to-transfinite transfer.** Identify which finite repair-rank phenomena survive under specific ultrafilters that are not normal. Normal ultrafilters collapse the spectrum to $1, 2, \kappa$, but non-normal ultrafilters may realize intermediate transversal ranks.
2. **Restricted portfolio repair.** Given an allowed family of sensors, compute the least subfamily whose alias relation has $\text{tcov}_U(E) \leq \theta$. Round 2 handled unrestricted default singleton detectors; restricted libraries should produce new invariants.
3. **Definable repair ranks.** If the transversals or repair maps must be definable, low-complexity, or generated by a fixed language, the abstract $\text{res}_U(E)$ may be unattainable. This is a promising descriptive-set-theoretic refinement.
4. **Closed-set interaction.** Combine closed Lyapunov ranks with repair ranks: when an observation of a closed bad set is ambiguous, the minimal repair rank needed to recover the closed alarm law should be computable from alias classes along the derivative tree.

8. Round 3 summary

This round found a unifying equation for sensor repair after arbitrary fusion. Given the alias relation E left by the fused sensor,

$$\text{tcov}_U(E) \leq \text{res}_U(E) \leq \text{tcov}_U(E) + 1,$$

and for infinite cardinal repair ranks,

$$\boxed{\text{res}_U(E) = \text{tcov}_U(E).}$$

Thus the amount of additional alphabet needed for faithful observation is the number of partial alias transversals needed to cover a large set. Globally, with bit budgets μ_n and effective ideal $L = H \vee J$, the exact spinal equation is

$$\boxed{\{n : 2^{\mu_n} < \text{res}_{U_n}(E_n)\} \in L.}$$

This recovers the earlier normal phase diagram and the Round 2 default-breaking cliff as special cases. Its most useful application is operational: after any sensor fusion, compute the alias-class repair rank. That rank is the true remaining information cost of faithful observation.

Spinal Nullity Consolidated Report
Round 4 Addendum

Completeness Collapse of the Repair Spectrum and the Selector-Defect Phase

Continuing the spinal nullity program at singular infinity
April 29, 2026 - exploratory draft, appended to the consolidated report

Status note. This addendum continues the same draft research program. The main result is a correction and sharpening of the Round 3 repair-rank picture. Round 3 proved that repair capacity is controlled by large covers by partial alias transversals. This round observes that, under the standing local hypothesis of a nonprincipal κ -complete ultrafilter on κ , that apparent cardinal spectrum collapses locally. The only local repair sizes are 1, 2, and κ . The finite computations below are sanity checks of finite shadows and also show why finite threshold models misleadingly display intermediate ranks.

1 Why this round looked here

Round 3 found the transversal repair law. For an alias relation E left by a fused sensor,

$$\text{tcov}_U(E) \leq \text{res}_U(E) \leq \text{tcov}_U(E) + 1,$$

and infinite repair ranks satisfy $\text{res}_U(E) = \text{tcov}_U(E)$. The natural next question was whether non-normal ultrafilters could realize many intermediate values of $\text{tcov}_U(E)$ between 1 and κ .

The exceptional finding is that the answer is *no* under the current hypotheses. Intermediate local repair ranks are finite-shadow artifacts. κ -completeness itself collapses every subcritical large cover to one large transversal.

New law. For any κ -complete ultrafilter U on κ and any alias relation E on κ ,

$$\text{tcov}_U(E) \in \{1, \kappa\}, \quad \text{res}_U(E) \in \{1, 2, \kappa\}.$$

Normality is not needed for the collapse. Normality is only needed for the simpler test saying that the bad κ phase is exactly the existence of one U -large alias class.

This creates a new phase distinction. Outside normal ultrafilters, an alias relation may have no U -large class and still have full repair rank κ , because it admits no U -large partial transversal. This is a *diffuse selector-defect* phase: the aliasing is not concentrated in one default fiber, but it is still unrecoverable by any subcritical repair alphabet.

2 Essential multiplicity and transversal covers

Let U be a nonprincipal κ -complete ultrafilter on κ , and let E be an equivalence relation on κ .

Definition 1 (Partial transversal and essential multiplicity). *A set $T \subseteq \kappa$ is a partial E -transversal if*

$$|T \cap [\alpha]_E| \leq 1$$

for every $\alpha < \kappa$. Define the essential alias multiplicity

$$\text{mult}_U(E) = \min \{ \mu : \exists A \in U \forall C \in \kappa/E, |A \cap C| \leq \mu \}.$$

Theorem 1 (Multiplicity-transversal identity). *For every equivalence relation E ,*

$$\boxed{\text{tcov}_U(E) = \text{mult}_U(E)}.$$

Proof sketch. If $A \in U$ is covered by μ partial transversals, then every E -class meets A in at most μ points. Hence $\text{mult}_U(E) \leq \text{tcov}_U(E)$.

Conversely, suppose $A \in U$ and each E -class meets A in at most μ points. Choose, class by class, an injection from $A \cap C$ into μ . The inverse images of the colors are μ partial E -transversals whose union is A . Thus $\text{tcov}_U(E) \leq \text{mult}_U(E)$. $\square$

This identity says that Round 3's transversal cover is not a mysterious covering invariant: it is the least uniform bound on how many representatives per alias class must remain on a U -large set.

3 The completeness-collapse theorem

The collapse uses one elementary but powerful consequence of κ -completeness.

Lemma 1 (Subcritical union absorption). *Let $\mu < \kappa$ and let $A_i \subseteq \kappa$ for $i < \mu$. If*

$$\bigcup_{i < \mu} A_i \in U,$$

then $A_i \in U$ for some $i < \mu$.

Proof sketch. If every A_i were U -small, then every complement $\kappa \setminus A_i$ would be in U . By κ -completeness,

$$\bigcap_{i < \mu} (\kappa \setminus A_i) \in U,$$

so the union would be U -small, contradiction. $\square$

Theorem 2 (Completeness collapse of local repair). *For every equivalence relation E on κ ,*

$$\boxed{\text{tcov}_U(E) = 1 \quad \text{or} \quad \text{tcov}_U(E) = \kappa}.$$

Consequently,

$$\boxed{\text{res}_U(E) \in \{1, 2, \kappa\}}.$$

More precisely:

$$\text{res}_U(E) = \begin{cases} 1, & \text{Sing}(E) \in U, \\ 2, & \text{Sing}(E) \notin U \text{ and there is a } U\text{-large partial } E\text{-transversal,} \\ \kappa, & \text{there is no } U\text{-large partial } E\text{-transversal.} \end{cases}$$

Here

$$\text{Sing}(E) = \{\alpha < \kappa : [\alpha]_E = \{\alpha\}\}.$$

Proof sketch. If $\text{tcov}_U(E) = \mu < \kappa$, then some U -large set is a union of μ partial transversals. By the subcritical union lemma, one of those partial transversals is itself U -large. Hence $\text{tcov}_U(E) = 1$. If no U -large partial transversal exists, at least κ transversals are needed; κ singleton transversals always cover all of κ , so $\text{tcov}_U(E) = \kappa$.

The repair-symbol statement follows from Round 3's theorem $\text{tcov}_U(E) \leq \text{res}_U(E) \leq \text{tcov}_U(E) + 1$. If $\text{Sing}(E) \in U$, no repair is needed and $\text{res}_U(E) = 1$. If $\text{Sing}(E) \notin U$ but a U -large partial transversal exists, one extra dump color isolates that transversal, so $\text{res}_U(E) = 2$. If no such transversal exists, $\text{tcov}_U(E) = \kappa$ and the infinite-cardinal equality gives $\text{res}_U(E) = \kappa$. $\square$

Correction to the search direction. The earlier idea that arbitrary non-normal ultrafilters might realize intermediate local repair ranks is false for κ -complete ultrafilters on κ . What non-normality can create is subtler: a full-rank bad phase without a single U -large alias class.

4 The selector-defect phase

Define the selector predicate

$$\text{Sel}_U(E) \iff \exists T \in U \text{ (} T \text{ is a partial } E\text{-transversal)}.$$

The local phase diagram is now:

Phase	Condition	Repair size
Faithful	$\text{Sing}(E) \in U$	1
Selectable aliasing	$\text{Sing}(E) \notin U$ and $\text{Sel}_U(E)$	2
Selector-defective aliasing	not $\text{Sel}_U(E)$	κ

Under normality, the selector-defective phase is exactly the principal/default phase. Indeed, the normal selector lemma says a map is either constant on a U -large set or injective on a U -large set. Therefore, for normal U ,

$$\neg \text{Sel}_U(E_f) \iff \exists y \ f^{-1}(\{y\}) \in U.$$

Without normality, the implication from no large fiber to a large transversal can fail. Then there may be aliasing that is locally diffuse, in the sense that no single alias class is U -large, but still has full repair rank κ . This is the new blind spot exposed by this round.

5 Completeness-reduced Nyquist law on the spine

Return to the singular spine

$$X = \prod_{n < \omega} \kappa_n, \quad \lambda = \sup_{n < \omega} \kappa_n,$$

with effective tolerance ideal $L = H \vee J$. Let E_n be the local alias relation after sensor fusion at coordinate n .

Define three coordinate sets:

$$\begin{aligned} F &= \{n : \text{Sing}(E_n) \in U_n\}, \\ S &= \{n : \text{Sing}(E_n) \notin U_n \text{ and } \text{Sel}_{U_n}(E_n)\}, \end{aligned}$$

$$D = \{n : \neg \text{Sel}_{U_n}(E_n)\}.$$

The set D is the selector-defect support.

Law 1 (Completeness-reduced Nyquist equation). *With bit budgets μ_n , robust faithful observation after repair is possible iff*

$$\boxed{\{n \in S : \mu_n = 0\} \cup \{n \in D : 2^{\mu_n} < \kappa_n\} \in L.}$$

Coordinates in F need no repair. Coordinates in S need exactly one bit. Coordinates in D need critical local width κ_n .

This is the Round 3 Nyquist law after using κ_n -completeness to collapse the local rank spectrum. The nontrivial design problem no longer lives in intermediate local cardinal ranks; it lives in reducing the selector-defect support D modulo the effective ideal L .

6 New design invariant: selector-defect support

For a sensor portfolio Γ , let $E_{\Gamma,n}$ be the alias relation of the fused sensor at coordinate n . Define

$$\text{sdef}_L(\Gamma) = \{n < \omega : \neg \text{Sel}_{U_n}(E_{\Gamma,n})\} / L.$$

The operational target is:

$$\text{make } \text{sdef}_L(\Gamma) = 0.$$

If this is achieved, then one repair bit outside an L -small set suffices. If not, every coordinate remaining in selector defect requires critical-width repair.

For normal ultrafilters, sdef_L reduces to the old default-fiber support. For arbitrary κ_n -complete ultrafilters, it detects diffuse nonselectability as well. This is a sharper audit invariant than observed alphabet size, entropy, or number of sensors.

7 Finite experiments and sanity checks

Finite nonprincipal κ -complete ultrafilters do not exist. The experiments therefore compare two finite shadows:

- a co-large threshold model, where intermediate ranks appear because many small sets can union to become large;
- a principal-ultrafilter model, where the repair spectrum collapses to 1 or 2, illustrating the union-absorption mechanism but not the uniform full-rank κ phase.

7.1 Deterministic threshold profiles

Let $k = 64$ and allow a small remainder $r = 4$, so the target large set has size 60. For an alias profile m^b meaning b classes of size m , the exact finite equations from Round 3 give:

Alias profile	classes	max class	tcov _r	res _r
1 ⁶⁴	64	1	1	1
2 ³²	32	2	2	2
4 ¹⁶	16	4	4	4
8 ⁸	8	8	8	8
16 ⁴	4	16	15	16
32 ²	2	32	30	31
64 ¹	1	64	60	61

These intermediate numbers are real for finite co-large thresholds, but they are exactly what κ -completeness forbids at a transfinite coordinate: a subcritical union of small partial transversals cannot become large.

7.2 Random partitions: threshold ranks versus principal collapse

Each trial assigns k points uniformly to b observed aliases. The threshold columns use co-large size $k - r$. The principal columns declare a set large iff it contains point 0.

k	aliases b	r	mean threshold tcov _r	mean threshold res _r	principal res = 1	principal mean res
32	4	2	9.43	10.16	0.0%	2.000
32	8	2	5.76	6.33	1.8%	1.982
64	8	4	9.82	10.52	0.0%	2.000
64	16	4	5.95	6.55	1.6%	1.984
128	16	8	10.01	10.70	0.0%	2.000
128	32	8	6.08	6.64	2.3%	1.977

All rows used 1000 random trials. The threshold model displays many intermediate ranks. The principal-ultrafilter shadow collapses to 1 when the distinguished point is already singleton and 2 otherwise. This supports the diagnosis: intermediate finite ranks come from failure of subcritical union absorption.

7.3 Experiment verdict

The finite tests do not prove the transfinite theorem, but they identify the source of the earlier illusion. Round 3's finite capacity equations are correct for finite threshold systems. Round 4 says that, at a genuine κ -complete ultrafilter coordinate, the threshold-style accumulation mechanism disappears. A large subcritical union already contains a large member.

8 Applications suggested by this round

Sensor-bank design. The local audit question should be: does the fused alias relation admit a large partial transversal? If yes, one repair bit suffices unless the sensor is already faithful. If no, no subcritical repair alphabet can work. This is more decisive than estimating entropy or counting observed symbols.

Privacy and re-identification. A population can fail to be re-identifiable in two different ways. The obvious way is a large default anonymity class. The new way is diffuse selector defect: no one alias class

is large, but there is still no large selectable set of representatives. Such a system may look diversified while still requiring critical-width side information for faithful identification.

Monitoring and verification. The observation state of a coordinate is a three-phase label: faithful, one-bit selectable, or selector-defective. Global monitoring budgets are then computed modulo $L = H \vee J$. This makes the repair-budget law easy to implement as a quotient set-cover problem over coordinate phases.

Finite engineering. Finite systems still need the exact Round 3 capacity equations, because finite threshold largeness allows intermediate ranks. The transfinite law is nevertheless useful as a warning: intermediate repair curves may be artifacts of finite sampling unless the underlying large-set semantics permits subcritical unions of small pieces to become large.

Set theory. The new structural property to study is kernel-selectivity:

$$\forall f : \kappa \rightarrow Y, \quad (\forall y \ f^{-1}(\{y\}) \notin U) \Rightarrow \exists T \in U \ f \restriction T \text{ is injective.}$$

Normal ultrafilters have this property. Characterizing which κ -complete ultrafilters have it may be the right local set-theoretic problem behind sensor repair.

9 Open targets after this round

1. **Kernel-selective ultrafilters.** Determine whether kernel-selectivity is equivalent to normality, follows from weaker selectivity principles, or separates known large-cardinal ultrafilter types.
2. **Diffuse selector-defect examples.** Construct or rule out explicit kernels with no large fiber and no large partial transversal for natural non-normal κ -complete ultrafilters.
3. **Spinal selector-defect minimization.** Given a restricted sensor library, compute the least portfolio whose selector-defect support lies in $L = H \vee J$.
4. **Finite-to-transfinite calibration.** Develop tests distinguishing genuine finite intermediate repair structure from threshold artifacts that disappear under κ -complete semantics.

10 Round 4 summary

This round sharpened the Round 3 repair law. The new local equations are

$$\boxed{\text{tcov}_U(E) = \text{mult}_U(E)}$$

and, using κ -completeness,

$$\boxed{\text{tcov}_U(E) \in \{1, \kappa\}, \quad \text{res}_U(E) \in \{1, 2, \kappa\}.$$

The resulting phase diagram is:

$$\boxed{\text{res}_U(E) = \begin{cases} 1, & \text{Sing}(E) \in U, \\ 2, & \text{Sing}(E) \notin U \text{ but a large partial transversal exists,} \\ \kappa, & \text{no large partial transversal exists.} \end{cases}}$$

The exceptional conceptual change is that non-normality does not create intermediate local repair cardinals under the standing completeness assumptions. It can instead create *diffuse selector-defect*:

full critical repair rank without a single large default class. Therefore the most useful next invariant is not an intermediate local rank but the spinal support of selector-defective coordinates modulo the effective ideal $H \vee J$.

Fubini Dimension Loss and Diffuse Selector Defect

Continuing the spinal nullity program at singular infinity

April 30, 2026 - exploratory draft, appended to the consolidated report

Status note. This addendum continues the draft research program. The main point is an explicit construction of the diffuse selector-defect phase isolated in Round 4. The construction is not exotic: it is just finite Fubini product structure. A projection of a two-dimensional Fubini coordinate has no large default fiber and has nonprincipal observed behavior, but it still has no large selector and therefore requires full critical-width repair. The finite computations are sanity checks of the corresponding finite threshold shadow.

1 Why this round looked here

Round 4 left a concrete open target: construct or rule out diffuse selector-defect examples. Recall the phase distinction for an alias relation E under a κ -complete ultrafilter U on κ :

$$\text{res}_U(E) = \begin{cases} 1, & \text{Sing}(E) \in U, \\ 2, & \text{Sing}(E) \notin U \text{ but } \text{Sel}_U(E), \\ \kappa, & \neg \text{Sel}_U(E). \end{cases}$$

Normal ultrafilters collapse the last case to a large default class. The blind spot was whether full repair rank can occur without any large alias class.

Exceptional finding. Yes. The canonical example is a Fubini product coordinate. If the hidden local state is a pair (i, j) distributed by $U \otimes U$ and the sensor reports only i , then:

- no reported value i has a large fiber;
- the observed pushforward is nonprincipal;
- nevertheless no large partial transversal exists;
- hence the repair alphabet must have size κ .

In words: projecting away one Fubini dimension creates full critical repair cost even though there is no default collapse.

2 Large sections and true Rudin-Keisler quotients

Let U be an ultrafilter on a set X , let $f : X \rightarrow Y$, and let $V = f_*U$ be the pushforward ultrafilter on Y .

Definition 1 (Large section). A map $f : X \rightarrow Y$ has a U -large section if there is $T \in U$ such that $f \upharpoonright T$ is one-to-one. Equivalently, the alias relation E_f admits a U -large partial transversal.

Theorem 1 (Sectioned quotient criterion). *For a map $f : X \rightarrow Y$, the following are equivalent.*

- (a) $\text{Sel}_U(E_f)$.
- (b) *There is $B \in V$ and a right inverse $s : B \rightarrow X$ with $f \circ s = \text{id}_B$ and $s[B] \in U$.*
- (c) *After restricting to large sets, f is an isomorphism between U and V .*

Proof sketch. If $T \in U$ is a partial transversal, put $B = f[T]$ and let $s = (f \upharpoonright T)^{-1}$. Then $B \in V$ and $s[B] = T \in U$. Conversely, the image of any large right inverse is a large partial transversal. The last clause follows because membership in U can be tested after intersecting with T , and membership in V can be tested after intersecting with B . $\square$

Thus Round 4's repair trichotomy can be read as a quotient law:

$$\text{res}_U(f) = \begin{cases} 1, & f \text{ is already singleton-fibered on a large set,} \\ 2, & f_*U \text{ is a sectioned nonprincipal quotient,} \\ \kappa, & f_*U \text{ is an unsectioned, or true, quotient.} \end{cases}$$

The new construction below shows that true nonprincipal quotients occur naturally.

3 The Fubini projection theorem

Let U be a nonprincipal κ -complete ultrafilter on κ . Define the finite Fubini powers recursively. $U^{[1]} = U$, and for $A \subseteq \kappa^{m+1}$,

$$A \in U^{[m+1]} \iff \{\alpha < \kappa : A_\alpha \in U^{[m]}\} \in U,$$

where $A_\alpha = \{\vec{\beta} : (\alpha, \vec{\beta}) \in A\}$.

For $0 \leq r \leq m$, let

$$\pi_{m,r} : \kappa^m \rightarrow \kappa^r$$

be the projection onto the first r Fubini coordinates. Let $E_{m,r}$ be the corresponding alias relation.

Theorem 2 (Fubini diffuse selector-defect theorem). *For $0 < r < m$:*

- (a) $(\pi_{m,r})_*U^{[m]} = U^{[r]}$, so the observed quotient is nonprincipal.
- (b) No fiber of $\pi_{m,r}$ is $U^{[m]}$ -large.
- (c) No $U^{[m]}$ -large partial $E_{m,r}$ -transversal exists.

Consequently $E_{m,r}$ is diffuse selector-defective and

$$\text{res}_{U^{[m]}}(E_{m,r}) = \kappa.$$

Proof sketch. The pushforward identity is the definition of the Fubini product. A fiber over a fixed first-coordinate tuple fixes at least one U -coordinate, so it is small. For a partial transversal T , each tail section over a base point in κ^r has at most one element in κ^{m-r} , hence is $U^{[m-r]}$ -small. Therefore the set of base points with large tail section is empty, so $T \notin U^{[m]}$. Round 4's completeness-collapse theorem then gives repair rank κ . $\square$

Since $|\kappa^m| = \kappa$ for finite m , any bijection $b_m : \kappa \rightarrow \kappa^m$ transports $U^{[m]}$ to a κ -complete nonprincipal ultrafilter on the original kind of local coordinate. Thus this is a genuine local example inside the standing framework, not a change of universe.

4 Hidden-dimension Nyquist law

The preceding theorem yields a compact local law.

Law 1 (Fubini hidden-dimension Nyquist law). *For a finite Fubini coordinate $U^{[m]}$ and the projection $\pi_{m,r}$,*

$$\text{res}_{U^{[m]}}(\pi_{m,r}) = \begin{cases} 1, & r = m, \\ \kappa, & r < m. \end{cases}$$

If $0 < r < m$, the bad phase is diffuse: the pushforward is nonprincipal and no observed value is large. If $r = 0$, the bad phase is concentrated: the sensor is trivial and the pushforward is principal. Both require critical-width repair.

Equivalently, for a proper Fubini projection, a bit budget μ can repair faithfully only when

$$2^\mu \geq \kappa.$$

At the transfinite coordinate scale, losing any positive finite Fubini dimension costs the full local cardinal. Finite dimension is visible structurally, but its repair threshold collapses to the same critical cardinal because $\kappa^d = \kappa$ for every finite $d > 0$.

5 Parallel rescue: two defective sensors can fuse to faithfulness

For $W = U \otimes U$ on κ^2 , let $\pi_0(i, j) = i$ and $\pi_1(i, j) = j$. The first projection is diffuse selector-defective by the theorem. The second is also selector-defective: any column fiber is small, and a partial column-transversal has pairwise disjoint row sections, so it cannot have U -many U -large rows.

Yet the fused sensor (π_0, π_1) is the identity. Therefore

$$\text{res}_W(\pi_0) = \kappa, \quad \text{res}_W(\pi_1) = \kappa, \quad \text{res}_W(\pi_0, \pi_1) = 1.$$

This is a sharper zero-bit synergy phenomenon than the earlier support-cover form. Each marginal sensor is maximally unrecoverable by subcritical repair, but their parallel fusion is perfectly faithful.

The lesson is that repair rank is not subadditive in the naive information sense. A sensor can be useless for repair alone and decisive in a portfolio because it cuts exactly across another sensor's alias classes.

6 Global spinal dimension-loss equation

Suppose coordinate n carries a hidden finite Fubini state $(\kappa_n)^{m_n}$ with ultrafilter $U_n^{[m_n]}$, and the available local sensor exposes the first r_n Fubini coordinates. Let $L = H \vee J$ be the effective tolerance ideal.

Law 2 (Spinal Fubini dimension-loss equation). *Robust faithful repair with bit budgets μ_n is possible exactly when*

$$\boxed{\{n < \omega : r_n < m_n \text{ and } 2^{\mu_n} < \kappa_n\} \in L.}$$

The selector-defect support is simply

$$D_{\text{Fub}} = \{n : r_n < m_n\}.$$

Coordinates with $r_n = m_n$ need no repair; every proper Fubini projection needs critical local width outside the tolerated set L .

This gives a concrete design rule: for Fubini-structured hidden state, it is not enough that every observed fiber be small or that the observed quotient be nonprincipal. Unless the portfolio observes all Fubini dimensions, the coordinate remains in critical repair phase.

7 Finite experiments and sanity checks

Finite nonprincipal ultrafilters do not exist, so the tests use a finite threshold Fubini shadow. On $[k]^d$, a set is large if recursively at least t first-coordinate sections are large, with the one-dimensional base condition $|A| \geq t$.

For a projection $[k]^m \rightarrow [k]^r$, the hidden tail dimension is $d = m - r$. A repair color is unique only for points whose color is unique inside their projection fiber. The finite threshold analogue gives the exact calibration

$$q_{\min}(k, t; m, r) = \begin{cases} 1, & r = m, \\ t^{m-r}, & r < m \text{ and } k = t, \\ t^{m-r} + 1, & r < m \text{ and } k > t. \end{cases}$$

The extra $+1$ is the finite dump color. It disappears at the cardinal level because $\kappa + 1 = \kappa$.

k	t	m	r	lost d	exact $q_{\min}$	constructive check
5	3	2	1	1	4	pass
6	4	2	1	1	5	pass
6	3	3	2	1	4	pass
6	3	3	1	2	10	pass
8	5	3	2	1	6	pass
8	5	4	2	2	26	pass

The constructive code chooses a threshold-large tail box of size t^{m-r} , gives those tail points distinct colors, and sends every other tail point to one dump color. The singleton set of the fused sensor is then threshold-large.

Random colorings are usually inefficient near the threshold. In 200 random trials per row, the observed success rates were:

k	t	(m, r)	q	lower t^{m-r}	success rate
5	3	(2,1)	3	3	0.000
5	3	(2,1)	4	3	0.050
5	3	(2,1)	5	3	0.380
6	4	(2,1)	4	4	0.000
6	4	(2,1)	5	4	0.005
6	4	(2,1)	6	4	0.045
6	3	(3,2)	3	3	0.000
6	3	(3,2)	4	3	0.000
6	3	(3,2)	6	3	0.535
6	3	(3,1)	9	9	0.000
6	3	(3,1)	10	9	0.000
6	3	(3,1)	18	9	0.000

The deterministic checks verify the threshold equation. The random rows add a practical warning: near-minimal repair is not usually found by unstructured random coding; it needs a deliberate tail-coordinate code.

8 Applications suggested by this round

Sensor-bank design. A marginal sensor can look healthy because every fiber is small and the output ultrafilter is nonprincipal, yet it can still be maximally unrecoverable. The audit question should be: does the quotient have a large section? For Fubini projections the answer is no unless all hidden dimensions are observed.

Privacy and re-identification. Diffuse anonymity can be stronger than default-class anonymity. A database shard key may have no dominant shard, but if it forgets an intra-shard coordinate, faithful re-identification still requires a full critical-width key. Conversely, two individually defective marginal attributes can jointly re-identify perfectly.

Distributed diagnostics. Observing only a subsystem marginal of a product-like state can leave a true quotient. No amount of subcritical side information repairs the lost local dimension; the missing coordinate must be sensed directly or encoded by a critical-width channel.

Verification. A monitor of a marginal process cannot certify path-level typicality in a Fubini hidden state. Safety certificates may still be possible, but positivity or faithfulness certificates require observing all Fubini dimensions outside the effective ideal $H \vee J$.

Set theory. Finite Fubini powers are canonical test objects for kernel-selectivity. They show that non-normal κ -complete ultrafilters can carry true nonprincipal quotients: quotient maps with no large default fiber and no large section.

9 Open targets after this round

1. **Kernel-selectivity hierarchy.** Determine which κ -complete ultrafilters have no true nonprincipal quotients. Normal measures do; Fubini products do not.
2. **Fubini dimension algebra.** Extend the projection law from initial finite Fubini powers to arbitrary finite coordinate projections, iterated products of different ultrafilters, and mixed normal/non-normal coordinates.
3. **Portfolio rescue number.** For a family of defective marginal sensors, compute the least subfamily whose fused code has a large section or is singleton-fibered on a large set.
4. **Finite code synthesis.** Turn the threshold formula $q_{\min} = t^d + 1$ into constructive near-minimal code design for finite safety monitors and privacy audits.

10 Round 5 summary

This round resolves one of the main Round 4 open targets by giving an explicit diffuse selector-defect construction. For $0 < r < m$,

$$(\pi_{m,r})_* U^{[m]} = U^{[r]}, \quad \forall y \ \pi_{m,r}^{-1} \{y\} \notin U^{[m]}, \quad \neg \text{Sel}_{U^{[m]}}(E_{m,r}),$$

and hence

$$\boxed{\text{res}_{U^{[m]}}(\pi_{m,r}) = \kappa.}$$

The most useful new law is the hidden-dimension Nyquist equation:

$$\text{res}_{U^{[m]}}(\pi_{m,r}) = 1 \text{ if } r = m, \quad \text{res}_{U^{[m]}}(\pi_{m,r}) = \kappa \text{ if } r < m.$$

The exceptional conceptual change is that a nonprincipal quotient with no large fibers can still be a true quotient, not a sectioned quotient. Such a quotient looks diversified from the observed side but cannot be repaired by any subcritical alphabet. In global spinal form, for Fubini-structured coordinates the faithful-repair condition is

$$\{n : r_n < m_n \text{ and } 2^{\mu_n} < \kappa_n\} \in H \vee J.$$

This is directly applicable to sensor design, privacy/re-identification, distributed diagnostics, and verification: marginal observation is not faithful unless every hidden Fubini dimension is either observed or paid for at critical local width.

Round 6: Dimension-Cover Nyquist and Adaptive No-Discount

A continuation of the spinal-nullity sensor/repair program

April 30, 2026 – exploratory draft, not peer reviewed

Round aim. Round 5 showed that a finite Fubini coordinate has a hidden-dimension Nyquist law: losing even one Fubini dimension forces full κ -width repair, while observing all dimensions is faithful. This round turns that local projection fact into a portfolio and adaptive-query law. The new claim is that, inside a genuine κ -complete Fubini coordinate, *dimension coverage* is the correct invariant. Bits, finite-valued adaptive choices, and later marginal correlations do not buy a discount unless every hidden dimension is actually observed on a large set.

1. Local Fubini coordinate and dimension portfolios

Let U be a κ -complete nonprincipal ultrafilter on κ , and let $U^{[m]}$ denote the m -fold finite Fubini product on κ^m . For $D \subseteq m = \{0, \dots, m-1\}$, write

$$\pi_D : \kappa^m \rightarrow \kappa^D$$

for the coordinate projection. A local sensor portfolio Γ is represented, in this Fubini model, by a family of dimension sets $D_\gamma \subseteq m$. Its fused observation is equivalent to

$$\Sigma_\Gamma = \pi_{D_\Gamma}, \quad D_\Gamma = \bigcup_{\gamma \in \Gamma} D_\gamma,$$

and its hidden residual dimension set is

$$M_\Gamma = m \setminus D_\Gamma.$$

Recall that $\text{res}_{U^{[m]}}(\Sigma_\Gamma)$ is the least repair alphabet size needed to refine Σ_Γ so that the refined sensor is faithful on a $U^{[m]}$ -large set.

Law (Fubini dimension-cover Nyquist law). *For every finite m and every projection portfolio Γ ,*

$$\boxed{\text{res}_{U^{[m]}}(\Sigma_\Gamma) = \begin{cases} 1, & M_\Gamma = \emptyset, \\ \kappa, & M_\Gamma \neq \emptyset. \end{cases}}$$

Equivalently: all hidden dimensions covered means no repair is needed; one uncovered Fubini dimension means every subcritical repair alphabet $\theta < \kappa$ is powerless.

Why this is exceptional. The law says that finite Fubini coordinates have no graded local cardinal scale between 1 and κ . A sensor may observe $m-1$ out of m dimensions and still be maximally unrecoverable. In finite engineering shadows there can be partial discounts; at a true κ -complete coordinate those discounts collapse.

2. Proof sketch and the new obstruction

The key local obstruction is stronger than “large fiber” collapse. It is a no-large-transversal fact.

Lemma (No large partial transversal for a missed dimension). *If $D \subsetneq m$, every partial π_D -transversal is $U^{[m]}$ -small.*

Sketch. A partial π_D -transversal meets each fiber of π_D in at most one point. If a dimension $j \notin D$ is missed, then after fixing all coordinates except j , the transversal behaves like a graph in the missed coordinate. In the innermost case a graph has singleton vertical sections, hence small. Inducting through the finite Fubini product shows that the set of prefixes with large tail section is never U -large. Thus the whole transversal is $U^{[m]}$ -small.

Now suppose a repair map $\tau : \kappa^m \rightarrow \theta$ has $\theta < \kappa$. Since $U^{[m]}$ is κ -complete, some color class $\tau^{-1}\{c\}$ is $U^{[m]}$ -large. If (π_D, τ) were faithful on a large singleton-fiber set, then its singleton-fiber set inside this one color would give a $U^{[m]}$ -large partial π_D -transversal, contradicting the lemma. Hence no subcritical alphabet repairs a missed dimension. The alphabet κ repairs by coding one missed coordinate; because m is finite, κ also codes all missed coordinates.

3. Global spinal dimension equation

Return to the countable spine. At coordinate n , let the local hidden state have m_n Fubini dimensions and let a sensor γ observe dimensions $D_{\gamma,n} \subseteq m_n$. Put

$$D_{\Gamma,n} = \bigcup_{\gamma \in \Gamma} D_{\gamma,n}, \quad M_{\Gamma,n} = m_n \setminus D_{\Gamma,n}.$$

Let μ_n be the available repair bits and let $L = H \vee J$ be the effective ideal combining coordinate noise and monitoring tolerance.

Law (Spinal dimension-cover equation). *Robust faithful observation by the fused portfolio Γ plus repair bits μ_n is possible exactly when*

$$\boxed{\{n : M_{\Gamma,n} \neq \emptyset \text{ and } 2^{\mu_n} < \kappa_n\} \in L.}$$

In particular, when all repair alphabets are subcritical, this reduces to

$$\boxed{\{n : D_{\Gamma,n} \neq m_n\} \in L.}$$

Thus a sensor portfolio is adequate iff it covers every hidden dimension on all but an L -small set of spine coordinates. The resulting design invariant is the quotient hypergraph cover number

$$\boxed{\text{dcov}_L(\mathcal{S}) = \min \{|\Gamma| : \{n : D_{\Gamma,n} \neq m_n\} \in L\} .}$$

This refines the previous nonprincipal-support cover theorem: not only must a coordinate be nonprincipal after fusion; in a Fubini coordinate, every hidden Fubini dimension must be covered.

4. Adaptive no-discount theorem

The same obstruction rules out a tempting loophole: choose sensors adaptively. Consider a finite local decision protocol P that queries coordinate dimensions one at a time, with each next query chosen from the finite set m as a function of the transcript so far. Let $R_P(x) \subseteq m$ be the set of dimensions actually read when the hidden state is $x \in \kappa^m$. The transcript includes the values read.

Theorem (Adaptive no-discount). *For finite coordinate-query protocols on $(\kappa^m, U^{[m]})$,*

$$\boxed{\text{res}(P) = \begin{cases} 1, & \{x : R_P(x) = m\} \in U^{[m]}, \\ \kappa, & \{x : R_P(x) = m\} \notin U^{[m]}. \end{cases}}$$

Consequently, every protocol of fixed depth $d < m$ has residual repair cost κ .

Sketch. There are only finitely many read-dimension patterns. By ultrafilter completeness, one pattern class is large inside any large set. If a large pattern reads all dimensions, the transcript is injective there. If every large pattern misses at least one dimension, the dimension-cover law applies on a large piece and rules out every subcritical repair alphabet. Adaptivity changes which dimensions are read on which branch, but it does not convert an uncovered large branch class into faithful information.

Interpretation. At a true Fubini coordinate, active diagnostics cannot trade clever branching for hidden-dimension coverage. The only successful adaptive policy is one that reads all hidden dimensions on a large set. This is a qualitative Nyquist theorem for transfinite active sensing.

5. Finite experiments: exact shadows and a warning

Finite nonprincipal ultrafilters do not exist, so the tests below use finite threshold Fubini shadows. For a finite product $\prod_{i < m} k_i$ with thresholds t_i , a set is threshold-large if at least t_i sections are large at each Fubini level. For an observed dimension set D , the exact finite repair thickness is

$$q_{\min}^{\text{fin}}(D) = \min_T \max_{\text{threshold-large } y \in \pi_D[X]} |T \cap \pi_D^{-1}\{y\}|.$$

This is the finite analogue of the least repair alphabet: choose a large set T , then color points inside each observed fiber injectively.

Grid k	thresholds t	observed D	simple product	exact $q_{\min}^{\text{fin}}$
(3, 3)	(2, 2)	$\emptyset$	4	4
(3, 3)	(2, 2)	$\{0\}$	2	2
(3, 3)	(2, 2)	$\{1\}$	2	2
(3, 3)	(2, 2)	$\{0, 1\}$	1	1
(3, 2, 2)	(2, 2, 1)	$\{2\}$	4	2
(3, 2, 2)	(2, 2, 1)	$\{0, 2\}$	2	1

The first four rows match the simple hidden-product prediction. The last two rows are the anomaly: a late observed coordinate with weak threshold 1 behaves like an extra finite repair palette.

The last two rows are important. Finite threshold models with weak later thresholds can look more recoverable than the transfinite theorem allows. The reason is that a later observed coordinate with threshold 1 can carry graph-coded information while still being threshold-large. In $U^{[m]}$, singleton graph sections are U -small, so this palette trick disappears. This is a useful warning: finite simulations should be calibrated with high section thresholds or interpreted through the exact fiber-thickness equation, not through naive dimension counting alone.

6. Portfolio simulations

The following random simulations used the dimension-cover equation directly. Each sensor independently observes each hidden dimension with probability p at each nonignored coordinate. A trial succeeds when all nonignored coordinates have every hidden dimension covered by the available sensor bank; the average minimum size is computed over successful trials.

coords	ignored	sensors	dims/coord	p	success rate	avg. min cover
10	2	8	2	0.35	57.5%	3.73
10	2	8	3	0.35	51.0%	4.55
12	3	10	3	0.30	44.0%	5.30
12	3	10	4	0.30	38.5%	5.96
16	4	14	3	0.25	56.0%	6.42
16	4	14	4	0.25	37.5%	7.32

The trend matches the law: adding hidden dimensions makes subcritical repair sharply harder unless the sensor bank grows or the effective ideal L ignores more coordinates. The quotient-cover formulation gives an implementable design test:

```

for n in spine_coordinates:
    covered[n] = union(D[sensor][n] for sensor in Gamma)
    bad[n] = (covered[n] != all_dimensions[n]) and (2**mu[n] < kappa[n])
faithful = {n for n in spine_coordinates if bad[n]} in L

```

7. Applications

Sensor-bank design. The local design unit is no longer merely “nonprincipal information.” For Fubini-structured hidden state, each sensor contributes a subset of hidden dimensions, and the engineering problem is weighted hypergraph cover modulo $H \vee J$. Subcritical repair bits should be budgeted only after dimension coverage has been checked.

Privacy and re-identification. A privacy mechanism remains transfinally nonrecoverable if it leaves at least one hidden Fubini dimension uncovered on an L -positive coordinate set and the attacker has only subcritical repair. Conversely, independent marginal releases that cover all dimensions destroy this protection even when each marginal alone is maximally unrecoverable.

Active diagnostics. Adaptive protocols should be audited by the large-set coverage of their query paths. A decision tree of depth below the hidden dimension count cannot be faithful; a branch policy is valuable only when the set of full-dimension transcripts is large.

Finite verification. Finite threshold simulations can understate unrecoverability if weak later coordinates act as graph palettes. The exact finite quantity to compute is $q_{\min}^{\text{fin}}(D)$, the minimum large-set fiber thickness. This suggests a practical calibration rule: use high section thresholds or compute the fiber-thickness Bellman problem before extrapolating to the transfinite law.

Set theory target. The new local invariant is kernel dimension-cover. A natural next theorem target is a Tukey/Rudin-Keisler classification of finite Fubini sensor kernels by their dimension-cover rank, with the conjectural principle that subcritical maps cannot reduce positive Fubini dimension.

8. Round 6 summary

The main new equation is the spinal dimension-cover equation

$$\{n : M_{\Gamma,n} \neq \emptyset \text{ and } 2^{\mu_n} < \kappa_n\} \in H \vee J.$$

It says that robust faithful observation fails exactly on the effective-positive set of coordinates where some Fubini dimension is still hidden and the repair alphabet is subcritical.

The conceptual advance is the *adaptive no-discount* principle: finite-valued active sensing cannot compensate for an unobserved Fubini dimension. The finite experiments support the cover law while also revealing a finite-shadow trap: weak finite thresholds may permit graph-coded recovery through later observed coordinates, a phenomenon killed by genuine ultrafilter largeness.

Next target. Replace coordinate projections by arbitrary finite Fubini-definable equivalence relations and define a dimension-rank $\dim_U(E)$ such that

$$\text{res}_U(E) = 1 \quad \text{iff} \quad \dim_U(E) = 0, \quad \text{res}_U(E) = \kappa \quad \text{iff} \quad \dim_U(E) > 0.$$

If this can be proved beyond coordinate projections, it would give a true dimension theory for transfinite sensor kernels.

Round 7: Selector Dimension and the One-Bit Skin of Fubini Kernels

A continuation of the spinal-nullity sensor/repair program

Exploratory research addendum

April 30, 2026 – draft, not peer reviewed

Round aim. Round 6 ended with the target of replacing coordinate projections by arbitrary finite Fubini-definable equivalence relations. I first tried the most tempting binary law: a kernel should have residual cost 1 when it has dimension 0, and residual cost κ otherwise. The exceptional finding of this round is that this binary target is false for arbitrary kernels. There is a genuine intermediate “selector skin” phase: nontrivial aliasing may be repairable by exactly one bit even though it is not already faithful. The useful invariant is therefore not ordinary fiber dimension alone, but a transversal-chromatic invariant.

1 The binary kernel-dimension target fails

Let $X = \kappa^m$, let $\mathcal{U}^{[m]}$ be the finite Fubini product of a κ -complete nonprincipal ultrafilter $\mathcal{U}$, and let E be an equivalence relation on X . For a repair map $\tau : X \rightarrow \Theta$, write

$$\text{Iso}_E(\tau) = \{x \in X : [x]_E \cap \tau^{-1}\{\tau(x)\} = \{x\}\}.$$

Then τ repairs the kernel on a large set exactly when $\text{Iso}_E(\tau) \in \mathcal{U}^{[m]}$. Define

$$\text{res}_{\mathcal{U}^{[m]}}(E) = \min\{|\Theta| : \exists \tau : X \rightarrow \Theta \text{ Iso}_E(\tau) \in \mathcal{U}^{[m]}\}.$$

Warning 1 (one-bit counterexample to the naive dimension law). *For $m = 1$, let $p : \kappa \rightarrow \kappa$ be a fixed-point-free involution and let E_p have classes $\{\alpha, p(\alpha)\}$. Then $\text{Sing}(E_p) = \emptyset$, so the quotient is not faithful on any large set. But choose one representative from each pair, call the selector T . Both T and $\kappa \setminus T$ are partial transversals; since $\mathcal{U}$ is an ultrafilter, one of them is $\mathcal{U}$ -large. Thus a binary repair bit, marking that large selector, makes the sensor faithful on a large set:*

$$\text{res}_{\mathcal{U}}(E_p) = 2.$$

The same construction works inside a finite Fubini coordinate by pairing one coordinate dimension and taking a $\mathcal{U}$ -large side of the pair selector. Hence the proposed law “nonzero kernel dimension implies full κ -repair” is too coarse.

The lesson is important. The obstruction is not merely that an alias class has more than one point. The obstruction is that the alias relation has no large partial transversal. Pairing aliases are nontrivial but selectable; projection aliases over a missed Fubini dimension are not selectable.

2 Transversal-chromatic Nyquist law

Definition 1 (large partial transversal and transversal chromatic number). *A set $T \subseteq X$ is a partial E -transversal if $|T \cap [x]_E| \leq 1$ for every $x \in X$. Define the large-transversal predicate*

$$\text{PT}_{\mathcal{U}^{[m]}}(E) \iff \exists T \in \mathcal{U}^{[m]} \text{ } T \text{ is a partial } E\text{-transversal}.$$

The transversal chromatic number is

$$\chi_{\mathcal{U}^{[m]}}^{\text{tr}}(E) = \min\{|\Theta| : \exists \tau : X \rightarrow \Theta \text{ Iso}_E(\tau) \in \mathcal{U}^{[m]}\}.$$

Thus χ^{tr} is the minimum repair alphabet size, stated as a coloring invariant of the kernel.

Theorem 1 (Transversal-chromatic Nyquist collapse). *For every equivalence relation E on κ^m , with $m < \omega$,*

$$\chi_{\mathcal{U}^{[m]}}^{\text{tr}}(E) = \text{res}_{\mathcal{U}^{[m]}}(E) = \begin{cases} 1, & \text{Sing}(E) \in \mathcal{U}^{[m]}, \\ 2, & \text{Sing}(E) \notin \mathcal{U}^{[m]} \text{ and } \text{PT}_{\mathcal{U}^{[m]}}(E), \\ \kappa, & \neg \text{PT}_{\mathcal{U}^{[m]}}(E). \end{cases}$$

Equivalently, every successful subcritical repair alphabet collapses to a one-bit repair.

Proof sketch. The equality between χ^{tr} and res is by definition of $\text{Iso}_E(\tau)$. If $\text{Sing}(E) \in \mathcal{U}^{[m]}$, the one-symbol repair map works. If $T \in \mathcal{U}^{[m]}$ is a partial transversal, the two-color map $\tau = 0$ on T and $\tau = 1$ off T has $T \subseteq \text{Iso}_E(\tau)$, so two symbols work. If $\theta < \kappa$ and some $\tau : X \rightarrow \theta$ works, then $A = \text{Iso}_E(\tau) \in \mathcal{U}^{[m]}$. Because $\mathcal{U}^{[m]}$ is κ -complete, one color class $A \cap \tau^{-1}\{i\}$ is large. That color class is a partial E -transversal. Therefore subcritical repair implies $\text{PT}_{\mathcal{U}^{[m]}}(E)$. Finally, $|\kappa^m| = \kappa$, so an injective κ -coloring repairs every kernel. $\square$

This is the sharper law hidden behind the earlier repair equations. The full-width phase is not “large fibers”; it is the absence of a large independent selector for the alias graph. All finite, countable, and generally subcritical repair alphabets are equivalent: if any one works, one bit works.

3 Selector dimension vector

The correct local dimension object is a two-coordinate phase vector:

$$\text{phase}_{\mathcal{U}}(E) = (a_U(E), d_U(E)),$$

where

$$a_U(E) = 0 \iff \text{Sing}(E) \in U, \quad d_U(E) = 0 \iff \text{PT}_U(E).$$

Since a singleton-class large set is itself a large partial transversal, only three phases can occur:

$$F = (0, 0), \quad S = (1, 0), \quad D = (1, 1).$$

They mean:

phase	meaning	repair cost
F	faithful on a large set	1
S	selector skin: aliased but large-selectable	2
D	selector defect: no large partial transversal	κ

Coordinate projections from Round 6 only show the endpoints:

$$\text{phase}(\ker \pi_D) = \begin{cases} F, & D = m, \\ D, & D \subsetneq m. \end{cases}$$

The fixed-point-free involution kernel realizes the missing phase S . Thus the dimension theory for arbitrary kernels must distinguish aliasing from selector defect:

$$\text{aliasing} \neq \text{full unrecoverability}, \quad \text{selector defect} = \text{full unrecoverability}.$$

4 Fusion and the global spinal law

For a sensor portfolio Γ , let $E_{\Gamma,n}$ be the fused local kernel at coordinate n ; equivalently, it is the intersection of the component kernels. Put

$$r_{\Gamma,n} = \chi_{\mathcal{U}_n^{[m_n]}}^{\text{tr}}(E_{\Gamma,n}) \in \{1, 2, \kappa_n\}.$$

Let μ_n be the available repair bits and let $L = H \vee J$ be the effective ideal combining noise and monitoring tolerance.

Law 1 (Spinal transversal-chromatic Nyquist equation). *The fused portfolio plus repair bits is robustly faithful exactly when*

$$\{n : 2^{\mu_n} < r_{\Gamma,n}\} \in L.$$

Equivalently, if

$$S_{\Gamma} = \{n : r_{\Gamma,n} = 2\}, \quad D_{\Gamma} = \{n : r_{\Gamma,n} = \kappa_n\},$$

then robust faithful observation is possible exactly when

$$\{n \in S_{\Gamma} : \mu_n = 0\} \cup \{n \in D_{\Gamma} : 2^{\mu_n} < \kappa_n\} \in H \vee J.$$

This refines the Fubini dimension-cover law. In pure projection portfolios, the selector-skin set S_{Γ} is empty, so the equation collapses to the Round 6 dimension-cover equation. For arbitrary kernels, a new design possibility appears: some coordinates are not faithful but need only one repair bit.

Fusion can move kernels between all three phases. Two maximally defective projections on κ^2 , π_0 and π_1 , satisfy

$$\text{phase}(\ker \pi_0) = D, \quad \text{phase}(\ker \pi_1) = D, \quad \text{phase}(\ker \pi_0 \cap \ker \pi_1) = F.$$

A pairing kernel satisfies $\text{phase} = S$. If it is fused with a projection that already observes the paired coordinate, it may become F ; if it is fused only with unrelated coarse sensors, it remains S and needs exactly one bit.

5 Finite experiments

Finite nonprincipal ultrafilters do not exist. The finite tests below use threshold Fubini shadows. A subset of $\prod_i k_i$ is large if it has at least t_i large sections at each Fubini level. For a finite equivalence relation E , define the exact finite repair thickness

$$q_{\min}^{\text{fin}}(E) = \min_{T \text{ threshold-large}} \max_C |T \cap C|,$$

where C ranges over the equivalence classes. The case $q_{\min}^{\text{fin}} = 1$ is the finite analogue of a large partial transversal.

Grid	thresholds	kernel	classes	large sets	exact $q_{\min}^{\text{fin}}$
(3, 3)	(2, 2)	equality	9	256	1
(3, 3)	(2, 2)	pair last coordinate	6	256	1
(3, 3)	(2, 2)	project first coordinate	3	256	2
(3, 3)	(2, 2)	diagonal mod 3	3	256	2
(3, 3)	(2, 2)	universal collapse	1	256	4
(4, 4)	(3, 2)	pair last coordinate	8	41261	1
(4, 4)	(3, 3)	pair last coordinate	8	6125	2
(4, 4)	(3, 2)	project first coordinate	4	41261	2
(4, 4)	(3, 3)	project first coordinate	4	6125	3

The table separates the phenomena. Pairing aliases can have a large selector, so the finite repair thickness can be 1. Projection aliases require thickness equal to the hidden threshold. Strong finite thresholds may suppress selector behavior even for pairings; this is a finite-shadow artifact, not a contradiction of the ultrafilter theorem, because a true ultrafilter must choose one side of every binary selector partition.

Random finite partitions show why finite shadows can be misleadingly graded. The following table gives 200 trials for each row.

Grid and thresholds	random classes	observed distribution of $q_{\min}^{\text{fin}}$	trials
(3, 3), (2, 2)	3 classes	$q = 2 : 198, q = 3 : 2$	200
(3, 3), (2, 2)	4 classes	$q = 1 : 128, q = 2 : 72$	200
(3, 3), (3, 2)	4 classes	$q = 2 : 178, q = 3 : 22$	200
(3, 3), (3, 3)	4 classes	$q = 3 : 69, q = 4 : 88, q = 5 : 40, q = 6 : 3$	200

Finite threshold models have many intermediate repair thicknesses. The transfinite κ -complete model collapses this entire subcritical range: if a finite or $< \kappa$ -sized repair palette succeeds, one color class is large and becomes a one-bit selector witness.

Experiment skeleton.

```

qmin = min(
    max(len(T & C) for C in equivalence_classes)
    for T in all_subsets(X)
    if threshold_large(T)
)
# qmin == 1  <=>  finite large partial transversal exists.

```

6 Applications

Sensor-bank design. The design target is not just to reduce alias-class size. Each coordinate should be classified as faithful F , selector skin S , or selector defect D . The cost schedule is then zero bits, one bit, or critical width. This gives an optimization problem:

$$\min_{\Gamma, \mu} \text{cost}(\Gamma, \mu) \quad \text{subject to} \quad \{n : 2^{\mu_n} < \chi_{\mathcal{U}_n}^{\text{tr}}(E_{\Gamma, n})\} \in H \vee J.$$

Privacy and re-identification. Finite multiplicity is not robust privacy in this setting. A mechanism that only pairs records may look anonymous, but a one-bit selector can recover a large faithful slice. Durable transfinite privacy requires selector defect: no large partial transversal on an effective-positive coordinate set.

Active diagnostics. If a local kernel is in phase S , one yes/no probe can choose a large representative skin and make the channel faithful. If it is in phase D , no finite-valued active probe can help unless it changes the kernel enough to create a large transversal or reaches critical width.

Verification. For monitoring systems, the alias graph should be audited for large independent selectors. A large independent selector means binary de-aliasing is enough; absence of one is the true certificate of full unrecoverability.

Finite calibration. Finite simulations should report $q_{\min}^{\text{fin}}$, not only average fiber size or entropy. Intermediate finite values are useful engineering data, but the transfinite extrapolation is discontinuous: all successful subcritical palettes collapse to the selector-skin phase.

7 Round 7 summary and next target

The main new equation is

$$\text{res}_U(E) = \chi_U^{\text{tr}}(E) = \begin{cases} 1, & \text{Sing}(E) \in U, \\ 2, & \text{Sing}(E) \notin U \text{ and } \text{PT}_U(E), \\ \kappa, & \neg \text{PT}_U(E). \end{cases}$$

The exceptional discovery is the one-bit skin phase. It refutes the naive binary dimension target for arbitrary kernels and replaces it with a selector-dimensional law. Projection kernels have no selector skin, which is why the Round 6 dimension-cover law was binary; arbitrary kernels need the third phase.

The next target is now sharper: develop structural criteria for $\text{PT}_U(E)$ for definable finite Fubini kernels. In graph language, this asks when the alias graph of E has a $\mathcal{U}^{[m]}$ -large independent set. In set-theoretic language, it asks for a Rudin-Keisler/Tukey classification of kernels by their transversal chromatic number under fusion.

Round 8: Bounded-Multiplicity Rescue and the Anti- k -Anonymity Law

A continuation of the spinal-nullity sensor/repair program

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Round aim. Round 7 isolated the selector-skin phase: a nonfaithful kernel can be repaired by one bit exactly when its alias relation has a large partial transversal. The next blind spot is structural: when can one recognize this phase without already knowing a transversal? The main finding of this round is that, under a κ -complete ultrafilter, all subcritical multiplicity collapses. If an alias relation is bounded-to-one on any large set, then it is already one-bit repairable. Thus full-width defect is not high average ambiguity; it is the absence of every large subcritical-multiplicity thinning.

1. Essential alias degree

Let W be a κ -complete nonprincipal ultrafilter on a set X of size κ , and let E be an equivalence relation on X . For $A \subseteq X$, write

$$\Delta_A(E) = \sup\{|A \cap C| : C \in X/E\}.$$

Define the W -essential alias degree

$$\text{edeg}_W(E) = \min\{\mu \leq \kappa : \exists A \in W \ \Delta_A(E) \leq \mu\}.$$

A partial transversal is exactly a set T with $\Delta_T(E) \leq 1$.

Theorem (Essential multiplicity collapse). *For every equivalence relation E on X ,*

$$\boxed{\text{edeg}_W(E) \in \{1, \kappa\}.$$

More precisely,

$$\text{edeg}_W(E) = 1 \iff \exists T \in W \text{ such that } T \text{ is a partial } E\text{-transversal,}$$

and if no large partial transversal exists, then $\text{edeg}_W(E) = \kappa$.

Proof sketch. If a large partial transversal exists, then the essential degree is 1. Conversely, suppose $A \in W$ and $\Delta_A(E) \leq \mu < \kappa$. By choice, color each class $A \cap C$ injectively with colors below μ . The color classes are partial transversals and they partition A into fewer than κ many pieces. Since W is κ -complete, one color class is W -large. Thus subcritical multiplicity implies a large transversal. If no large transversal exists, no subcritical bound can occur on a large set, so the minimum is κ .

Combining this with Round 7 gives the local repair equation in a more diagnostic form:

$$\boxed{\text{res}_W(E) = \begin{cases} 1, & \text{Sing}(E) \in W, \\ 2, & \text{Sing}(E) \notin W \text{ and } \text{edeg}_W(E) = 1, \\ \kappa, & \text{edeg}_W(E) = \kappa. \end{cases}}$$

The exceptional point is the discontinuity: there is no stable repair phase corresponding to 2-to-one, k -to-one, countable-to-one, or any other fixed subcritical multiplicity. All of them collapse to the one-bit selector skin.

2. Bounded-multiplicity rescue

Corollary (Bounded-to-one sensors are one-bit repairable). *If there are $A \in W$ and $\mu < \kappa$ such that every E -class meets A in at most μ points, then*

$$\text{res}_W(E) \leq 2.$$

If the kernel is not already faithful on a large set, then $\text{res}_W(E) = 2$.

Law (Defect-breaker fusion law). *Let E and B be local sensor kernels on X . If B is bounded-to-one on a W -large set with bound $< \kappa$, then the fused kernel*

$$E \wedge B = E \cap B$$

is not selector-defective:

$$\boxed{\text{res}_W(E \wedge B) \leq 2.}$$

Thus a bounded-multiplicity auxiliary sensor is a universal full-defect breaker: it converts any fused coordinate from the κ -repair phase into either the faithful phase or the one-bit selector-skin phase.

Proof sketch. Classes of $E \wedge B$ are contained in classes of B . Hence the same large bounded-multiplicity set witnesses subcritical essential degree for $E \wedge B$. Essential multiplicity collapse then supplies a large partial transversal.

This gives a sharper engineering principle. Projection kernels are maximally defective because a missed Fubini dimension creates classes of critical width. But if a second critical-width channel hashes the hidden dimension into uniformly bounded buckets, the full defect disappears even if the hash is not injective.

3. Residual hash-Nyquist law

Let the local hidden state be $X = \kappa^m$ with finite Fubini ultrafilter $U^{[m]}$. Fix $D \subseteq m$ and put $R = m \setminus D$. The projection π_D leaves the residual coordinates R hidden. Let a residual hash

$$h : \kappa^R \rightarrow Y$$

be fused with π_D , so the sensor is

$$\Sigma_{D,h}(x) = (x \upharpoonright D, h(x \upharpoonright R)).$$

Let E_h be the kernel of h on the residual block, with its finite Fubini ultrafilter $U^{[R]}$.

Law (Residual hash-Nyquist law). *For separated residual hashes,*

$$\boxed{\text{res}_{U^{[m]}}(\ker \Sigma_{D,h}) = \begin{cases} 1, & \text{Sing}(E_h) \in U^{[R]}, \\ 2, & \text{Sing}(E_h) \notin U^{[R]} \text{ and } \text{edeg}_{U^{[R]}}(E_h) = 1, \\ \kappa, & \text{edeg}_{U^{[R]}}(E_h) = \kappa. \end{cases}}$$

In particular, if h is $< \kappa$ -to-one on a $U^{[R]}$ -large residual set, then the projection defect is reduced to at most one bit.

Proof sketch. Sections over the observed coordinates D reduce the problem to the residual kernel E_h . A large partial transversal for E_h lifts to a large partial transversal for $\ker \Sigma_{D,h}$ by using the same selector in almost every residual section. Conversely, any large transversal for the fused kernel has residual sections giving a large transversal for E_h on a large set of D -coordinates. The three cases then follow from essential multiplicity collapse.

A finite-valued hash cannot usually do this: if $|Y| < \kappa$, κ -completeness pushes some hash fiber to be large. The useful intermediate design is therefore a critical-width but bounded-to-one hash. It does not fully decode the residual dimension, yet with one extra bit it is faithful on a large set.

4. Global spinal degree-cover equation

Return to the countable spine, with effective ideal $L = H \vee J$. At coordinate n , let $E_{\Gamma,n}$ be the fused kernel of a portfolio Γ , and put

$$\delta_{\Gamma,n} = \text{edeg}_{U_n}(E_{\Gamma,n}) \in \{1, \kappa_n\}.$$

Define

$$F_{\Gamma} = \{n : \text{Sing}(E_{\Gamma,n}) \in U_n\}, \quad S_{\Gamma} = \{n : n \notin F_{\Gamma} \text{ and } \delta_{\Gamma,n} = 1\}, \\ D_{\Gamma} = \{n : \delta_{\Gamma,n} = \kappa_n\}.$$

Then $F_{\Gamma}, S_{\Gamma}, D_{\Gamma}$ are the faithful, selector-skin, and selector-defect supports.

Law (Spinal essential-degree repair equation). *With repair bit budgets μ_n , robust faithful observation is possible exactly when*

$$\boxed{\{n \in S_{\Gamma} : \mu_n = 0\} \cup \{n \in D_{\Gamma} : 2^{\mu_n} < \kappa_n\} \in H \vee J.}$$

Equivalently,

$$\boxed{\{n : 2^{\mu_n} < \text{res}_{U_n}(E_{\Gamma,n})\} \in H \vee J.}$$

The new contribution is that the full-defect support can be recognized by essential degree:

$$D_{\Gamma} = \{n : \text{edeg}_{U_n}(E_{\Gamma,n}) = \kappa_n\}.$$

This suggests a portfolio invariant that generalizes the Round 6 dimension cover:

$$\text{bdcov}_L(\mathcal{S}) = \min\{|\Gamma| : \{n : \text{edeg}_{U_n}(E_{\Gamma,n}) = \kappa_n\} \in L\}.$$

It is the least number of sensors needed so that one repair bit per remaining relevant coordinate suffices. For projection-only portfolios, bdcov_L reduces to hidden-dimension cover. With bounded-multiplicity hashes available, it can be strictly smaller: one hash of the residual block can destroy full defect without observing the residual coordinates injectively.

5. Finite experiments

Finite threshold models are not ultrafilters, so they can show graded multiplicity that the transfinite theorem collapses. They are still useful for detecting which finite effects are artifacts. For a finite Fubini threshold grid, define

$$q_{\min}^{\text{fin}}(E) = \min_{T \text{ threshold-large}} \max_{C \in X/E} |T \cap C|.$$

The value $q_{\min}^{\text{fin}} = 1$ is the finite analogue of a large partial transversal.

Grid	thresholds	kernel	classes	exact $q_{\min}^{\text{fin}}$
(3, 4)	(2, 2)	row projection	3	2
(3, 4)	(2, 2)	row + hidden pairs	6	1
(3, 4)	(2, 2)	row + hidden triples	6	1
(3, 4)	(2, 2)	universal collapse	1	4
(4, 4)	(3, 3)	row projection	4	3
(4, 4)	(3, 3)	row + hidden pairs	8	2
(4, 4)	(3, 3)	row + hidden triples	8	2
(4, 4)	(3, 3)	universal collapse	1	9
(3, 6)	(2, 3)	row projection	3	3
(3, 6)	(2, 3)	row + hidden pairs	9	1
(3, 6)	(2, 3)	row + hidden triples	6	2
(4, 5)	(3, 3)	row + hidden pairs	12	1
(4, 5)	(3, 3)	row + hidden triples	8	2

The deterministic pattern has a simple finite explanation. For a two-dimensional row projection plus a hidden-column block hash with B block classes per row, the finite thickness is

$$q_{\min}^{\text{fin}} = \left\lceil \frac{t_{\text{col}}}{B} \right\rceil$$

when the blocks have enough capacity. Finite thresholds may therefore fail to select one side of every bounded alias class, because unlike an ultrafilter they need not choose one cell of a finite partition as large.

Random bounded-class partitions on the $(4, 4)$ grid with thresholds $(3, 3)$ show the same finite artifact. There were 200 exact brute-force trials per row.

max class size	observed distribution of $q_{\min}^{\text{fin}}$	average
2	$q = 1 : 187, q = 2 : 13$	1.065
3	$q = 1 : 61, q = 2 : 139$	1.695
4	$q = 1 : 8, q = 2 : 191, q = 3 : 1$	1.965
5	$q = 1 : 2, q = 2 : 162, q = 3 : 36$	2.170
8	$q = 2 : 45, q = 3 : 132, q = 4 : 19, q = 5 : 4$	2.910

Experiment skeleton:

$$\begin{aligned} q_{\min} &= \min\{\max_C |T \cap C| : T \text{ is threshold-large}\}, \\ \text{bounded class size} < \kappa &\implies \text{transfinite } q_{\min} \text{ collapses to } 1. \end{aligned}$$

The tests match the theory in shape but also show the finite-shadow trap: finite thresholds retain graded q -values, while κ -complete ultrafilters collapse every successful subcritical bound to a large selector.

6. Applications

Privacy and anti- k -anonymity. Fixed finite k -anonymity is selector-skin, not durable transfinite privacy. If every record is hidden among at most $k < \kappa$ candidates on a large region, one extra bit can select a faithful large slice. Durable privacy requires essential alias degree κ : no large subset may have any subcritical multiplicity bound.

Sensor-bank design. A full critical-width decoder is not the only way to escape projection defect. A bounded-to-one residual hash, followed by one bit, is enough. This suggests a cheaper two-stage architecture: first reduce every critical alias class to bounded buckets, then use a binary selector skin.

Distributed diagnostics. A diagnostic channel can be designed as a defect breaker rather than a full identifier. It only needs to make residual aliases bounded-to-one on a large set; the final one-bit probe chooses the large selector.

Verification and monitoring. Audits should report essential alias degree, not merely entropy or average class size. Average small fibers do not matter if every large set still contains critical aliasing. Conversely, uniformly bounded fibers imply one-bit repair even if the sensor is not injective.

Set theory. The invariant $\text{edeg}_W(E)$ gives a compact Rudin-Keisler/Tukey diagnostic for kernels: every kernel is either essentially transversal or essentially critical. The next structural problem is to classify which definable finite Fubini kernels force $\text{edeg} = \kappa$.

7. Round 8 summary and next target

The main new equation is the essential multiplicity collapse

$$\boxed{\text{edeg}_W(E) \in \{1, \kappa\}.$$

Thus every subcritical alias multiplicity on a large set collapses to a one-bit selector:

$$\boxed{\exists A \in W \ \Delta_A(E) < \kappa \implies \text{res}_W(E) \leq 2.}$$

The most useful engineering law is the defect-breaker fusion equation

$$\boxed{\text{edeg}_W(B) < \kappa \implies \text{res}_W(E \wedge B) \leq 2.}$$

For residual hidden dimensions, this yields the hash-Nyquist principle: a critical-width bounded-to-one hash of the residual block plus one bit can replace full coordinate recovery.

Next target. Determine exact criteria for selector defect in definable finite Fubini kernels. The leading conjectural shape is a dichotomy between kernels that admit a bounded-multiplicity large thinning and kernels that contain a projection-like critical fiber obstruction on every large set. If true, it would turn the abstract $\text{edeg} = \kappa$ condition into a structural finite-dimensional obstruction theorem.

Round 9: Fubini Selector Matching and the No-Teleport Law

A continuation of the spinal-nullity sensor/repair program

April 30, 2026 – exploratory draft, not peer reviewed

Round aim. Round 8 showed that essential alias multiplicity collapses: a kernel is either essentially transversal or essentially critical. The open structural question was how to recognize selector defect in finite Fubini kernels. The main discovery here is an exact two-level criterion. In a Fubini coordinate, a large selector is the same thing as a large family of row-wise injective selectors whose observed palettes are pairwise disjoint. This reveals a blind spot: a sensor can be injective inside almost every row and still be maximally unrecoverable because different rows reuse the same code palette.

1 Two-level Fubini selector matching

Let $\mathcal{V}$ and $\mathcal{U}$ be nonprincipal countably complete ultrafilters on sets I and K , and let

$$\mathcal{V} \otimes \mathcal{U} = \{A \subseteq I \times K : \{i \in I : A_i \in \mathcal{U}\} \in \mathcal{V}\}, \quad A_i = \{j : (i, j) \in A\}.$$

For a sensor $f : I \times K \rightarrow Y$, let E_f be its alias relation:

$$(i, j) E_f (i', j') \iff f(i, j) = f(i', j').$$

Write $f_i(j) = f(i, j)$. A set $T \subseteq I \times K$ is a partial E_f -transversal iff $f \upharpoonright T$ is injective.

Definition 1 (large disjoint-palette matching). *A $\text{DPM}_{\mathcal{V}, \mathcal{U}}(f)$ -witness is a set $A \in \mathcal{V}$ and sets $B_i \in \mathcal{U}$, for $i \in A$, such that*

1. $f_i \upharpoonright B_i$ is injective for every $i \in A$;
2. the palettes $P_i = f_i[B_i]$ are pairwise disjoint: $P_i \cap P_{i'} = \emptyset$ whenever $i \neq i'$ in A .

Write $\text{DPM}_{\mathcal{V}, \mathcal{U}}(f)$ when such a witness exists.

Theorem 1 (Fubini selector-matching law). *For every sensor $f : I \times K \rightarrow Y$,*

$$\text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_f) \iff \text{DPM}_{\mathcal{V}, \mathcal{U}}(f).$$

Equivalently, a two-level Fubini kernel has a large partial transversal exactly when it admits a large disjoint-palette matching.

Proof sketch. If $T \in \mathcal{V} \otimes \mathcal{U}$ is a partial E_f -transversal, put

$$A = \{i : T_i \in \mathcal{U}\} \in \mathcal{V}, \quad B_i = T_i \quad (i \in A).$$

Since f is injective on T , each row restriction $f_i \upharpoonright B_i$ is injective, and no two different rows can use the same observed value on their selected sets. Hence the palettes are pairwise disjoint. Conversely, if A and B_i 's witness DPM , then

$$T = \bigcup_{i \in A} \{i\} \times B_i$$

is $(\mathcal{V} \otimes \mathcal{U})$ -large and f is injective on T . Thus T is a large partial transversal. $\square$

Combining this with the Round 7/8 repair calculus gives

$$\text{res}_{\mathcal{V} \otimes \mathcal{U}}(E_f) = \begin{cases} 1, & \text{Sing}(E_f) \in \mathcal{V} \otimes \mathcal{U}, \\ 2, & \text{Sing}(E_f) \notin \mathcal{V} \otimes \mathcal{U} \text{ and } \text{DPM}_{\mathcal{V}, \mathcal{U}}(f), \\ \kappa, & \neg \text{DPM}_{\mathcal{V}, \mathcal{U}}(f), \end{cases}$$

where κ is the local critical width. Thus the three phases are now structurally visible: faithful, selector skin, and selector defect.

2 Two primitive causes of selector defect

The matching law separates selector defect into two mechanisms that were previously conflated.

Row-collapse defect. For $\mathcal{V}$ -many rows, no $\mathcal{U}$ -large subset is injectively labeled by the row sensor. The row projection

$$f(i, j) = i$$

has this defect: every row is constant, so a row-wise partial transversal has at most one point in that row, never a $\mathcal{U}$ -large row section.

Palette-collision defect. Row-wise large injective selectors exist, but their image palettes cannot be made pairwise disjoint on a $\mathcal{V}$ -large family of rows. The column projection

$$f(i, j) = j$$

has this defect. Each row is locally injective, but any selected palette $B_i \in \mathcal{U}$ is $\mathcal{U}$ -large in the same code space K . Two $\mathcal{U}$ -large palettes intersect, so no two rows can be assigned disjoint palettes, let alone $\mathcal{V}$ -many rows.

This explains why losing either Fubini dimension is fatal, but for different reasons:

$$f(i, j) = i \quad \text{fails inside rows,} \quad f(i, j) = j \quad \text{fails by cross-row palette collision.}$$

The identity sensor $f(i, j) = (i, j)$ avoids both defects: each row is injective and the row tag separates palettes.

3 The namespace law

Let $g_i : K \rightarrow Y$ be row-local sensors. Compare the unnamespaced sensor

$$f(i, j) = g_i(j)$$

with the namespaced sensor

$$\hat{f}(i, j) = (i, g_i(j)).$$

Law 1 (Namespace law). *For the namespaced sensor,*

$$\text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_{\hat{f}}) \iff \{i \in I : \text{PT}_{\mathcal{U}}(E_{g_i})\} \in \mathcal{V}.$$

Therefore adding the row namespace removes the cross-row palette-collision obstruction and reduces two-level selector repair to the row-local selector problem.

Proof sketch. If $\text{PT}_{\mathcal{U}}(E_{g_i})$ holds on a $\mathcal{V}$ -large set A , choose $B_i \in \mathcal{U}$ on which g_i is injective. Then $\hat{f}_i[B_i] \subseteq \{i\} \times Y$, so palettes from different rows are automatically disjoint. The matching law gives $\text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_{\hat{f}})$. The converse follows by projecting any large $\hat{f}$ -transversal to its row sections. $\square$

A sharp special case is the reused-code law. If $g_i = g$ for all i , and $g_*\mathcal{U}$ is nonprincipal, then

$$f(i, j) = g(j)$$

after local row selection still reuses one nonprincipal observed ultrafilter. Any two large selected palettes are $g_*\mathcal{U}$ -large and must intersect. Hence

$$\text{PT}_{\mathcal{U}}(E_g) \text{ may hold, but } \text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_f) \text{ fails.}$$

Local identifiers without a namespace are Fubini-defective.

Corollary 1 (Critical namespace equation). *For the column sensor with a namespace code $c : I \rightarrow \Theta$,*

$$f_c(i, j) = (c(i), j),$$

one has

$$\text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_{f_c}) \iff c \text{ is injective on some } A \in \mathcal{V}.$$

Consequently, if the outer coordinate has critical width κ , then

$$\min\{|\Theta| : \exists c : I \rightarrow \Theta \text{ with } \text{PT}(E_{f_c})\} = \kappa.$$

A subcritical namespace cannot repair column-projection palette collision.

4 Subcritical no-teleport law

The namespace equation is a special case of a more general permanence theorem for selector defect.

Theorem 2 (Subcritical no-teleport). *Let $\mathcal{W}$ be a κ -complete ultrafilter on X , and let E be selector-defective:*

$$\text{edeg}_{\mathcal{W}}(E) = \kappa.$$

If $c : X \rightarrow \Theta$ has $|\Theta| < \kappa$, then the fused kernel remains selector-defective:

$$\text{edeg}_{\mathcal{W}}(E \wedge \ker(c)) = \kappa.$$

Equivalently, no subcritical auxiliary observation can move a true defect coordinate into the one-bit selector-skin phase.

Proof sketch. Suppose $T \in \mathcal{W}$ were a partial transversal for $E \wedge \ker(c)$. Partition T into the fewer than κ many sets $T_\theta = T \cap c^{-1}(\{\theta\})$. By κ -completeness and ultrafilterness, some $T_\theta \in \mathcal{W}$. Inside one c -fiber, $E \wedge \ker(c)$ agrees with E , so T_θ is a $\mathcal{W}$ -large partial E -transversal, contradiction. $\square$

Thus subcritical feature engineering is not merely insufficient for faithful decoding; it cannot even create the one-bit selector skin. The exact local capacity law can be written as

$$\text{crit}_{\mathcal{W}}(E) = \begin{cases} 1, & \text{Sing}(E) \in \mathcal{W}, \\ 2, & \text{Sing}(E) \notin \mathcal{W} \text{ and } \text{edeg}_{\mathcal{W}}(E) = 1, \\ \kappa, & \text{edeg}_{\mathcal{W}}(E) = \kappa. \end{cases}$$

A total extra transcript alphabet of size $< \text{crit}_{\mathcal{W}}(E)$ cannot make the coordinate faithful. If $|X| = \kappa$, the boundary is sharp: an injective κ -valued code makes the fused kernel faithful.

This also covers finite or transfinite adaptive protocols whose total transcript set has size $< \kappa$. After flattening the protocol to its transcript map, the theorem says that adaptive low-bandwidth querying cannot teleport a selector-defect coordinate out of the critical phase.

5 Recursive matching tree and spinal capacity law

For a finite Fubini product $\mathcal{U}_0 \otimes \cdots \otimes \mathcal{U}_{m-1}$, the two-level theorem recursively unfolds into a matching tree.

For $m = 1$, a palette witness for $f : X_0 \rightarrow Y$ is a set $B \in \mathcal{U}_0$ such that $f \upharpoonright B$ is injective, with terminal palette $f[B]$. For $m + 1$, write $x = (a, z)$, and $f_a(z) = f(a, z)$. A palette-tree witness consists of

1. an outer set $A \in \mathcal{U}_0$;
2. for each $a \in A$, a tail palette-tree witness T_a for f_a ;
3. pairwise disjoint terminal palettes $f_a[T_a]$ for $a \in A$.

Then

$$\text{PT}_{\mathcal{U}_0 \otimes \dots \otimes \mathcal{U}_m}(E_f) \iff \text{there is a Fubini palette-tree witness for } f.$$

This is formally repeated sectioning, but it is useful: selector defect is failure of recursive palette separation, not merely large average fiber size.

Globally, let $L = H \vee J$ be the effective ideal. At coordinate n , let $E_{\Gamma,n}$ be the current fused kernel and let ν_n be the total extra alphabet capacity available at that coordinate, including added features, repair bits, and finite adaptive transcripts. Define

$$c_n = \text{crit}_{U_n}(E_{\Gamma,n}) \in \{1, 2, \kappa_n\}.$$

The capacity-permanence equation is

$$\boxed{\text{robust faithful observation is possible} \iff \{n : \nu_n < c_n\} \in H \vee J.}$$

The new content is the permanence clause: if $c_n = \kappa_n$, no cascade of subcritical auxiliary features can lower c_n outside the tolerated ideal. Critical coordinates require critical-width information or must be ignored by $H \vee J$.

6 Finite experiments

Finite nonprincipal ultrafilters do not exist, so these are finite shadows. In a $k_0 \times k_1$ grid with thresholds (t_0, t_1) , a set is large if at least t_0 rows contain at least t_1 selected cells. The finite matching criterion says:

large partial transversal exists $\iff$ some t_0 rows can be assigned t_1 labels each, with no label reused.

I checked this against brute-force subset search on random finite sensors.

Grid, thresholds	sensor	row labels	labels	skin?	note
(4, 5), (2, 3)	identity	5–5	20	yes	full namespace
(4, 5), (2, 3)	row projection	1–1	4	no	row collapse
(4, 5), (2, 3)	column projection	5–5	5	no	palette collision
(4, 5), (2, 3)	sum mod 5	5–5	5	no	palette collision
(4, 5), (2, 3)	namespaced column	5–5	20	yes	namespace rescue
(4, 5), (2, 2)	column projection	5–5	5	yes*	finite artifact
(5, 6), (3, 3)	column projection	6–6	6	no	collision
(5, 6), (2, 3)	column projection	6–6	6	yes*	finite artifact

Rows marked yes* are weak-threshold artifacts: finite thresholds may allow two disjoint large-enough palettes in a finite set. A true ultrafilter does not allow two disjoint large palettes.

Grid, thresholds	labels	trials	skin yes	skin no	mismatches
(3, 3), (2, 2)	4	200	131	69	0
(3, 4), (2, 2)	5	160	159	1	0
(4, 3), (2, 2)	5	160	158	2	0
(3, 4), (2, 3)	6	160	35	125	0

The zero-mismatch column means brute-force search for a large partial transversal agreed exactly with the row-palette matching criterion.

I also tested the finite no-teleport shadow. Let E be row aliasing on a $k \times c$ grid. A q -valued auxiliary code refines classes to row-color cells, so every fused partial transversal has size at most kq . If the finite largeness threshold is T , then no $q < \lceil T/k \rceil$ auxiliary code can rescue the defect. A cyclic code with $q = \lceil T/k \rceil$ succeeds when enough columns are available.

rows	cols	T	$q_{\min}$	tested q	trials	random rescues	constructed rescue
6	10	25	5	4	300	0	yes
8	8	40	5	4	300	0	yes
10	6	51	6	5	300	0	yes
12	5	49	5	4	300	0	yes

This finite inequality is the shadow of the transfinite theorem: a union of fewer than κ failed selector slices is still failed, so subcritical transcripts cannot create a large selector.

7 Applications

Sensor-bank design. A local sensor audit should check two independent properties: row-local injective selectors and cross-row palette separation. A sensor can pass the first test and fail the second. Repair budgets should be assigned by the matching-explicit capacity equation, not by average fiber size or entropy.

Namespaces and distributed identifiers. Local identifiers reused across shards are Fubini-defective even when each shard is internally injective. Adding the shard/row namespace is not cosmetic metadata; it is a Nyquist dimension. If there are κ relevant shards, any subcritical namespace leaves a large collision class.

Privacy and re-identification. Pairwise aliases are one-bit vulnerable, but palette-collision aliases can be critical-width robust. Adding a namespace can destroy this protection by making row palettes disjoint. Durable privacy should distinguish multiplicity anonymity from palette-collision anonymity.

Active diagnostics. If failure is row-collapse, improve row-local resolution. If failure is palette collision, add namespaces or disjoint codebooks. These are different interventions; adding more local precision without separating palettes may leave the coordinate selector-defective.

Verification and monitoring. Monitoring systems should report why repair fails: local collapse, palette collision, or subcritical transcript capacity. This gives a diagnostic certificate for why finite-valued repair cannot produce faithful observation.

8 Round 9 summary and next target

The main new equation is the **Fubini selector-matching law**

$$\text{PT}_{\mathcal{V} \otimes \mathcal{U}}(E_f) \iff \exists A \in \mathcal{V} \exists \langle B_i \in \mathcal{U} : i \in A \rangle [f_i \upharpoonright B_i \text{ injective and } f_i[B_i] \cap f_{i'}[B_{i'}] = \emptyset].$$

The exceptional conceptual discovery is the palette-collision obstruction: local injectivity on large row sections is not enough; the selected row palettes must also be globally disjoint.

The second new law is the **subcritical no-teleport theorem**

$$\text{edeg}_{\mathcal{W}}(E) = \kappa \text{ and } |\Theta| < \kappa \implies \text{edeg}_{\mathcal{W}}(E \wedge \ker(c)) = \kappa$$

for every $c : X \rightarrow \Theta$. Thus true selector defect is absorbing under subcritical feature fusion.

Together they yield the spinal capacity-permanence equation

$$\{n : \nu_n < \text{crit}_{U_n}(E_{\Gamma,n})\} \in H \vee J, \quad \text{crit}_{U_n}(E_{\Gamma,n}) \in \{1, 2, \kappa_n\}.$$

The next target is a Tukey/Rudin-Keisler classification of palette-collision kernels: which definable Fubini sensors can be namespaced or codebook-separated by structural maps weaker than full identity, and which require genuinely critical-width namespaces?

Round 10 Addendum: Collision Graphs and Namespace Chromatic Collapse

Spinal nullity, Fubini sensors, and the hidden cost of global keys

April 30, 2026 — exploratory continuation, not peer reviewed

Purpose. Rounds 5–9 isolated a selector-defect phase for Fubini sensors: rowwise information may be locally enough, but palette collisions between rows can still destroy every large partial transversal. Round 9 gave a matching criterion for two-level Fubini sensors and a subcritical no-teleport law. This round sharpens that picture: palette collision is a graph/hypergraph obstruction, but in the true κ -complete setting all subcritical chromatic structure collapses to one question: *does there exist a large independent selector skin?*

Key discovery. Finite systems can be rescued by using many small namespace colors. A true κ -complete coordinate cannot: any subcritical coloring has one large color fiber, so subcritical namespace coloring helps iff a large collision-free row set already existed. Therefore finite chromatic numbers are mostly finite-shadow artifacts; the transfinite invariant is an independence/critical-namespace dichotomy.

1. Two-level Fubini collision graph

Let $W = \mathcal{V} \otimes \mathcal{U}$ be a two-level Fubini ultrafilter on $I \times J$, where $\mathcal{V}$ is κ_I -complete on I and $\mathcal{U}$ is κ_J -complete on J . Let

$$f : I \times J \rightarrow Y, \quad f(i, j) = f_i(j),$$

be a sensor and let E_f be its alias equivalence relation.

Definition 1 (Row selector and collision graph). *A row-selector packet is a family $B = \langle B_i : i \in A \rangle$ with $A \in \mathcal{V}$, $B_i \in \mathcal{U}$, and $f_i \upharpoonright B_i$ injective. Its palette collision graph G_B has vertex set A and*

$$i \mathrel{G_B} i' \iff f_i[B_i] \cap f_{i'}[B_{i'}] \neq \emptyset.$$

A set $A_0 \subseteq A$ is collision-free iff it is independent in G_B .

Round 9's matching law can be rephrased as a graph law:

$$\text{PT}_W(E_f) \iff \exists A_0 \in \mathcal{V} \exists \langle B_i \in \mathcal{U} : i \in A_0 \rangle \left[\begin{array}{l} f_i \upharpoonright B_i \text{ is injective for each } i, \text{ and} \\ A_0 \text{ is independent in } G_B \end{array} \right].$$

Thus the row palettes do not merely need to be good locally; they need a large global independent set.

2. Chromatic collapse law

A tempting finite-engineering idea is to add a row namespace $c : I \rightarrow \Theta$ and observe

$$F_c(i, j) = (c(i), f_i(j)).$$

In finite systems, this is graph coloring: if c separates every collision edge, then row palettes no longer collide. The transfinite situation is harsher.

Theorem 1 (Subcritical chromatic collapse). *Assume $|\Theta| < \kappa_I$. If the fused namespace sensor F_c has a W -large partial transversal, then the original sensor f already has a W -large partial transversal on a single large color fiber. Equivalently,*

$$\exists c : I \rightarrow \Theta \ (|\Theta| < \kappa_I \ \& \ \text{PT}_W(E_{F_c})) \iff \text{PT}_W(E_f).$$

Proof sketch. Let $T \subseteq I \times J$ be a W -large partial transversal for F_c . For $W = \mathcal{V} \otimes \mathcal{U}$, there is $A \in \mathcal{V}$ such that each row section T_i is $\mathcal{U}$ -large. Since $A = \bigcup_{\theta \in \Theta} (A \cap c^{-1}(\{\theta\}))$ and $|\Theta| < \kappa_I$, κ_I -completeness implies that one color fiber $A_\theta = A \cap c^{-1}(\{\theta\})$ belongs to $\mathcal{V}$. On A_θ , the namespace is constant, so any alias separation by F_c is alias separation by f alone. Hence $T \cap (A_\theta \times J)$ is a W -large partial transversal for f . The reverse implication is immediate by taking a constant namespace. $\square$

Law 1 (U-chromatic dichotomy). *Define the large chromatic namespace number*

$$\chi_{\mathcal{V}}^{\text{ns}}(f) = \min\{|\Theta| : \exists c : I \rightarrow \Theta \text{ such that } F_c \text{ is one-bit repairable on a } W\text{-large set}\}.$$

Under the row-selector hypotheses above,

$$\boxed{\chi_{\mathcal{V}}^{\text{ns}}(f) \in \{1, \kappa_I\}.$$

The value is 1 exactly when a $\mathcal{V}$ -large independent row-selector skin exists; otherwise it is κ_I .

Interpretation. Finite graph theory sees chromatic numbers $2, 3, 4, \dots$. A κ -complete ultrafilter collapses every subcritical coloring into one large color class. Therefore the transfinite question is not “how many small colors color the collision graph?” but “is there one large independent color class?”

3. Critical namespace theorem

If rowwise injectivity is available but palette collision has no large independent set, a critical-width namespace is enough and is necessary.

Theorem 2 (Critical namespace theorem). *Assume that for $\mathcal{V}$ -many rows i , there is $B_i \in \mathcal{U}$ such that $f_i \upharpoonright B_i$ is injective. Then:*

$$\boxed{\text{row namespace width needed} = \begin{cases} 1, & \exists \text{ a } \mathcal{V}\text{-large independent row-selector skin,} \\ \kappa_I, & \text{otherwise.} \end{cases}}$$

A critical namespace $c : I \rightarrow I$ repairs the horizontal collision obstruction by assigning a distinct row name on a $\mathcal{V}$ -large set.

Proof sketch. The lower bound is the chromatic collapse theorem: no subcritical namespace can help if no large independent skin exists. For the upper bound, use $c(i) = i$ on a $\mathcal{V}$ -large set of rows with rowwise injective B_i . Then $(c(i), f_i(j))$ cannot collide between distinct rows, and rowwise injectivity eliminates within-row aliases. $\square$

This recovers the Round 9 slogan in sharper form:

A local key is not a global key unless it carries a large enough namespace.

Round 10 adds the missing qualifier:

A subcritical namespace is equivalent to choosing one large collision-free skin.

4. Orthogonal repair surface: vertical vs. horizontal defects

The Fubini defect has two orthogonal components.

- **Vertical defect:** within-row aliasing. This is the old local repair problem for each map $f_i : J \rightarrow Y$.
- **Horizontal defect:** palette collision across rows after vertical repair has made rows locally injective.

In a clean row-regular case, define

$$\rho_{\text{in}}(f) \in \{1, 2, \kappa_J\}$$

as the $\mathcal{V}$ -essential rowwise repair rank, and define

$$\rho_{\text{out}}(f) \in \{1, \kappa_I\}$$

as the horizontal namespace rank from the critical namespace theorem. Then the repair alphabet may be taken as a product of inner and outer repairs. For infinite critical ranks this gives the capacity surface

$$\boxed{\text{res}_{\mathcal{V} \otimes \mathcal{U}}(E_f) = \max\{\rho_{\text{in}}(f), \rho_{\text{out}}(f)\}} \quad (\text{clean row-regular case}).$$

For finite shadows, replace the maximum by the ordinary product of the finite inner palette and finite namespace; the collapse to maximum is an infinite-cardinal effect.

5. Global spinal capacity equation

For a spinal portfolio with two-level Fubini coordinates, let

$$\rho_n = \max\{\rho_{\text{in},n}, \rho_{\text{out},n}\}$$

be the coordinate repair rank after all fused sensors in the portfolio are applied. With effective ideal $L = H \vee J$ and bit budgets μ_n , the global law becomes

$$\boxed{\text{robust faithful observation is possible} \iff \{n : 2^{\mu_n} < \rho_n\} \in L.}$$

This is the same Nyquist shape as earlier rounds, but ρ_n now separates two hidden causes of failure: inner fiber aliasing and outer palette collision.

6. Finite experiments: why finite shadows mislead

The finite analogue used row palettes and their collision graphs. In a finite threshold model, a partial transversal exists iff the graph has an independent set of size at least t . A finite namespace of size q can separate all rows iff $q \geq \chi(G)$. Hence finite graphs can be namespace-repairable even when no threshold-large independent set exists. This is exactly the finite illusion that disappears under κ -complete ultrafilters.

rows	palette	symbols	threshold	trials	threshold PT / namespace rescue
5	2	5	3	60	0 / 60
6	2	6	3	60	2 / 58
6	3	7	3	60	0 / 60
7	2	7	4	60	0 / 60
7	3	8	4	60	0 / 60

All exact coloring checks in these trials had zero mismatches against the formula “namespace of size q works iff $q \geq \chi(G)$.” The common α, χ pairs were dense-collision cases such as $(1, 5)$, $(2, 5)$, $(1, 6)$, and $(1, 7)$, where the largest independent set was below the threshold but a finite namespace could still color the graph.

Deterministic complete-graph shadows make the trap obvious:

$$\alpha(K_N) = 1, \quad \chi(K_N) = N.$$

For $N = 6, 8, 10$ with thresholds 3, 4, 5, respectively, there is no large partial transversal, while a finite namespace of size N separates all rows. In the transfinite theorem, N is replaced by κ_I : the needed namespace is critical, not merely “several colors.”

7. Applications

1. **Distributed identifiers.** A row-local identifier is not a global identifier. If many shards reuse the same local keyspace, the collision graph measures whether a small namespace prefix is enough. At transfinite scale, subcritical prefixes help only when one large shard set was already collision-free.
2. **Privacy and re-identification.** Anonymization by deleting namespace information is dangerous when a critical namespace is the only thing preventing palette collision. Conversely, adding many small tags can create a misleading finite impression of re-identifiability that would collapse under an ultrafilter-style large-system model.
3. **Sensor-bank audits.** Rowwise tests must be followed by a palette-collision audit. Passing every local test does not imply faithful global observation.
4. **Verification.** In modular systems, local proof names or local state labels can collide across modules. The new equation says verification needs either a large collision-free module skin or critical namespace capacity.
5. **Database engineering.** The law formalizes the familiar warning that composite keys need a shard/table/tenant namespace. The novel part is the sharp phase transition: subcritical namespace width is equivalent to finding one large independent shard family.

8. Next target

The next natural target is a *hypergraph version*. For m -level Fubini products, a palette collision may involve several context dimensions at once. The expected invariant is a large independent set in a collision hypergraph, with the same collapse:

$$\text{subcritical hypergraph coloring helps} \iff \text{a large independent selector skin already exists.}$$

If true, the final sensor theory becomes a dimension-by-dimension defect calculus:

$$\rho_n = \max_{d < m_n} \rho_{n,d},$$

where each hidden dimension contributes either a selector-skin cost, a critical namespace cost, or an inner critical repair cost.

9. Summary of Round 10

- Palette collision can be represented by a row collision graph after rowwise injective selectors are chosen.
- The transfinite namespace invariant is not the finite chromatic number. It collapses to 1 or κ_I .
- A subcritical namespace cannot repair lack of a large independent selector skin.
- A critical namespace repairs horizontal collision whenever vertical rowwise injectivity is already available.
- The resulting clean capacity surface is

$$\text{res}_{\mathcal{V} \otimes \mathcal{U}}(E_f) = \max\{\rho_{\text{in}}(f), \rho_{\text{out}}(f)\}.$$

- Globally, robust faithful observation keeps the same Nyquist form:

$$\{n : 2^{\mu_n} < \rho_n\} \in H \vee J.$$

Round 11 Addendum: Critical Dimension Ladders

Fubini block projection, lost coordinates, and scale-filtered repair

April 30, 2026 — exploratory continuation, not peer reviewed

Purpose. Round 10 showed that finite chromatic namespace phases collapse at a true κ -complete coordinate: a subcritical namespace helps only if a large independent selector skin already existed. The next target was the hypergraph version for higher Fubini products. This round identifies a cleaner phenomenon that should be treated first: once a coordinate is itself a finite Fubini block with unequal critical sizes, repair is no longer just $1, 2, \kappa$. It becomes a *critical dimension ladder*. The possible large repair scales are the sizes of lost Fubini blocks.

Key discovery. Single-cardinal coordinates collapse subcritical repair into the selector-skin phase. Multi-cardinal finite Fubini blocks create genuine intermediate transfinite repair costs, but only at critical coordinate scales:

$$1, 2, \kappa_{d_0}, \kappa_{d_1}, \dots, \prod_{d \in M} \kappa_d.$$

Thus there are no arbitrary chromatic middle phases, but there can be a structured ladder of *dimension-critical* phases.

1. Finite Fubini block setup

Let

$$Z_M = \prod_{d \in M} K_d, \quad |K_d| = \kappa_d,$$

where M is finite and each K_d carries a nonprincipal κ_d -complete ultrafilter U_d . Let U_M denote the finite Fubini product ultrafilter on Z_M . For finite M ,

$$|Z_M| = \prod_{d \in M} \kappa_d = \max_{d \in M} \kappa_d$$

when the κ_d are infinite, with the empty product equal to 1.

Now split a larger coordinate into an observed block Y and a hidden Fubini block Z_M :

$$X = Y \times Z_M, \quad W = V \otimes U_M,$$

where V is any ultrafilter on the observed block. The pure block projection is

$$\pi_Y : Y \times Z_M \rightarrow Y.$$

Definition 1 (Repair rank for a projection). *For an alias relation E induced by a sensor f , write $\text{res}_W(E)$ for the least alphabet size $|\Theta|$ for which there is a repair map $\tau : X \rightarrow \Theta$ such that (f, τ) is injective on a W -large set. Equivalently, (f, τ) has a W -large singleton-fiber domain.*

This definition is the direct faithful version of the earlier one-bit/partial-transversal repair calculus. If one only asks for a large partial transversal, an additional binary selector turns it into faithful observation unless the fused sensor is already faithful.

2. Lost-block Nyquist theorem

Theorem 1 (Fubini lost-block Nyquist law). *For the block projection $\pi_Y : Y \times Z_M \rightarrow Y$,*

$$\text{res}_W(E_{\pi_Y}) = |Z_M| = \prod_{d \in M} \kappa_d.$$

Thus the repair cost of a lost finite Fubini block is exactly the cardinality of that block. In particular, if $M \neq \emptyset$ and the hidden cardinals are increasing, the cost is $\max_{d \in M} \kappa_d$.

Proof sketch. The upper bound is immediate: use $\tau(y, z) = z$. Then (π_Y, τ) is injective everywhere.

For the lower bound, suppose $|\Theta| < |Z_M|$ and $\tau : Y \times Z_M \rightarrow \Theta$. If (π_Y, τ) were injective on a W -large set T , then for V -many y , the section

$$T_y = \{z : (y, z) \in T\}$$

would be U_M -large. Every U_M -large subset of the finite Fubini block has cardinality $|Z_M|$. But injectivity of (π_Y, τ) on T forces $\tau_y : z \mapsto \tau(y, z)$ to be injective on T_y , which requires $|\Theta| \geq |T_y| = |Z_M|$, a contradiction. $\square$

Interpretation. A lost dimension is not paid for by entropy or by finite chromatic complexity. It is paid for at its critical Fubini scale. Losing a small outer coordinate costs the outer cardinal. Losing a larger inner coordinate costs the larger inner cardinal. Losing several finite dimensions costs the maximum of their cardinals, equivalently their finite product.

3. Critical dimension ladder

For a coordinate sensor $f : X \rightarrow Y_0$ and a finite set of candidate namespace dimensions $D \subseteq m$, write

$$f_D = (f, \pi_D),$$

where π_D names the coordinates in D . Define the coordinate-namespace cost

$$\text{cost}(D) = \prod_{d \in D} \kappa_d$$

with $\text{cost}(\emptyset) = 1$.

Definition 2 (Dimension-critical repair profile). *The dimension-critical profile of f is*

$$\delta_W(f) = \min\{\text{cost}(D) : \text{PT}_W(E_{f_D})\}.$$

If no such D is specified, $D = m$ always works by naming every local dimension, so the invariant is defined for finite products.

Law 1 (Critical dimension ladder). *Coordinate-namespace repair lives on the finite ladder*

$$\delta_W(f) \in \left\{ \prod_{d \in D} \kappa_d : D \subseteq m \right\}.$$

For increasing infinite cardinals this is the finite set

$$\{1\} \cup \{\kappa_d : d < m\},$$

up to repetitions and the final maximum convention. Therefore the only stable transfinite middle phases are dimension-critical phases. Finite chromatic phases do not survive, but finite Fubini dimension scales do.

Warning 1 (Scope of the ladder law). *The invariant $\delta_W(f)$ is a coordinate-namespace invariant: it measures the cost of repairing by naming dimensions, possibly followed by a one-bit selector. A completely arbitrary repair map of size θ may encode a more clever quotient than a coordinate projection. The exact arbitrary-repair spectrum should be studied separately. The lost-block theorem above is exact for pure projections, and the ladder law is exact for coordinate-namespace repair design.*

4. Scale-filtered visibility

The previous result can be read as a scale law. A repair alphabet of size θ can fully name only dimensions whose critical size is at most θ . Let

$$V_\theta = \{d < m : \kappa_d \leq \theta\}.$$

Then a θ -sized coordinate namespace can only remove aliasing that is already resolvable after adding π_{V_θ} . In the pure projection case this gives the exact condition

$$\boxed{\text{res}_W(E_{\pi_O}) \leq \theta \iff m \setminus O \subseteq V_\theta.}$$

Equivalently,

$$\boxed{\text{res}_W(E_{\pi_O}) = \max_{d \notin O} \kappa_d}$$

where the maximum over the empty set is 1.

5. Global spinal law with dimension-critical coordinates

For a spinal system whose n th coordinate contains a finite Fubini block, let

$$\rho_n \in \{1, 2\} \cup \{\kappa_{n,d} : d < m_n\}$$

be the post-fusion repair scale at coordinate n :

- 1 if the fused sensor is already faithful on a large set;
- 2 if it has a selector skin but is not already faithful;
- $\kappa_{n,d}$ if the least remaining coordinate-namespace repair requires naming a dimension of critical size $\kappa_{n,d}$.

With effective recurrence/noise ideal $L = H \vee J$ and bit budgets μ_n , the global Nyquist law keeps the same form:

$$\boxed{\text{robust faithful observation is possible} \iff \{n : 2^{\mu_n} < \rho_n\} \in L.}$$

The discovery is not a new global shape but a richer local alphabet of possible ρ_n values. Earlier single-cardinal coordinates had $\rho_n \in \{1, 2, \kappa_n\}$. Finite Fubini coordinates have a finite critical ladder.

6. Finite experiments

The complete-threshold finite shadow treats “large” as “all values”. In that model, a projection must be repaired on the whole finite product, so the exact finite analogue of the lost-block theorem is immediate: the repair alphabet must have at least one value per hidden tuple.

coordinate sizes	observed dimensions D	predicted $q_{\min} = \prod_{d \notin D} k_d$
(3, 4)	$\emptyset, (0), (1), (0, 1)$	12, 4, 3, 1
(2, 5)	$\emptyset, (0), (1), (0, 1)$	10, 5, 2, 1
(2, 3, 4)	$\emptyset, (0), (1), (2), (0, 1), (0, 2), (1, 2), (0, 1, 2)$	24, 12, 8, 6, 4, 3, 2, 1
(3, 4, 5)	same pattern	60, 20, 15, 12, 5, 4, 3, 1
(2, 3, 5, 7)	first eight subsets in lexicographic order	210, 105, 70, 42, 30, 35, 21, 15

These checks are simple but useful: they show the exact coordinate-subset formula one should expect when finite thresholds are tightened all the way to ultrafilter-like full largeness.

Loose finite thresholds, however, are dangerous shadows. In a two-dimensional finite threshold model, a set is large when at least t_I rows each contain at least t_J columns. If one observes columns and hides rows, then a finite repair of size

$$q_{\min} = \left\lceil \frac{t_I t_J}{k_J} \right\rceil$$

can often suffice by spreading selected row witnesses across many observed columns. This is impossible in the ultrafilter version because a subcritical partition of a large coordinate has a large cell.

k_I	k_J	t_I	t_J	finite $q_{\min}$	observing columns	complete analogue
8	20	5	3		1	8
10	10	6	4		3	10
12	40	7	4		1	12
6	6	4	3		2	6
7	9	5	4		3	7
20	8	10	5		7	20

Three-dimensional threshold shadows can be even more misleading. Observing only the last coordinate, the crude total-capacity lower bound gives:

k_0	k_1	k_2	t_0	t_1	t_2	finite lower q / complete hidden product
5	5	30	3	3	2	1/25
8	6	50	4	3	2	1/48
10	8	100	5	4	3	1/80
6	6	12	4	4	2	3/36

Lesson. Loose finite thresholds can make a hidden dimension appear cheaply repairable by distributing witnesses across many observed values. The transfinite ultrafilter model forbids this: subcritical spreading collapses into a large cell, so lost dimensions demand their critical cardinal.

7. Applications

1. **Hierarchical identifiers.** In tenant/shard/table/key systems, losing a low-level key costs the low-level keyspace; losing a high-cardinality object identifier costs the object-ID scale. The repair budget follows the largest lost block, not the number of missing fields.
2. **Sensor-bank engineering.** Sensors should be audited by hidden dimension, not just by output entropy. A channel with many finite labels can still be blind to a critical hidden dimension.

3. **Privacy.** Redacting a namespace field may create a specific re-identification threshold. The ladder says which namespace removals are cheap to reverse and which require critical-width auxiliary information.
4. **Distributed diagnostics.** If failures are indexed by module, shard, and local state, a diagnostic channel must either observe each unresolved critical layer or provide a repair alphabet at that layer's scale.
5. **Database composite keys.** The exact projection formula is a transfinite version of a practical rule: if a composite key loses fields M , then the missing-key repair space is the product of the missing field domains. The new point is the collapse of all smaller repair schemes under ultrafilter largeness.

8. Next target

The next target is the fully general hypergraph theorem. Given a finite Fubini coordinate and a sensor f , each candidate selector packet induces a collision hypergraph whose vertices are contexts at one or more dimensions. The conjectural extension of Round 10 and this round is:

*Subcritical hypergraph coloring below a dimension's critical size helps
if and only if a large independent selector skin already exists at that dimension.*

The expected final local invariant is a layered profile

$$\vec{\rho}(f) = (\rho_0(f), \dots, \rho_{m-1}(f)),$$

where each $\rho_d(f)$ is 1, 2, or κ_d , and the coordinate repair scale is

$$\rho(f) = \max_{d < m} \rho_d(f).$$

This would unify selector skins, palette collision, lost dimensions, and critical namespaces into one dimension-by-dimension calculus.

9. Summary of Round 11

- Pure projection from $Y \times Z_M$ to Y has exact repair cost $|Z_M|$.
- Finite Fubini coordinates with unequal cardinal layers create a genuine critical repair ladder.
- The stable transfinite middle phases are not finite chromatic numbers; they are dimension-critical cardinals.
- The global spinal Nyquist equation remains

$$\{n : 2^{\mu_n} < \rho_n\} \in H \vee J,$$

but the local values of ρ_n now come from a dimension ladder.

- Finite loose-threshold experiments show why engineering-scale simulations can underestimate critical repair costs.

Round 12 Addendum: Canonical Kernel Profiles

Top-hidden-dimension law, selector skins, and dimension-first diagnostics

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Purpose. Round 11 proved the exact lost-block law for pure projections: if a finite Fubini coordinate hides a block M , the repair alphabet must have size $\prod_{d \in M} \kappa_d$. This round asks the next natural question: when a sensor is not literally a projection, can its alias kernel still be reduced to a hidden-coordinate profile? The answer proposed here is a new canonization target. Under a finite normal-Fubini canonization principle, every stable repair cost is controlled by either a one-bit selector skin or by the largest remaining hidden Fubini dimension.

Key discovery. The dimension ladder from Round 11 has a sharper design form. If a kernel canonizes to equality on an observed coordinate set D , then the repair cost is not an arbitrary chromatic invariant. It is

$$\widehat{c}(m \setminus D) = \begin{cases} 1, & \text{already faithful,} \\ 2, & \text{selector-skin only,} \\ \prod_{d \in m \setminus D} \kappa_d, & m \setminus D \neq \emptyset. \end{cases}$$

For increasing infinite dimension cardinals, the last product is just

$$\kappa_{\max(m \setminus D)}.$$

Thus the highest-cardinality hidden coordinate dominates the repair scale. Lower hidden coordinates matter only after all larger hidden coordinates have been exposed or otherwise repaired.

1. Canonical kernels

Let

$$X_m = \prod_{d < m} K_d, \quad |K_d| = \kappa_d,$$

with finite Fubini product ultrafilter $W = U_0 \otimes \cdots \otimes U_{m-1}$. For $D \subseteq m$, let

$$E_D(x, y) \iff x \upharpoonright D = y \upharpoonright D.$$

This is the kernel of the coordinate projection π_D . Its hidden block is

$$M_D = m \setminus D.$$

Definition 1 (Canonical profile). *For an equivalence relation E on X_m , a pair (A, D) is a canonical profile for E if $A \in W$ and*

$$\forall x, y \in A, \quad xEy \iff x \upharpoonright D = y \upharpoonright D.$$

Write $\text{Can}_W(E)$ for the set of such pairs.

Definition 2 (Selector-skin profile). *A selector skin is the special case $D = m$ with $A \in W$ and $E \upharpoonright A$ equal to identity, while E is not already faithful without naming the skin. This is the one-bit phase from earlier rounds: the extra bit marks membership in the large transversal.*

The profile cost of (A, D) is

$$c(D) = \prod_{d \in m \setminus D} \kappa_d.$$

For infinite increasing finite blocks,

$$c(D) = \begin{cases} 1, & D = m, \\ \kappa_{\max(m \setminus D)}, & D \neq m. \end{cases}$$

2. Canonical Profile Principle

Conjecture 1 (Finite normal-Fubini canonical profile principle). *For every sensor $f : X_m \rightarrow Y$ and kernel E_f , there is a W -large set A and a coordinate set $D \subseteq m$ such that $E_f \upharpoonright A = E_D \upharpoonright A$. Equivalently,*

$$\text{Can}_W(E_f) \neq \emptyset.$$

This principle is the natural higher-dimensional continuation of the one-coordinate normal selector theorem: at one measurable coordinate, a function is large-constant or large-injective. At finite Fubini depth, the expected canonical alternatives are: depend on no coordinates, depend on some coordinates, or become injective on a selector skin.

Warning 1 (Status). *The principle above is a theorem target, not a completed proof in this note. The laws below are proved conditional on this canonization principle, while the pure projection lost-block theorem from Round 11 remains unconditional.*

3. Conditional repair law

Define the canonical repair scale

$$\rho_W(E) = \min \left\{ \prod_{d \in m \setminus D} \kappa_d : (A, D) \in \text{Can}_W(E) \right\},$$

with the convention that the empty product is 1. If the minimum is obtained only by a selector skin and E is not already faithful as an observed sensor, replace 1 by 2.

Theorem 1 (Canonical kernel repair law). *Assume the finite normal-Fubini canonical profile principle for E . Then*

$\text{res}_W(E) = \begin{cases} 1, & E \text{ is already faithful on a } W\text{-large observed domain,} \\ 2, & E \text{ has a } W\text{-large selector skin but is not already faithful,} \\ \min_{(A,D) \in \text{Can}_W(E), D \neq m} \prod_{d \in m \setminus D} \kappa_d, & \text{otherwise.} \end{cases}$

For increasing infinite dimensions, the nontrivial infinite case is

$\text{res}_W(E) = \kappa_{h(E)}, \quad h(E) = \min_{(A,D) \in \text{Can}_W(E), D \neq m} \max(m \setminus D),$

where the minimum chooses the profile whose largest hidden dimension is as small as possible.

Proof sketch. For a canonical pair (A, D) , repairing by the hidden coordinates $m \setminus D$ makes $(f, \pi_{m \setminus D})$ injective on A . This gives the upper bound $\prod_{d \in m \setminus D} \kappa_d$.

For the lower bound, suppose a repair alphabet Θ has size below that product. On the canonical large set, every fixed observed D -tuple has a hidden block of size $\prod_{d \in m \setminus D} \kappa_d$ in the finite Fubini sense. A smaller alphabet cannot injectively name a W -large collection of hidden-block values; by the same Fubini pigeonhole used in Round 11, a subcritical partition has a large cell. Thus the fused sensor still has recurrent aliasing. If $D = m$, the hidden product is 1; when the original sensor is not already faithful, one binary mark is needed to isolate the large transversal from its small complement. $\square$

4. Top-hidden-dimension law

For increasing infinite critical sizes,

$$\prod_{d \in M} \kappa_d = \kappa_{\max M}$$

for every nonempty finite hidden set M . Therefore only the largest hidden dimension controls the infinite repair scale.

Law 1 (Top-hidden-dimension law). *Let M be the current hidden coordinate set of a canonical profile. Then*

$$\rho(M) = \begin{cases} 1 \text{ or } 2, & M = \emptyset, \\ \kappa_{\max M}, & M \neq \emptyset. \end{cases}$$

Moreover, observing a coordinate $d \in M$ changes the infinite repair scale iff $d = \max M$:

$$\rho(M \setminus \{d\}) = \rho(M) \quad \text{for } d < \max M.$$

Only once the top hidden dimension is removed does the repair scale drop to the next rung.

Design meaning. If a sensor is missing dimensions of sizes $\kappa_0 < \kappa_1 < \kappa_2$, then repairing κ_0 or κ_1 while κ_2 remains hidden does not change the critical repair scale. The rational diagnostic query is to expose the highest hidden cardinal first.

5. Global spinal form

At spinal coordinate n , suppose the fused kernel has canonical hidden profile $M_n \subseteq m_n$, with selector bit status $s_n \in \{1, 2\}$ when $M_n = \emptyset$. Define

$$\rho_n = \begin{cases} s_n, & M_n = \emptyset, \\ \prod_{d \in M_n} \kappa_{n,d}, & M_n \neq \emptyset. \end{cases}$$

For increasing finite blocks this is $\rho_n = \kappa_{n, \max M_n}$ in the nonempty case. With effective ideal $L = H \vee J$ and bit budgets μ_n ,

$$\text{robust faithful observation is possible} \iff \{n : 2^{\mu_n} < \rho_n\} \in L.$$

The marginal value of adding a sensor σ at coordinate n is not its entropy. It is the rung drop

$$\kappa_{n, \max M_n} \longrightarrow \kappa_{n, \max (M_n \setminus O_{\sigma,n})},$$

where $O_{\sigma,n}$ is the set of hidden dimensions the sensor resolves. A sensor has zero first-order critical value if it misses the current top hidden dimension.

6. Finite experiments

Python experiments checked the exact finite projection formula and then deliberately tested loose threshold shadows to identify where finite intuition misleads.

sizes	observed D	predicted hidden product	exact max fiber	match
(4, 5)	$\emptyset$	20	20	yes
(4, 5)	(0)	5	5	yes
(4, 5)	(1)	4	4	yes
(4, 5)	(0, 1)	1	1	yes
(3, 4, 5)	$\emptyset$	60	60	yes
(3, 4, 5)	(0)	20	20	yes
(3, 4, 5)	(1)	15	15	yes
(3, 4, 5)	(2)	12	12	yes
(3, 4, 5)	(0, 1)	5	5	yes
(3, 4, 5)	(0, 2)	4	4	yes
(3, 4, 5)	(1, 2)	3	3	yes
(3, 4, 5)	(0, 1, 2)	1	1	yes

Across all tested coordinate subsets for sizes (4, 5), (3, 4, 5), and (2, 3, 5, 7) there were 28 projection checks and 0 failures.

The loose threshold shadow used $k = (4, 5)$ and threshold $t = (3, 4)$: a set is finite-large when at least three rows each contain at least four columns. The minimum max-fiber width on such a large set can be smaller than the full hidden product.

function	full exact width	threshold min width
constant	20	12
outer projection i	5	4
inner projection j	4	3
pair (i, j)	1	1
$(i, j \bmod 2)$	3	2

Random finite functions from 4×5 into 7 labels produced threshold widths

$$\{2 : 127, 3 : 72, 4 : 1\}$$

in 200 trials. These extra finite values are not evidence for transfinite chromatic phases. They are finite spreading artifacts: a threshold-large set can distribute witnesses across many observed labels, whereas a κ -complete ultrafilter forces a subcritical partition to have a large cell.

7. Applications

1. **Sensor-bank design.** Audit sensors by the largest unresolved hidden dimension. A sensor that resolves many small dimensions but misses the top hidden one may leave the critical repair cost unchanged.
2. **Active diagnostics.** Query order should be dimension-first: identify the current top hidden layer, then spend diagnostic bandwidth there before querying lower layers.
3. **Privacy and re-identification.** In hierarchical records, hiding the highest-cardinality namespace dominates anonymity. Revealing low-cardinality fields may not materially reduce the critical ambiguity if the top hidden namespace remains hidden.
4. **Composite keys.** The repair cost of a missing composite-key field set is governed by the largest missing domain. In transfinite form, no subcritical hash of the missing keyspace can replace naming that top field on a large set.

5. **Distributed systems.** For tenant/shard/object/local-state identifiers, global uniqueness is bottlenecked by the highest hidden namespace. Local object tags cannot repair a missing tenant namespace if the tenant layer is the top hidden coordinate.

8. Next targets

The next high-value target is to prove, refute, or correctly weaken the Canonical Profile Principle. A practical route is:

1. prove the two-level case for normal $V \otimes U$ kernels;
2. isolate the palette-collision obstruction as the only obstacle to increasing D ;
3. show that subcritical collision colorings collapse to selector skins;
4. iterate the argument through finite Fubini depth;
5. define a publication-ready canonical profile invariant $\text{core}_W(E)$.

A possible final local invariant is

$$\text{core}_W(E) = (s(E), M(E)),$$

where $s(E) \in \{1, 2\}$ records faithful versus selector-skin behavior after all coordinate dimensions are resolved, and $M(E)$ is the minimal hidden coordinate set left by canonization. Then

$$\text{res}_W(E) = \begin{cases} s(E), & M(E) = \emptyset, \\ \kappa_{\max M(E)}, & M(E) \neq \emptyset. \end{cases}$$

This would unify selector skins, palette collision, lost dimensions, and namespace Nyquist into one canonical-kernel calculus.

9. Summary of Round 12

- The new target is a finite normal-Fubini canonization theorem for sensor kernels.
- Conditional on canonization, arbitrary kernels reduce to coordinate profiles on a W -large set.
- Repair cost is 1 if already faithful, 2 for selector skin, and otherwise the product of hidden coordinate cardinals.
- For increasing finite Fubini blocks, the product is the largest hidden cardinal: the top-hidden-dimension law.
- Finite tests confirmed the exact projection formula and showed that loose threshold shadows create misleading intermediate widths.
- The most useful design rule is: resolve the largest hidden dimension first.

Round 13 Addendum: Tropical Repair Algebra

Critical-band cover, non-Archimedean Nyquist scales, and top-first diagnostics

April 30, 2026 — exploratory continuation, not peer reviewed

Purpose. Round 11 introduced the dimension ladder for finite Fubini blocks, and Round 12 proposed a canonical-kernel profile principle. This round extracts the algebra hidden inside those results. Once local dimensions have strongly increasing infinite cardinal sizes, finite products of hidden dimensions behave like a *maximum*, not like a sum. Therefore repair cost is a non-Archimedean/tropical quantity: the largest still-hidden dimension dominates all smaller hidden dimensions.

Key discovery. For a finite hidden block M ,

$$\rho(M) = \prod_{d \in M} \kappa_d$$

collapses, in infinite cardinal arithmetic with increasing dimension cardinals, to

$$\boxed{\rho(M) = \kappa_{\max M} \quad (M \neq \emptyset).}$$

Thus

$$\boxed{\rho(M \cup N) = \max\{\rho(M), \rho(N)\}.}$$

This is the **tropical repair law**. It says that independent hidden repair obligations combine by maximum critical scale. In practical language: do not spend effort resolving lower hidden coordinates while a higher hidden coordinate remains unresolved, unless those lower coordinates are needed for a later phase.

1. Setup: hidden blocks and repair valuations

Let

$$X_m = \prod_{d < m} K_d, \quad |K_d| = \kappa_d,$$

where $m < \omega$ and

$$\kappa_0 < \kappa_1 < \cdots < \kappa_{m-1}$$

are infinite cardinals satisfying the finite ladder growth condition

$$\prod_{e < d} \kappa_e < \kappa_d \quad (d < m).$$

For $D \subseteq m$, the projection kernel E_D observes D and hides

$$M_D = m \setminus D.$$

Define the hidden-block repair valuation

$$\rho(M) = \begin{cases} 1, & M = \emptyset, \\ \prod_{d \in M} \kappa_d, & M \neq \emptyset. \end{cases}$$

The associated top dimension is

$$\text{top}(M) = \begin{cases} -\infty, & M = \emptyset, \\ \max M, & M \neq \emptyset. \end{cases}$$

Law 1 (Tropical repair valuation). *For every finite $M \subseteq m$,*

$$\rho(M) = \begin{cases} 1, & M = \emptyset, \\ \kappa_{\text{top}(M)}, & M \neq \emptyset. \end{cases}$$

Consequently,

$$\rho(M \cup N) = \max\{\rho(M), \rho(N)\},$$

with the convention that 1 is the empty-hidden-block value.

Proof sketch. A finite product of infinite cardinals equals the maximum factor. The growth hypothesis ensures that this maximum is exactly the largest hidden dimension cardinal in the ladder. Taking the union of two hidden blocks simply takes the larger of their top dimensions. Hence product-composition of finite hidden blocks becomes max-composition of repair scales.

2. Fusion algebra

Projection sensors have an especially clean fusion law. If sensor σ hides M_σ and sensor τ hides M_τ , then their fusion observes the union of observed dimensions, so it hides

$$M_{\sigma \wedge \tau} = M_\sigma \cap M_\tau.$$

Therefore

$$\rho(\sigma \wedge \tau) = \rho(M_\sigma \cap M_\tau) \leq \min\{\rho(M_\sigma), \rho(M_\tau)\}.$$

For a portfolio Γ ,

$$M_\Gamma = \bigcap_{\sigma \in \Gamma} M_\sigma, \quad \rho(\Gamma) = \rho(M_\Gamma) = \kappa_{\max} M_\Gamma$$

when $M_\Gamma \neq \emptyset$.

Theorem 1 (Top-drop law). *Let $M \neq \emptyset$ and $t = \max M$. Exposing a hidden dimension $d \in M$ changes the repair scale iff $d = t$:*

$$\rho(M \setminus \{d\}) = \text{ho}(M) \iff d \neq t.$$

If $d = t$, then the repair scale drops to the next hidden dimension:

$$\rho(M \setminus \{t\}) = \begin{cases} 1, & M = \{t\}, \\ \kappa_{\max(M \setminus \{t\})}, & |M| > 1. \end{cases}$$

Interpretation. This is a sharp diagnostic priority rule. A sensor that exposes only lower hidden dimensions may be useful later, but it has zero immediate effect on the critical repair scale while the top hidden dimension remains invisible.

3. Critical-band cover law

Let θ be an available repair alphabet size. Define the critical band above θ by

$$B_\theta = \{d < m : \kappa_d > \theta\}.$$

A portfolio Γ with observed set D_Γ is repair-sufficient at capacity θ iff no hidden dimension above capacity remains:

$$\boxed{\rho(m \setminus D_\Gamma) \leq \theta \iff B_\theta \subseteq D_\Gamma.}$$

Thus repair design at a fixed budget is not a cover problem over all dimensions. It is a cover problem over the dimensions whose cardinal scale exceeds the budget.

Theorem 2 (Global critical-band Nyquist law). *Let $L = H \vee J$ be the effective ideal, and at coordinate n let the fused portfolio observe $D_{\Gamma,n} \subseteq m_n$. Given repair alphabets $\theta_n = 2^{\mu_n}$, robust faithful observation for projection-profile coordinates is possible exactly when*

$$\boxed{\{n : B_{n,\theta_n} \not\subseteq D_{\Gamma,n}\} \in L,}$$

where

$$B_{n,\theta_n} = \{d < m_n : \kappa_{n,d} > \theta_n\}.$$

Equivalently,

$$\{n : \rho_n(m_n \setminus D_{\Gamma,n}) > 2^{\mu_n}\} \in H \vee J.$$

This is the same global shape as previous Nyquist laws, but with a more economical local interpretation: for each bit budget, only the above-budget dimensions must be covered. Below-budget dimensions may remain hidden without causing failure at that repair scale.

4. Conditional extension to canonical kernels

Round 12 proposed the canonical-kernel target: every finite normal-Fubini sensor kernel becomes, on a large set, equality on a coordinate profile D , possibly after a selector-skin reduction. Under that target, the tropical law extends beyond pure projections.

Theorem 3 (Conditional tropical canonization law). *Assume a kernel E has a large canonical profile D and no one-bit selector skin that improves it. Then*

$$\text{res}_W(E) = \rho(m \setminus D) = \begin{cases} 1, & D = m, \\ \kappa_{\max(m \setminus D)}, & D \neq m. \end{cases}$$

If a selector skin exists but the original observation is not already faithful, the local repair cost is 2 instead. Thus the stable local repair scales are

$$1, \quad 2, \quad \kappa_{d_0}, \quad \kappa_{d_1}, \quad \dots$$

not arbitrary intermediate chromatic widths.

Warning 1 (Status boundary). *The projection laws above are direct cardinal-arithmetic consequences of the lost-block theorem. The extension to arbitrary finite normal-Fubini kernels depends on the canonical-profile principle from Round 12. The main new target is now to prove or refute that canonization principle.*

5. Finite and symbolic experiments

The following checks used a finite symbolic ladder

$$(2, 5, 11, 127, 20000, 10^9),$$

chosen so that each entry exceeds the product of all previous entries. This lets finite products preserve the same order-level behavior that infinite cardinal products collapse to exactly.

Experiment	Checks	Failures
Top-shadow order checks for all hidden subsets	64	0
Marginal top-drop checks	192	0
Random weighted portfolio band-cover checks	500	0
Exact finite projection fiber checks	16	0

Representative random portfolio rows from the band-cover test:

Dims	Sensors	Target top	Required dimensions	Min cost	Chosen sensors
4	9	≤ 0	$\{1, 2, 3\}$	6	$\{7, 8\}$
4	6	≤ 0	$\{1, 2, 3\}$	4	$\{3\}$
5	9	≤ 2	$\{3, 4\}$	6	$\{1, 3\}$
5	4	≤ 1	$\{2, 3, 4\}$	9	$\{1, 3\}$
5	6	≤ 4	$\emptyset$	0	$\emptyset$

The exact finite projection checks verified that the repair alphabet for a projection equals the maximum fiber size, namely the product of the hidden finite coordinates. The symbolic ladder checks verified the order-level form of the top-dimension collapse. The portfolio checks compared brute-force repair feasibility against the predicted critical-band cover condition and found no mismatches.

6. Applications

1. **Sensor-bank design.** At a fixed repair budget, only sensors that cover above-budget hidden dimensions can change the success/failure phase. This converts design into weighted set cover on the critical band B_θ , not on the full hidden profile.
2. **Active diagnostics.** The top-drop law says that the next best diagnostic query should target the largest remaining hidden dimension unless a cheaper query also covers it. Lower-dimensional probes are deferred until the dominant dimension is exposed.
3. **Privacy and re-identification.** An anonymization scheme remains protected at the scale of its largest hidden dimension. Revealing many low-cardinality attributes may not change the transfinite re-identification phase, but revealing the largest hidden namespace can cause a phase drop.
4. **Composite keys and distributed identifiers.** A composite key has a critical missing field. Missing a high-cardinality tenant, region, or namespace field dominates all lower fields. Adding the top namespace is the only action that changes the repair tier immediately.
5. **Verification.** Monitoring budgets should be stated as cardinal bands. A monitor is sufficient at band θ exactly when all failure modes with hidden dimension above θ are observed modulo the effective ideal $H \vee J$.

7. Summary of Round 13

The new useful law is:

$$\rho(M \cup N) = \max\{\rho(M), \rho(N)\}.$$

The new design law is:

$$\rho(m \setminus D_\Gamma) \leq \theta \iff B_\theta \subseteq D_\Gamma.$$

The new diagnostic slogan is:

$$\text{repair scale is controlled by the largest still-hidden dimension.}$$

This turns the finite Fubini dimension ladder into a non-Archimedean information geometry. Repair costs do not add; they take maxima. Sensor fusion does not gradually smooth the whole hidden block; it removes top dimensions one at a time, causing sharp phase drops.

Round 14 Addendum: Hidden-Debt Frontiers and Critical-Band Cost Curves

Spinal Nullity Research Log

April 30, 2026

Status and purpose

This addendum continues the sensor and repair-rank program after the tropical repair algebra. The standing local picture is a finite Fubini coordinate with strongly increasing hidden dimensions

$$\kappa_0 < \kappa_1 < \cdots < \kappa_{m-1}, \quad \prod_{i < d} \kappa_i < \kappa_d.$$

Under the projection and canonical-profile cases isolated in the previous rounds, if a portfolio observes the dimension set $D \subseteq m$, then the unresolved repair scale is

$$\rho(m \setminus D) = \begin{cases} 1, & m \setminus D = \emptyset, \\ \kappa_{\max(m \setminus D)}, & m \setminus D \neq \emptyset. \end{cases}$$

Round 13 emphasized the tropical law $\rho(M \cup N) = \max\{\rho(M), \rho(N)\}$. This round turns that into an optimization and design calculus: the hidden dimensions form a Boolean debt vector, fusion is bitwise debt-erasure, and the best achievable repair scale is a step function whose breakpoints are exactly critical-band set-cover costs.

1 Hidden-debt algebra

Definition 1 (Hidden-debt vector). *For a portfolio Γ , define*

$$D_\Gamma = \bigcup_{\sigma \in \Gamma} D_\sigma, \quad h_\Gamma(d) = \begin{cases} 1, & d \notin D_\Gamma, \\ 0, & d \in D_\Gamma. \end{cases}$$

The vector $h_\Gamma \in 2^m$ records dimension debt: 1 means the dimension is still hidden.

Law 1 (Hidden-debt fusion law). *For two portfolios Γ, Λ ,*

$$h_{\Gamma \cup \Lambda} = h_\Gamma \wedge h_\Lambda$$

coordinatewise. The repair scale is the highest surviving debt bit:

$$\rho(\Gamma) = \begin{cases} 1, & h_\Gamma = 0, \\ \kappa_{\max\{d: h_\Gamma(d)=1\}}, & h_\Gamma \neq 0. \end{cases}$$

Proof sketch. A dimension is hidden after fusion exactly when it is hidden from every fused component; hence hidden sets intersect and hidden bits AND. The tropical repair equation then reads the largest dimension still hidden. $\square$

Law 2 (Marginal invisibility). *Suppose the current highest hidden dimension is $r = \max\{d : h_\Gamma(d) = 1\}$. Adding a sensor σ changes the repair scale below κ_r if and only if it clears dimension r and no other dimension of scale κ_r remains hidden. Under strictly increasing one-dimension-per-scale ladders, this reduces to*

$$\rho(\Gamma \cup \{\sigma\}) < \rho(\Gamma) \iff r \in D_\sigma.$$

Lower-dimensional coverage has no immediate effect until all higher debt has been cleared.

This is the practical version of the tropical law: a sensor can be rich in lower-level information and still be worthless for the current repair bottleneck.

2 Critical-band frontier theorem

For a repair budget threshold θ , define the critical band

$$B_\theta = \{d < m : \kappa_d > \theta\}.$$

For a cost function w on sensors, set

$$C(\theta) = \min\{w(\Gamma) : B_\theta \subseteq D_\Gamma\},$$

where $w(\Gamma)$ may be cardinality, money, latency, energy, model calls, human review time, or any additive engineering cost.

Theorem 1 (Repair frontier inverse law). *Let*

$$\Phi(c) = \min\{\rho(\Gamma) : w(\Gamma) \leq c\}$$

be the best repair scale attainable under cost c . Then

$$\Phi(c) \leq \theta \iff C(\theta) \leq c.$$

Equivalently, the entire cost-performance frontier is the inverse of the critical-band cover function.

Proof sketch. By the critical-band law,

$$\rho(\Gamma) \leq \theta \iff B_\theta \subseteq D_\Gamma.$$

So a portfolio of cost at most c achieves threshold θ exactly when the top band B_θ has a set-cover solution of cost at most c . $\square$

Proposition 1 (Staircase Nyquist law). *The function $\Phi(c)$ is a step function. It can drop only at costs $C(\kappa_r)$, and its drops are dimension drops. Partial coverage of a critical band does not partially lower the transfinite repair rank.*

This is the exceptional point of the round: in the true ultrafilter/Fubini model, information return is not smooth. It is a staircase. You pay enough to cover the next critical band or the repair scale does not move.

3 Global spinal version

Let $L = H \vee J$ be the effective coordinate ideal combining noise tolerance and recurrence tolerance. At spinal coordinate n , let m_n be its dimension set, let $D_{\Gamma,n}$ be the dimensions observed there by portfolio Γ , and let $\theta_n = 2^{\mu_n}$ be the available local repair alphabet.

Law 3 (Global hidden-debt law). *For a selected portfolio Γ ,*

$$\{n : \rho_n(\Gamma) > \theta_n\} \in L \iff \{n : B_{n,\theta_n} \not\subseteq D_{\Gamma,n}\} \in L.$$

In hidden-debt notation,

$$\left\{ n : \bigvee_{d \in B_{n,\theta_n}} h_{\Gamma,n}(d) = 1 \right\} \in L.$$

Thus robust faithful observation is exactly critical-band debt clearance modulo L .

Define the global critical-band cover cost

$$C_L(\theta_\bullet) = \min \{w(\Gamma) : \{n : B_{n,\theta_n} \not\subseteq D_{\Gamma,n}\} \in L\}.$$

Then C_L is the exact minimum sensor cost needed to make the repair budget sequence $\theta_\bullet$ sufficient modulo the tolerated coordinates. This is the global optimization target suggested by the theory.

4 Finite experiments

I ran three finite-shadow checks. These are not proofs of the transfinite results; they test the combinatorial algebra behind the hidden-debt and critical-band formulation.

Experiment	Checks	Failures
Local frontier inverse law	1,925	0
Global L -threshold law, finite $ \text{bad} \leq q$ ideal	8,120	0
Exact greedy-failure search	found counterexample	–

The local test randomly generated weighted sensor portfolios over 3 to 8 dimensions and compared two quantities: brute-force best repair rank under a cost cap, and the critical-band cover cost. They matched in every case. The global test randomly generated sensors over several spinal coordinates, modeled L as “ignore at most q bad coordinates”, and verified

$$\rho_n(\Gamma) > \theta_n \iff B_{n,\theta_n} \not\subseteq D_{\Gamma,n}$$

inside the global failure count. Again there were no mismatches.

Warning 1 (Greedy is not the theorem). *The repair algebra turns design into critical-band set cover, but it does not make naive greedy selection exact. A random finite weighted example produced optimal cost 7 while a gain-per-cost greedy rule spent 10. The critical band had five dimensions and sensors included:*

<i>sensor</i>	<i>cost</i>	<i>covered dimensions</i>
5	2	$\{1, 2\}$
6	3	$\{0, 1, 4\}$
8	5	$\{0, 2, 3, 4\}$

Greedy chose sensors 5, 6, 8 for cost 10. The optimal cover is sensors 5 and 8: cost $2 + 5 = 7$, covering $\{1, 2\} \cup \{0, 2, 3, 4\} = \{0, 1, 2, 3, 4\}$. The lesson is: the theory gives the exact objective; the finite engineering optimizer may still need exact search, integer programming, or approximation.

The warning is useful because it prevents an overinterpretation of the discovery. The transfinite theorem supplies the right target, not a magic greedy algorithm.

5 Applications

1. **Sensor-bank design.** Score each available sensor by which high-cardinality hidden dimensions it clears. Ignore lower-band coverage until the current critical band is coverable.
2. **Active diagnostics.** A diagnostic question should be chosen by the highest still-hidden dimension it can clear, not by total information bits measured in a finite shadow.
3. **Privacy and re-identification.** Privacy risk drops only when the adversary lacks at least one high-cardinality dimension. Conversely, re-identification becomes cheap once the top critical band is covered.
4. **Composite keys and distributed IDs.** A composite identifier is bottlenecked by the largest namespace component not yet included. Adding small namespaces cannot compensate for a missing large namespace.
5. **Verification.** Monitoring resources should be allocated to clear the largest unresolved fault dimension. Many low-level monitors may certify local detail while leaving the global repair rank unchanged.

Round 14 summary

The new law is the hidden-debt frontier calculus:

$$h_{\Gamma \cup \Lambda} = h_{\Gamma} \wedge h_{\Lambda}, \quad \rho(\Gamma) = \kappa_{\max\{d: h_{\Gamma}(d)=1\}},$$

with the inverse frontier equation

$$\Phi(c) \leq \theta \iff C(\theta) \leq c.$$

The strongest practical slogan is:

sensor value is staircase-shaped: clear the top debt band or the repair rank does not move.

This turns the tropical repair algebra into a concrete design pipeline: choose a repair budget, form the critical band above that budget, cover that band modulo $H \vee J$, and only then spend effort on lower bands.

Round 15 Addendum: Fault-Tolerant Hidden Debt and Redundancy Nyquist Laws

Spinal Nullity Research Log

April 30, 2026

Status and purpose

This addendum continues the finite-Fubini sensor and repair-rank program. The previous round turned hidden dimensions into a debt vector and showed that, for strongly increasing dimension cardinals,

$$\rho(M) = \begin{cases} 1, & M = \emptyset, \\ \kappa_{\max M}, & M \neq \emptyset, \end{cases}$$

so repair scale is controlled by the largest still-hidden dimension. This round adds sensor unreliability. The new question is not just: which dimensions are covered? It is: which dimensions stay covered after permitted sensor failures? The answer is an ideal-positive coverage law.

1 Fault ideals and robust debt

Let $m = \{0, \dots, m-1\}$ be the finite set of hidden dimensions, with

$$\kappa_0 < \kappa_1 < \dots < \kappa_{m-1}, \quad \prod_{i < d} \kappa_i < \kappa_d.$$

A sensor σ reveals a dimension set $D_\sigma \subseteq m$. A selected portfolio is $\Gamma \subseteq \mathcal{S}$. Sensor failure is modeled by an ideal $\mathcal{F}$ on the sensor library $\mathcal{S}$: a failure set F is allowed when $F \in \mathcal{F}$. After F fails, the surviving observed dimensions are

$$D_{\Gamma \setminus F} = \bigcup_{\sigma \in \Gamma \setminus F} D_\sigma.$$

For a dimension d , define its selected cover set

$$C_\Gamma(d) = \{\sigma \in \Gamma : d \in D_\sigma\}.$$

Definition 1 (Fault-hidden debt). *Dimension d is robustly covered by Γ against $\mathcal{F}$ iff every allowed failure set leaves at least one selected sensor seeing d . Equivalently,*

$$h_\Gamma^\mathcal{F}(d) = 1 \iff C_\Gamma(d) \in \mathcal{F}.$$

Here $h_\Gamma^\mathcal{F}(d) = 1$ means d is hidden debt under the failure model.

Theorem 1 (Ideal-positive coverage law). *For every failure ideal $\mathcal{F}$,*

$$d \text{ is robustly covered} \iff C_\Gamma(d) \notin \mathcal{F}.$$

Consequently the worst-case repair scale over all allowed failures is

$$\rho_{\mathcal{F}}(\Gamma) = \sup_{F \in \mathcal{F}} \rho(m \setminus D_{\Gamma \setminus F}) = \begin{cases} 1, & \{d : C_\Gamma(d) \in \mathcal{F}\} = \emptyset, \\ \kappa_{\max\{d : C_\Gamma(d) \in \mathcal{F}\}}, & \text{otherwise.} \end{cases}$$

Proof sketch. If $C_\Gamma(d) \in \mathcal{F}$, the adversary may fail exactly the selected sensors that see d , so d becomes hidden. If some allowed $F \in \mathcal{F}$ hides d , then $C_\Gamma(d) \subseteq F$; since $\mathcal{F}$ is downward closed, $C_\Gamma(d) \in \mathcal{F}$. Thus robust coverage is exactly $\mathcal{F}$ -positive coverage. The repair equation then reads off the largest dimension not robustly covered. $\square$

2 The redundancy Nyquist law

The most common fault model is: up to q selected sensors may fail. Then

$$\mathcal{F}_q = [\mathcal{S}]^{\leq q}.$$

Let

$$c_\Gamma(d) = |C_\Gamma(d)|$$

be the coverage multiplicity of dimension d .

Law 1 (Redundancy Nyquist law). *For q tolerated sensor failures,*

$$\rho_q(\Gamma) = \begin{cases} 1, & \forall d < m \ c_\Gamma(d) \geq q + 1, \\ \kappa_{\max\{d : c_\Gamma(d) \leq q\}}, & \text{otherwise.} \end{cases}$$

Equivalently, to make repair budget θ sufficient, every critical dimension above θ must be covered at least $q + 1$ times:

$$\rho_q(\Gamma) \leq \theta \iff \forall d \in B_\theta, ; c_\Gamma(d) \geq q + 1,$$

where

$$B_\theta = \{d < m : \kappa_d > \theta\}.$$

This is a clean reliability version of the hidden-debt frontier. Ordinary coverage is the case $q = 0$. One fault requires double coverage, two faults require triple coverage, and so on. But the tropical bottleneck remains: redundant low-dimensional coverage does not lower the repair rank while a higher dimension has coverage multiplicity at most q .

Proposition 1 (Marginal reliability gate). *Let*

$$r = \max\{d : c_\Gamma(d) \leq q\}$$

be the current highest under-covered dimension. Adding a new sensor can lower ρ_q only if it covers r and raises $c_\Gamma(r)$ above q . Coverage of dimensions below r is latent until all higher under-covered dimensions are made $(q + 1)$ -covered.

3 Fault-tolerant frontier inverse

For additive cost w , define the q -redundant critical-band cover cost

$$C_q(\theta) = \min\{w(\Gamma) : \forall d \in B_\theta, c_\Gamma(d) \geq q + 1\}.$$

Let

$$\Phi_q(c) = \min\{\rho_q(\Gamma) : w(\Gamma) \leq c\}$$

be the best worst-case repair scale attainable under cost c while tolerating q sensor failures.

Theorem 2 (Fault-tolerant frontier inverse law).

$$\Phi_q(c) \leq \theta \iff C_q(\theta) \leq c.$$

Thus robust sensor design is exactly redundant critical-band set cover.

Interpretation. This is a Nyquist law for reliability: if a hidden dimension has critical scale above the repair budget, then it must not merely be observed; it must be observed with enough independent multiplicity to survive the allowed failure ideal.

4 Global spinal version

Let $L = H \vee J$ be the effective coordinate ideal. At spinal coordinate n , let m_n be the finite Fubini dimension set, let $\mathcal{F}_n$ be the allowed sensor-failure ideal, and let $\theta_n = 2^{\mu_n}$ be the available repair alphabet. Then robust faithful observation under sensor failures is governed by

$$\{n : \exists d \in B_{n,\theta_n} \text{ with } C_{\Gamma,n}(d) \in \mathcal{F}_n\} \in L.$$

For the q_n -fault model this becomes

$$\{n : \exists d \in B_{n,\theta_n} \text{ with } c_{\Gamma,n}(d) \leq q_n\} \in L.$$

So the global law is: every dimension whose cardinality exceeds the local repair budget must have fault-positive sensor coverage on all but L -small many coordinates.

5 Finite experiments

I ran four finite-shadow checks. These test the combinatorial core of the theorem, not the large-cardinal assumptions.

Experiment	Checks	Failures
q -fault redundancy law	3,000	0
General ideal-positive coverage law	1,200	0
Fault-tolerant frontier inverse law	12,504	0
Global L -threshold law with q -faults	1,500	0

The q -fault test brute-forced every failure subset of size at most q and compared the worst hidden top dimension with the closed-form multiplicity equation

$$\max\{d : c_\Gamma(d) \leq q\}.$$

The general-ideal test generated random downward-closed failure ideals and verified the exact criterion

$$d \text{ robustly covered} \iff C_\Gamma(d) \notin \mathcal{F}.$$

The frontier test compared brute-force best repair scale under a cost budget with the redundant critical-band cover inverse. The global test used a finite ignored-coordinate ideal as a shadow of L and again found no mismatches.

6 Applications

1. **Sensor-bank reliability.** To tolerate q sensor outages, cover each top-band dimension at least $q + 1$ times. Extra redundancy below the top band is useful later but does not lower the current worst-case repair scale.
2. **Active diagnostics.** The next diagnostic probe should target the highest dimension whose coverage multiplicity is still below the required fault threshold.
3. **Distributed identifiers.** A namespace component that must survive q node failures needs $q + 1$ independent carriers. Duplicating smaller namespaces cannot compensate for an under-replicated larger namespace.
4. **Privacy and re-identification.** Defensive privacy can be viewed as deliberately keeping some high-cardinality dimension $\mathcal{F}$ -small in the adversary's sensor portfolio. Offensive re-identification needs fault-positive coverage of each critical top-band dimension.
5. **Verification and monitoring.** A safety monitor portfolio should be audited by critical-band redundancy, not by total number of alarms. Many alarms can leave the top fault dimension under-covered.

Round 15 summary

The new law is fault-tolerant hidden debt:

$$h_\Gamma^\mathcal{F}(d) = 1 \iff C_\Gamma(d) \in \mathcal{F},$$

with worst-case repair scale

$$\rho_\mathcal{F}(\Gamma) = \kappa_{\max\{d: C_\Gamma(d) \in \mathcal{F}\}}$$

when any such dimension exists. For q failures this becomes the redundancy Nyquist equation

$$\rho_q(\Gamma) \leq \theta \iff \forall d \in B_\theta, c_\Gamma(d) \geq q + 1.$$

The strongest practical slogan is:

critical dimensions need fault-positive coverage, not merely one observation.

Round 16 Addendum: Debt-Blocker Duality and Budgeted Failure Frontiers

Spinal Nullity Research Log

April 30, 2026

Status and purpose

This addendum continues the finite-Fubini sensor and repair-rank program. Round 15 modeled failures by an ideal $\mathcal{F}$ and showed that a dimension is robustly covered exactly when its selected cover set is $\mathcal{F}$ -positive. This round grades the failure ideal by a cost or attack budget. The new object is not just whether a dimension can fail, but the least cost needed to expose it. This produces a clean duality: the repair rank after adversarial failures is governed by the cheapest complete erasure of a critical hidden dimension.

1 Budgeted failures and blocker capacity

Let $m = \{0, \dots, m-1\}$ be a finite Fubini dimension set with strongly increasing hidden scales

$$\kappa_0 < \kappa_1 < \dots < \kappa_{m-1}, \quad \prod_{i < d} \kappa_i < \kappa_d.$$

A sensor σ observes a subset $D_\sigma \subseteq m$. A selected portfolio is $\Gamma \subseteq \mathcal{S}$. For dimension d , define the selected cover set

$$C_\Gamma(d) = \{\sigma \in \Gamma : d \in D_\sigma\}.$$

Now assign each sensor a positive failure cost $a(\sigma) > 0$. For $F \subseteq \Gamma$, let

$$a(F) = \sum_{\sigma \in F} a(\sigma).$$

The adversary may remove any $F \subseteq \Gamma$ with $a(F) \leq b$.

Definition 1 (Debt-blocker capacity). *The blocker capacity of dimension d in portfolio Γ is*

$$\beta_\Gamma(d) = a(C_\Gamma(d)).$$

If $C_\Gamma(d) = \emptyset$, then $\beta_\Gamma(d) = 0$. Intuitively, $\beta_\Gamma(d)$ is the price of erasing every selected sensor that sees dimension d .

Theorem 1 (Debt-blocker duality). *Under a failure budget b , dimension d can be made hidden by the adversary iff*

$$\beta_\Gamma(d) \leq b.$$

Consequently the worst-case repair scale is

$$\rho_b(\Gamma) = \begin{cases} 1, & \{d : \beta_\Gamma(d) \leq b\} = \emptyset, \\ \kappa_{\max\{d : \beta_\Gamma(d) \leq b\}}, & \text{otherwise.} \end{cases}$$

Proof sketch. If $\beta_\Gamma(d) \leq b$, the adversary fails exactly $C_\Gamma(d)$; then no surviving selected sensor observes d . Conversely, if some budget-feasible F hides d , every selected sensor seeing d must lie in F , so $C_\Gamma(d) \subseteq F$ and therefore $\beta_\Gamma(d) \leq a(F) \leq b$. The repair equation then reads off the largest exposed hidden dimension, using the tropical dimension-ladder law from earlier rounds. $\square$

2 Critical-band reliability law

For a repair budget θ , recall the critical band

$$B_\theta = \{d < m : \kappa_d > \theta\}.$$

Dimensions outside B_θ may remain hidden without exceeding repair budget θ . Dimensions inside B_θ must survive the failure budget.

Law 1 (Critical-band blocker law).

$$\rho_b(\Gamma) \leq \theta \iff \min_{d \in B_\theta} \beta_\Gamma(d) > b,$$

with the convention that the minimum over the empty set is $+\infty$.

This is the budgeted version of the redundancy Nyquist law. The q -fault model is the special case $a(\sigma) = 1$ and $b = q$, where

$$\beta_\Gamma(d) = |C_\Gamma(d)|, \quad \rho_q(\Gamma) \leq \theta \iff \forall d \in B_\theta \ |C_\Gamma(d)| \geq q + 1.$$

Proposition 1 (Weakest-critical-link principle). *Let*

$$\beta_\Gamma(\theta) = \min_{d \in B_\theta} \beta_\Gamma(d).$$

Then the reliability of the whole critical band is exactly $\beta_\Gamma(\theta)$, not the sum or average of the cover capacities. Adding redundancy below the current weakest critical dimension does not change the worst-case repair scale.

3 Frontier inverse with two budgets

Sensor design now has two budgets: an acquisition budget and an adversarial failure budget. Let $w(\sigma) > 0$ be the cost to buy/select sensor σ . Define the budgeted robust band-cover cost

$$C_b(\theta) = \min \{w(\Gamma) : \forall d \in B_\theta, \beta_\Gamma(d) > b\}.$$

Let

$$\Phi_b(c) = \min \{\rho_b(\Gamma) : w(\Gamma) \leq c\}$$

be the best worst-case repair scale available with acquisition cost at most c while facing failure budget b .

Theorem 2 (Two-budget frontier inverse).

$$\Phi_b(c) \leq \theta \iff C_b(\theta) \leq c.$$

Proof sketch. The condition $\Phi_b(c) \leq \theta$ says some selected portfolio of acquisition cost at most c has no failure-exposable hidden dimension in the critical band B_θ . By the critical-band blocker law, this is exactly the requirement $\beta_\Gamma(d) > b$ for every $d \in B_\theta$, which is precisely the feasibility condition defining $C_b(\theta)$. $\square$

4 Global spinal version

Let $L = H \vee J$ be the effective coordinate ideal. At spinal coordinate n , let m_n be the finite Fubini dimension set, let $\theta_n = 2^{\mu_n}$ be the local repair alphabet, and let b_n be the local failure budget. For a portfolio Γ , define

$$\beta_{\Gamma,n}(d) = a_n(C_{\Gamma,n}(d)).$$

Then robust faithful observation under budgeted sensor failures is governed by

$$\left\{ n : \min_{d \in B_{n,\theta_n}} \beta_{\Gamma,n}(d) \leq b_n \right\} \in L.$$

Equivalently, outside the tolerated effective ideal, every dimension whose hidden cardinal exceeds the repair alphabet must have blocker capacity above the adversary's failure budget.

5 Finite experiments

I ran three finite-shadow checks. They test the exact finite blocker identities, not the transfinite ultrafilter hypotheses.

Experiment	Checks	Failures
Budgeted failure law	20,000	0
Two-budget frontier inverse law	858,590	0
q -fault special case	1,000	0

The first experiment brute-forced all failure sets of cost at most b and compared the worst hidden top dimension with

$$\max\{d : \beta_{\Gamma}(d) \leq b\}.$$

The second enumerated all sensor selections, acquisition budgets c , failure budgets b , and thresholds θ , checking

$$\Phi_b(c) \leq \theta \iff C_b(\theta) \leq c.$$

The third verified that the earlier q -fault redundancy law is exactly the unit-failure-cost specialization of blocker capacity.

6 Applications

1. **Reliability engineering.** Do not count total sensors. For each critical hidden dimension, compute the cost to erase all sensors that see it. The smallest such erasure cost is the real reliability margin.
2. **Sensor-bank procurement.** The design problem becomes a two-budget staircase: buy sensors so every top-band dimension has blocker capacity above the predicted outage or attack budget.
3. **Active diagnostics.** The next probe should target the highest dimension with the lowest blocker capacity. Probes below that dimension are latent until the weakest higher dimension is hardened.

4. **Privacy and re-identification.** A privacy defense can keep a high-cardinality identifier dimension cheap to erase from the adversary's view. A re-identification system must make each critical dimension expensive to erase.
5. **Distributed identifiers and composite keys.** A key component is robust only if every shard carrying that component cannot be removed under the failure budget. Smaller key components cannot compensate for a weak large component.
6. **Verification and monitoring.** Monitoring portfolios should be audited by blocker capacity on top-band hazards. A large number of low-criticality monitors can coexist with a catastrophic single weak critical band.

Round 16 summary

The new law is debt-blocker duality:

$$d \text{ is exposable under budget } b \iff \beta_{\Gamma}(d) = a(C_{\Gamma}(d)) \leq b.$$

Thus

$$\rho_b(\Gamma) = \kappa_{\max\{d: \beta_{\Gamma}(d) \leq b\}}$$

when such a dimension exists. The most useful design equation is

$$\rho_b(\Gamma) \leq \theta \iff \min_{d \in B_{\theta}} \beta_{\Gamma}(d) > b.$$

The practical slogan is:

a critical dimension is safe exactly when it is too expensive to erase.

Round 17. Random Portfolio Nyquist and Probabilistic Debt Erasure

Status. This addendum continues the sensor/repair program. Round 15 introduced fault-positive hidden-debt coverage; Round 16 introduced the blocker capacity of a dimension under budgeted attacks. This round asks a different design question: if sensors are acquired from a noisy or random source, how many sensors are needed before the highest repair bands disappear with high probability?

Exceptional new finding. The new law is a *random redundant Nyquist law*. In deterministic form, the earlier theory says that a critical dimension must be observed at least $q + 1$ times to survive q sensor failures. If sensors are sampled independently, that requirement becomes a binomial tail. Therefore the transfinite repair frontier has a coupon-collector-like acquisition threshold:

$$N_\theta(q, t) \approx \frac{\log |B_\theta| + q \log \log |B_\theta| + t}{p_\theta}.$$

Here $B_\theta = \{d : \kappa_d > \theta\}$ is the critical band above repair budget θ , and p_θ is the weakest coverage probability among dimensions in that band. The step nature of the tropical repair rank survives randomization: acquiring sensors below the threshold usually leaves the top hidden band in place; acquiring just beyond it usually erases the band.

17.1 Random redundant band-cover law

Fix one finite Fubini coordinate with hidden dimensions $d < m$ and increasing cardinal scales

$$\kappa_0 < \kappa_1 < \cdots < \kappa_{m-1}.$$

Let $B_\theta = \{d < m : \kappa_d > \theta\}$ be the set of dimensions whose unresolved debt would force repair above θ . Suppose N sensors are sampled, and sensor s observes dimension d with probability p_d , independently across sensors and dimensions. Let

$$C_N(d) = |\{s < N : s \text{ observes } d\}|.$$

To tolerate q arbitrary sensor failures, the dimension d is robustly covered exactly when

$$C_N(d) \geq q + 1.$$

Thus the probability that the resulting repair rank is at most θ is exactly

$$\Pr(\rho_q(\Gamma_N) \leq \theta) = \prod_{d \in B_\theta} \left[1 - \sum_{r=0}^q \binom{N}{r} p_d^r (1 - p_d)^{N-r} \right].$$

Equivalently, the random repair frontier is the first threshold whose critical band is redundantly covered:

$$\Phi_N^q = \min\{\theta : \forall d \in B_\theta, C_N(d) \geq q + 1\}.$$

Its distribution function is

$$\Pr(\Phi_N^q \leq \theta) = \Pr(\rho_q(\Gamma_N) \leq \theta).$$

17.2 Redundant coupon-collector Nyquist threshold

Assume the critical band has size $b = |B_\theta|$ and that every critical dimension has the same coverage probability p . Write

$$\lambda = Np.$$

The exact success probability is

$$P_{N,q}(b, p) = \left[1 - \sum_{r=0}^q \binom{N}{r} p^r (1-p)^{N-r} \right]^b.$$

In the usual sparse-acquisition regime, the lower binomial tail is well approximated by a Poisson tail. If

$$\lambda = \log b + q \log \log b + t,$$

then the expected number of undercovered dimensions is approximately

$$\frac{e^{-t}}{q!},$$

and the success probability follows the threshold curve

$$P_{N,q}(b, p) \approx \exp\left(-\frac{e^{-t}}{q!}\right).$$

This is the acquisition version of the redundancy Nyquist law: each additional tolerated failure shifts the threshold by roughly $(\log \log b)/p$ sensors.

For nonuniform coverage probabilities, define the Poissonized remaining debt

$$\Lambda_{N,q}(\theta) = \sum_{d \in B_\theta} e^{-Np_d} \sum_{r=0}^q \frac{(Np_d)^r}{r!}.$$

Then the practical transition is governed by

$$\Pr(\rho_q(\Gamma_N) \leq \theta) \approx e^{-\Lambda_{N,q}(\theta)}.$$

Thus the rarest high-cardinality dimensions dominate the random acquisition frontier.

17.3 Band-first acquisition principle

Because the repair objective is tropical, a sensor's value is not proportional to the number of dimensions it observes. A sensor has high marginal value only if it helps erase the current highest exposed band. Formally, if T is the current highest hidden dimension under q -fault tolerance, then an added sensor cannot reduce repair scale unless it observes at least one dimension in the current top critical band whose count is still at most q .

Law 1 (Band-first random acquisition). *For a fixed target repair budget θ , only the dimensions in B_θ enter the success event*

$$\rho_q(\Gamma_N) \leq \theta.$$

Coverage of dimensions outside B_θ has zero direct value for this target, although it may matter for a later, lower target. Therefore the optimal staged design is to erase exposed bands from highest cardinal scale downward.

Interpretation. This explains a common engineering trap: a sensor can appear information-rich because it observes many low-scale dimensions, while still doing nothing to lower the critical repair width. The correct objective is not average coverage. It is redundant coverage of the highest still-hidden debt band.

17.4 Finite experiments

The experiments are finite shadows of the random acquisition formulas. They are not proofs of the transfinite laws, but they test the exact finite probability equations and the band-first design principle.

Experiment	Checks/configurations	Main statistic	Failures
Exact binomial formula	27 configurations	max absolute error 0.0187	0 structural
Top-hidden CDF	4 configurations	max CDF error 0.0045	0 structural
Band-first policy	4 configurations	dominated uniform/low-first	0 reversals

Selected exact-formula rows:

m	N	q	exact	empirical	abs. err.
6	8	0	0.3497	0.3683	0.0187
6	14	1	0.3910	0.3927	0.0017
6	22	2	0.6491	0.6427	0.0064
10	8	1	0.0009	0.0007	0.0002
10	14	2	0.0082	0.0047	0.0036
14	22	0	0.9936	0.9930	0.0006

Selected threshold rows:

b	q	t	N	exact	empirical	asymptotic
50	0	-2	11	0.0025	0.0040	0.0006
50	0	2	33	0.9309	0.9285	0.8734
50	1	0	29	0.3073	0.3210	0.3679
50	2	2	48	0.7859	0.7890	0.9346
100	0	0	26	0.5622	0.5260	0.3679
100	1	2	45	0.8659	0.8615	0.8734
100	2	0	43	0.3439	0.3475	0.6065

The exact formulas match simulation within sampling error. The asymptotic curve is visibly rough for small bands, but it captures the sharp transition and improves as the critical band grows.

The band-first acquisition test compared three policies for the event that the top 30% of dimensions are q -fault covered:

m	N	q	band-first	uniform	low-first
20	12	1	0.950	0.147	0.000
20	18	1	0.998	0.546	0.006
30	16	1	0.992	0.243	0.000
30	24	2	0.999	0.341	0.000

This supports the design rule: if the goal is to lower repair rank, target the current highest debt band first.

17.5 Applications

- **Sensor-bank sizing.** The formula gives a direct way to size random or commodity sensor fleets: choose N so the critical band above the desired repair budget is redundantly covered with high probability.
- **Reliability engineering.** The term $q \log \log |B_\theta|$ is the price of tolerating q arbitrary sensor failures in a random acquisition model.
- **Active diagnostics.** A diagnostic policy should query high-cardinality unresolved dimensions first; low-band observations do not lower the current bottleneck.
- **Privacy and re-identification.** Many weak identifiers become dangerous once repeated observations redundantly cover the high-cardinality identifying dimensions. The same coupon-collector threshold estimates the onset of re-identification risk.
- **Composite keys and distributed IDs.** Randomly assigned partial keys should be audited by critical-band redundant coverage, not by average number of fields present.

17.6 Summary of this round

Round 17 converts deterministic hidden-debt and blocker laws into a probabilistic acquisition theory. The compact new equation is

$$\Pr(\rho_q(\Gamma_N) \leq \theta) = \prod_{d \in B_\theta} \left[1 - \sum_{r=0}^q \binom{N}{r} p_d^r (1 - p_d)^{N-r} \right].$$

The practical law is

$$N_\theta(q, t) \approx \frac{\log |B_\theta| + q \log \log |B_\theta| + t}{p_\theta}.$$

The conceptual discovery is that random sensor acquisition inherits the tropical staircase structure: cover the highest debt band redundantly, or the repair scale does not move.

Round 18: The Gumbel-Nyquist Window for Random Redundant Coverage

Spinal nullity, hidden debt, and probabilistic sensor acquisition

Draft continuation, not peer reviewed. Appended after Round 17.

18.1 Question from the previous frontier

Round 17 gave an exact finite product formula for random redundant coverage of a critical hidden-debt band. If

$$B_\theta = \{d : \kappa_d > \theta\}, \quad |B_\theta| = m,$$

and each newly acquired sensor independently observes each dimension in the band with probability p , then q -fault-tolerant clearance of the band is

$$C_N(d) \geq q + 1 \quad \text{for every } d \in B_\theta,$$

where $C_N(d) \sim \text{Bin}(N, p)$ is the number of acquired sensors that observe d . Therefore

$$\mathbb{P}(\rho_q(\Gamma_N) \leq \theta) = [1 - F_q(N, p)]^m, \quad F_q(N, p) = \mathbb{P}(\text{Bin}(N, p) \leq q). \quad (18.1)$$

The blind spot is the *shape* of the transition. Does the repair frontier drift smoothly, or is there a universal critical window?

18.2 Redundant Gumbel-Nyquist law

Define the finite critical centering $N_{m,q,p}(t)$ by

$$mF_q(N_{m,q,p}(t), p) \approx e^{-t}. \quad (18.2)$$

Then (18.1) gives the distributional law

$$\mathbb{P}(\rho_q(\Gamma_{N_{m,q,p}(t)}) \leq \theta) \approx \exp(-e^{-t}). \quad (18.3)$$

Thus the random repair frontier has a Gumbel-shaped transition window. The band is not gradually cleared. It remains effectively uncleared until the last poorly covered high-debt dimensions disappear, and then the repair scale drops in a narrow window.

Poissonized center. When p is small or when the binomial lower tail is Poisson-approximated, put $\lambda = Np$ and define $\lambda_{m,q}(t)$ by

$$me^{-\lambda} \sum_{r=0}^q \frac{\lambda^r}{r!} = e^{-t}. \quad (18.4)$$

Then

$$N_{m,q,p}(t) \approx \frac{\lambda_{m,q}(t)}{p}. \quad (18.5)$$

Expanding (18.4) gives the asymptotic acquisition law

$$\lambda_{m,q}(t) = \log m + q \log \log m - \log(q!) + t + o(1), \quad (18.6)$$

so

$$N_{m,q,p}(t) \approx \frac{\log m + q \log \log m - \log(q!) + t}{p}. \quad (18.7)$$

This sharpens the earlier Round 17 threshold by identifying the universal limiting shape and the redundancy tax.

Redundancy tax. Relative to zero-fault coverage, the extra acquisition needed for q tolerated failures is approximately

$$\Delta N_q \approx \frac{q \log \log m - \log(q!)}{p}. \quad (18.8)$$

So each additional tolerated failure is cheap compared with the main coupon-collector term $\log m/p$, but it is not free: it appears as a log-log surcharge.

18.3 Confidence quantile law

For a desired success confidence α at repair budget θ , the safest finite formula is not the asymptotic expression but the binomial quantile condition

$$\boxed{N_\alpha(\theta, q, p) = \min \{N : |B_\theta| F_q(N, p) \leq -\log \alpha\}.} \quad (18.9)$$

Then

$$\mathbb{P}(\rho_q(\Gamma_{N_\alpha}) \leq \theta) \approx \alpha. \quad (18.10)$$

This is the practical engineering form of the random redundant Nyquist law. It converts a desired confidence level directly into a number of random sensors to acquire.

Exceptional point. The discovery is that the random hidden-debt frontier is not merely a coverage threshold. It is a *Gumbel-Nyquist frontier*: after centering by the redundancy-corrected coupon-collector term, all finite bands have the same limiting transition curve $\exp(-e^{-t})$. The scale parameter is controlled by p , while the location is controlled by band size m and redundancy q .

18.4 Proof sketch

For a fixed dimension d , clearance fails exactly when $C_N(d) \leq q$. In the independent model this has probability $F_q(N, p)$. All dimensions in the band are cleared iff none of the m dimensions fails, hence (18.1). If N is chosen so that $mF_q(N, p) \rightarrow e^{-t}$ and $F_q(N, p) \rightarrow 0$, then

$$(1 - F_q(N, p))^m \rightarrow \exp(-e^{-t}).$$

For the Poissonized form, replace $\text{Bin}(N, p)$ by $\text{Pois}(\lambda)$. Its lower tail is

$$\mathbb{P}(\text{Pois}(\lambda) \leq q) = e^{-\lambda} \sum_{r=0}^q \frac{\lambda^r}{r!}.$$

Solving m times this tail equal to e^{-t} gives (18.4). Since the top term $\lambda^q/q!$ dominates the finite sum, (18.6) follows by taking logarithms.

18.5 Finite computations and sanity checks

The following checks used the exact binomial tail $F_q(N, p)$, not simulation. For each row, N is the least integer with $mF_q(N, p) \leq 1$, so the predicted success is close to $e^{-1} = 0.367879\dots$

m	q	p	N	exact success	error vs. e^{-1}
1000	0	0.01	688	0.370226	0.002347
1000	1	0.01	920	0.370307	0.002428
1000	2	0.01	1119	0.369974	0.002095
1000	3	0.01	1302	0.370078	0.002198
10000	0	0.01	917	0.370001	0.002122
10000	1	0.01	1171	0.370418	0.002539
10000	2	0.01	1387	0.368366	0.000487
10000	3	0.01	1585	0.368127	0.000247
100000	0	0.01	1146	0.369628	0.001749
100000	1	0.01	1418	0.371218	0.003339
100000	2	0.01	1649	0.370948	0.003068
100000	3	0.01	1859	0.368716	0.000836

Maximum error in the table was 0.003339. The important lesson is that the binomial-tail centering (18.2) is much more reliable at finite sizes than the rough asymptotic formula (18.7), especially for $q \geq 2$.

Ninety-five percent design examples. Using (18.9), the exact binomial quantile gives:

m	q	p	target α	required N	exact success
1000	0	0.01	0.95	983	0.950073
1000	1	0.01	0.95	1243	0.950378
1000	2	0.01	0.95	1464	0.950339
10000	1	0.01	0.95	1489	0.950451
10000	2	0.01	0.95	1723	0.950086
10000	2	0.02	0.95	858	0.950389
10000	2	0.05	0.95	339	0.951379

18.6 Applications

Sensor-bank sizing. To make repair rank at most θ with confidence α , estimate the size $m = |B_\theta|$ of the critical band, the per-sensor hit probability p , and the failure tolerance q , then use (18.9). This gives a direct procurement number.

Reliability engineering. The law separates three design levers: reduce m by deterministic targeted sensors, increase p by better random sensors, or reduce required q by hardening the sensors already acquired. These levers move different terms in (18.7).

Active diagnostics. If a diagnostic system can choose where to sample, it should spend targeted effort on dimensions still inside the highest uncleared band. Random sampling is efficient only after the high band has been narrowed enough that the Gumbel window is reachable.

Privacy and re-identification. For attackers acquiring random quasi-identifiers, (18.9) estimates how many independent data sources are needed before a protected high-cardinality identity band is cleared with high confidence. The same equation can guide defensive throttling: keep the attacker's effective p or N below the Gumbel window.

Composite keys and distributed identifiers. A distributed key system that relies on random components should not ask only for expected coverage. It should require redundant coverage of every high-cardinality component, because the top-hidden repair rank drops only when the last high-debt component crosses the $q + 1$ threshold.

18.7 Summary of Round 18

Round 18 upgrades the random portfolio threshold into a distributional law:

$$\mathbb{P}(\rho_q(\Gamma_N) \leq \theta) \approx \exp(-e^{-t}) \quad \text{when} \quad |B_\theta|F_q(N, p) \approx e^{-t}.$$

The new practical equation is the confidence quantile

$$N_\alpha = \min\{N : |B_\theta|F_q(N, p) \leq -\log \alpha\}.$$

The new conceptual slogan is:

random repair frontiers fall through a Gumbel-Nyquist window.

Round 19: Poisson Booster Completion for Random Sensor Portfolios

From Gumbel-Nyquist thresholds to hybrid random-plus-targeted repair

Draft continuation, not peer reviewed. Appended after Round 18.

19.1 The new blind spot: what remains after the Gumbel window?

Round 18 identified the Gumbel-Nyquist window for random redundant coverage. If

$$B_\theta = \{d : \kappa_d > \theta\}, \quad |B_\theta| = m,$$

and each random sensor observes each dimension in the critical band with probability p , then q -fault-tolerant clearance of the band is

$$C_N(d) \geq q + 1 \quad \text{for every } d \in B_\theta,$$

where $C_N(d) \sim \text{Bin}(N, p)$. The success probability is

$$\mathbb{P}(\rho_q(\Gamma_N) \leq \theta) = [1 - F_q(N, p)]^m, \quad F_q(N, p) = \mathbb{P}(\text{Bin}(N, p) \leq q). \quad (19.1)$$

The overlooked engineering question is not just whether random acquisition has already cleared the band. In real sensor design, one can usually inspect the remaining uncovered dimensions and add targeted deterministic boosters. This creates a hybrid random-plus-targeted completion problem.

Deficit field. Define the coverage deficit of a critical dimension by

$$\delta_N(d) = (q + 1 - C_N(d))_+, \quad (19.2)$$

and the total booster debt by

$$\Delta_N(\theta) = \sum_{d \in B_\theta} \delta_N(d). \quad (19.3)$$

A targeted booster that observes a chosen critical dimension reduces $\Delta_N(\theta)$ by one. Therefore, after N random sensors and b targeted boosters, the critical band is cleared exactly when

$$\Delta_N(\theta) \leq b. \quad (19.4)$$

19.2 Exact finite booster law

Let

$$p_r = \mathbb{P}(\text{Bin}(N, p) = r), \quad 0 \leq r \leq q.$$

For one critical dimension, the probability generating function of the deficit is

$$G_{N,p,q}(z) = 1 - F_q(N, p) + \sum_{r=0}^q p_r z^{q+1-r}. \quad (19.5)$$

Independence across dimensions gives the exact finite law

$$\mathbb{P}(\Delta_N(\theta) \leq b) = \sum_{j=0}^b [z^j] G_{N,p,q}(z)^m. \quad (19.6)$$

This is the finite random analogue of the hidden-debt frontier. Round 18 measured the chance that there are no deficits; Round 19 measures the whole distribution of the remaining repair punch list.

Heterogeneous version. If dimension d has random observation probability p_d , set

$$G_d(z) = 1 - F_q(N, p_d) + \sum_{r=0}^q \mathbb{P}(\text{Bin}(N, p_d) = r) z^{q+1-r}. \quad (19.7)$$

Then

$$\mathbb{P}(\Delta_N(\theta) \leq b) = \sum_{j=0}^b [z^j] \prod_{d \in B_\theta} G_d(z). \quad (19.8)$$

Thus the exact finite hybrid-completion problem is coefficient extraction from a product of small polynomials.

19.3 Poisson booster law

Let

$$D_N = |\{d \in B_\theta : C_N(d) \leq q\}| \quad (19.9)$$

be the number of deficient dimensions. If

$$mF_q(N, p) \rightarrow \lambda \quad (19.10)$$

then $D_N \Rightarrow \text{Pois}(\lambda)$ by the standard law of rare independent defects. The stronger booster statement needs multi-deficits to vanish:

$$m\mathbb{P}(\text{Bin}(N, p) \leq q-1) \rightarrow 0. \quad (19.11)$$

Under (19.10) and (19.11), almost every deficient dimension is short by exactly one cover, hence

$$\Delta_N(\theta) \Rightarrow \text{Pois}(\lambda). \quad (19.12)$$

At the Gumbel-Nyquist centering from Round 18,

$$mF_q(N, p) \approx e^{-t}, \quad (19.13)$$

so

$$\mathbb{P}(\text{band cleared after } b \text{ targeted boosters}) \approx e^{-e^{-t}} \sum_{j=0}^b \frac{e^{-tj}}{j!}. \quad (19.14)$$

This is the new useful law: the random phase does not leave a vague residue. It leaves an approximately Poisson-sized punch list.

Booster confidence quantile. For confidence α and booster budget b , let $\Lambda_{b,\alpha}$ be the largest λ satisfying

$$\mathbb{P}(\text{Pois}(\lambda) \leq b) \geq \alpha. \quad (19.15)$$

Then the random acquisition requirement with a post-random targeted booster budget b is approximately

$$N_{\alpha,b} \approx \frac{\log m + q \log \log m - \log(q!) - \log \Lambda_{b,\alpha}}{p}. \quad (19.16)$$

For $b = 0$, this collapses to the Round 18 confidence law. For $b > 0$, targeted final repair buys back part of the random coverage tail.

19.4 Experiments and finite checks

I tested the exact coefficient law and its Poisson shadow with finite computations.

Check	Configs	Failures	Error
Exhaustive Bernoulli matrices vs. coefficient law	78	0	$4.44 \cdot 10^{-16}$
Poisson approximation for deficient-dimension count	6	–	$8.85 \cdot 10^{-4}$
Exact deficit-polynomial evaluations	7	0	exact

The exhaustive check enumerated all Bernoulli coverage matrices for small m, N , computed the total deficit directly, and compared it with (19.6). No mismatch occurred.

Representative exact deficit-polynomial values:

m	N	p	q	boosters b	$\mathbb{P}(\Delta \leq b)$	low coefficients
50	18	.20	0	1	.772562	(.402954, .369607)
100	64	.10	1	1	.706367	(.382542, .323825)
100	75	.10	1	2	.986282	(.707569, .218916, .059797)
250	93	.08	1	1	.704734	(.376795, .327939)
500	204	.05	2	2	.860595	(.372576, .305666, .182353)
1000	181	.05	1	2	.891270	(.375959, .333014, .182297)
5000	546	.02	1	2	.894180	(.374003, .337570, .182608)

Representative Poisson checks for $D_N = |\{d : C_N(d) \leq q\}|$:

m	p	q	N	b	mF_q	exact $\mathbb{P}(D \leq b)$	Poisson
100	.10	1	64	2	.9563	.928440	.927556
250	.08	1	93	2	.9742	.924751	.924391
500	.05	2	204	3	.9863	.981956	.981838
1000	.05	1	181	3	.9778	.982401	.982343
5000	.02	1	546	3	.9834	.982025	.982013
5000	.02	2	651	4	.9942	.996433	.996429

The exact finite law was perfect in the exhaustive checks. The Poisson law became excellent as the critical-band defects became rare and numerous, which is exactly the intended Gumbel-Nyquist regime.

19.5 Applications and design interpretation

Sensor acquisition. A practical sensor program can be split into two phases: broad random acquisition to enter the Gumbel window, then targeted boosters for the few remaining deficient dimensions. The expected endgame is not proportional to the band size; at the correct center it is approximately Poisson-sized.

Reliability engineering. If q failures must be tolerated, random redundancy is useful until the remaining deficit list is short. After that, targeted hardening beats more undirected acquisition.

Privacy and re-identification. For an attacker, random leaked features create a random deficit field. The number of additional targeted leaks needed to identify all critical dimensions is approximately Δ_N . Thus privacy risk can be expressed as a Poisson tail rather than a vague “some identifiers are missing” statement.

Verification and monitoring. A monitoring system can budget random tests plus targeted recertifications. Equation (19.16) gives a confidence-calibrated tradeoff between random monitoring effort and deterministic follow-up work.

Discovery summary. Round 18 found the Gumbel transition for random repair-band clearance. Round 19 adds the post-transition residue law:

random acquisition creates a Poisson-sized targeted repair punch list.

This turns the hidden-debt theory into an operational two-stage algorithm: random coverage to the critical window, then targeted booster completion.

Round 20: Poisson Repair Frontiers and Booster Order Statistics

From fixed-band completion to the full distribution of the remaining repair scale

Draft continuation, not peer reviewed. Appended after Round 19.

20.1 The blind spot after booster completion

Round 19 solved the following fixed-band question: after N random sensors and b targeted boosters, what is the probability that a chosen critical band B_θ is fully cleared? The overlooked next question is stronger: what is the *entire remaining repair rank* after the targeted boosters are used optimally?

Let finite hidden dimensions be ordered by increasing repair scale

$$\kappa_0 < \kappa_1 < \cdots < \kappa_{M-1}.$$

A random sensor observes dimension d with probability p_d . After N random sensors, write

$$C_N(d) \sim \text{Bin}(N, p_d), \quad \delta_N(d) = (q + 1 - C_N(d))_+, \quad (20.1)$$

where q is the number of tolerated sensor failures and $\delta_N(d)$ is the remaining number of targeted boosters needed to make dimension d safe.

For a threshold index t , define the upper-tail debt

$$\Delta_N(> t) = \sum_{d>t} \delta_N(d). \quad (20.2)$$

Because the tropical repair law says that only the largest still-deficient dimension controls the repair scale, the optimal use of b targeted boosters is immediate: repair the highest deficient dimensions first. This gives the exact frontier law

$$\boxed{\rho_{q,b}(\Gamma_N) \leq \kappa_t \iff \Delta_N(> t) \leq b.} \quad (20.3)$$

This is the booster version of the hidden-debt frontier: a threshold is reached exactly when the total remaining debt above that threshold fits inside the booster budget.

Exchange proof. Suppose a booster is spent on a lower dimension while a higher dimension remains deficient. Swapping that booster upward cannot increase the largest remaining deficient dimension and may decrease it. Repeating this exchange yields the top-down rule. Hence (20.3) is optimal, not merely greedy.

20.2 Exact suffix-polynomial law

For one dimension d , define the deficit generating function

$$G_d(z) = 1 - F_{q,d} + \sum_{r=0}^q \mathbb{P}(\text{Bin}(N, p_d) = r) z^{q+1-r}, \quad (20.4)$$

where

$$F_{q,d} = \mathbb{P}(\text{Bin}(N, p_d) \leq q).$$

Independence of random sensor observations across dimensions gives, for every threshold t ,

$$\boxed{\mathbb{P}(\rho_{q,b}(\Gamma_N) \leq \kappa_t) = \sum_{j=0}^b [z^j] \prod_{d>t} G_d(z).} \quad (20.5)$$

Thus the entire repair-rank distribution is obtained by suffix coefficient extraction. The probability mass at exactly κ_t is

$$\mathbb{P}(\rho_{q,b} = \kappa_t) = \mathbb{P}(\rho_{q,b} \leq \kappa_t) - \mathbb{P}(\rho_{q,b} \leq \kappa_{t-1}), \quad (20.6)$$

with the evident boundary convention.

Useful risk formula. For any nondecreasing numerical severity $w_t = w(\kappa_t)$,

$$\mathbb{E}[w(\rho_{q,b})] = \sum_{t=0}^{M-1} w_t (F_t - F_{t-1}), \quad F_t := \mathbb{P}(\rho_{q,b} \leq \kappa_t). \quad (20.7)$$

This turns the abstract transfinite repair scale into a computable expected-loss curve once the cardinals are assigned operational severity weights.

20.3 Poisson frontier and order-statistic law

In the Gumbel-Nyquist regime, deficient dimensions are rare. Let

$$\Lambda_t(N) = \sum_{d>t} F_{q,d}. \quad (20.8)$$

If multi-deficits are negligible, so that almost every deficient dimension needs only one booster, then the suffix debt above t is approximately Poisson with mean $\Lambda_t(N)$. Therefore

$$\mathbb{P}(\rho_{q,b}(\Gamma_N) \leq \kappa_t) \approx e^{-\Lambda_t(N)} \sum_{j=0}^b \frac{\Lambda_t(N)^j}{j!}. \quad (20.9)$$

This is the *Poisson repair-frontier law*.

Equivalently, deficient dimensions form an approximate Poisson point process on the dimension ladder with intensity masses $F_{q,d}$. Targeted boosters delete the highest points of this process. Thus, in the unit-deficit regime,

$$\rho_{q,b} \text{ is controlled by the } (b+1)\text{-st highest deficient dimension.} \quad (20.10)$$

For $b = 0$ this is the highest random defect. For $b = 1$ it is the second-highest random defect, and so on.

Confidence frontier inverse. Let $\Lambda_{b,\alpha}$ be the largest λ satisfying

$$\mathbb{P}(\text{Pois}(\lambda) \leq b) \geq \alpha. \quad (20.11)$$

Then an α -confidence repair frontier is approximately

$$\rho_{q,b} \leq \kappa_t \text{ with confidence } \alpha \iff \Lambda_t(N) \leq \Lambda_{b,\alpha}. \quad (20.12)$$

For homogeneous $p_d = p$, this says that a suffix of length $s = M - t - 1$ is cleared at confidence α when

$$s F_q(N, p) \leq \Lambda_{b,\alpha}. \quad (20.13)$$

So each additional targeted booster does not merely increase success probability at one fixed threshold; it moves the whole repair frontier downward by deleting one more extreme point of the defect process.

20.4 Experiments and finite checks

I tested the exact suffix law, the top-down booster rule, and the Poisson frontier approximation.

Check	Configurations	Failures	Max error
Exhaustive Bernoulli matrices vs. suffix-polynomial law	216	0	$1.89 \cdot 10^{-14}$
Homogeneous Monte Carlo vs. exact suffix CDF	3	–	$4.49 \cdot 10^{-3}$
Heterogeneous Monte Carlo vs. exact suffix CDF	3	–	$3.74 \cdot 10^{-3}$
Poisson frontier approximation in rare-defect regime	20 suffix samples	–	$1.80 \cdot 10^{-4}$
Suffix CDF monotonicity checks	63 thresholds	0	exact

Representative exact-versus-simulation checks:

M	N	p range	q	b	trials	max error
16	18	.20	1	1	60000	.003739
24	26	.16	1	2	60000	.004487
30	42	.10	2	2	60000	.002346
18	24	.080–.240	1	1	70000	.003745
20	38	.075–.243	2	2	70000	.002203
25	36	.073–.227	1	3	70000	.000491

Representative Poisson frontier rows:

M	N	p	q	b	max suffix error
500	178	.05	1	0	$1.80 \cdot 10^{-4}$
500	178	.05	1	2	$7.27 \cdot 10^{-5}$
1000	230	.04	1	2	$7.98 \cdot 10^{-5}$
2000	387	.025	1	3	$4.80 \cdot 10^{-5}$
5000	550	.02	1	3	$9.70 \cdot 10^{-6}$

The finite tests support the new claim: once the fixed-band booster law is known, the full repair frontier is a suffix process. The Poisson approximation is strongest exactly where the Gumbel-Nyquist theory predicts it should be: many critical dimensions, rare defects, and mostly one-unit deficits.

20.5 Applications and design interpretation

Sensor-bank dashboards. Equations (20.5) and (20.9) give a complete risk curve, not just a pass/fail probability. A dashboard can report the probability that the remaining repair scale is below each cardinal band.

Two-stage acquisition. The operational algorithm is: acquire random sensors until the expected suffix defect intensity $\Lambda_t(N)$ crosses the desired confidence threshold, then spend targeted boosters from the top dimension downward. This is optimal for tropical repair because lower repairs have no value while a higher dimension remains deficient.

Privacy and re-identification. For an attacker, the remaining privacy protection after random feature leaks is the highest still-deficient hidden dimension after targeted follow-up leaks. The risk is not just the number of missing features; it is the order statistic of the missing critical dimensions.

Verification and monitoring. Monitoring can budget random checks plus targeted recertifications. The full CDF in (20.5) tells how likely the monitor is to force the system below any chosen repair-risk band.

Discovery summary. Round 19 showed that random acquisition leaves a Poisson-sized targeted punch list for a fixed critical band. Round 20 upgrades this to a full repair-rank distribution:

the post-booster repair scale is the suffix debt quantile of a Poisson defect frontier.

This turns the hidden-debt theory into a practical sequential design rule: random coverage creates the defect frontier; targeted boosters erase its highest points; the next highest surviving point is the residual repair scale.

Round 21 Addendum: Spinal Borel–Nyquist Summability

From local random repair frontiers to global almost-sure faithful observation

Status. Draft research addendum, not peer reviewed. This round continues the sensor/repair program after the Poisson repair-frontier law. Rounds 17–20 described a single coordinate or a single finite critical band: random acquisition creates a Gumbel threshold; inspection leaves a Poisson-sized punch list; targeted boosters erase the highest remaining debt. Round 21 asks the next blind-spot question: what happens across the entire countable singular spine?

21.1 The blind spot: local high confidence is not global safety

A repeated engineering mistake is to make every coordinate “very reliable” but not reliable fast enough. In spinal nullity the global success condition is not that each coordinate has high success probability. For the tame ideal $J = \text{Fin}$, the random set of failed coordinates must be finite. Therefore the local failure probabilities must be summable.

For coordinate n , let

$$F_n = \{\rho_{n,q,b_n}(\Gamma_{n,N_n}) > \theta_n\}$$

be the event that the random sensor portfolio, after b_n targeted boosters and tolerance q , still leaves local repair rank above the allowed threshold θ_n . Put

$$\varepsilon_n = \mathbb{P}(F_n).$$

The global failure set is

$$F = \{n : F_n \text{ occurs}\}.$$

Robust faithful observation modulo the effective ideal $L = H \vee J$ is the condition

$$F \in L.$$

For $H = \emptyset$ and $J = \text{Fin}$, this means only finitely many local coordinates may fail.

21.2 Spinal Borel–Nyquist law

Law 1 (Spinal Borel–Nyquist law, tame independent form). *Assume the local random acquisition/repair choices are independent across coordinates. For $J = \text{Fin}$,*

$$\mathbb{P}(F \in \text{Fin}) = 1 \iff \sum_{n < \omega} \varepsilon_n < \infty,$$

while, if $\sum_n \varepsilon_n = \infty$, then

$$\mathbb{P}(F \in \text{Fin}) = 0.$$

Equivalently,

global almost-sure faithful observation requires summable local Nyquist failure.

Proof sketch. This is the classical Borel–Cantelli dichotomy applied to the derived local failure events F_n . The first implication needs no independence: if $\sum \mathbb{P}(F_n) < \infty$, then only finitely many F_n occur almost surely. The divergent direction uses independence. The new content is not the probability lemma

itself, but its identification of the correct spinal design variable: the repair-frontier failure probabilities ε_n produced by the local Nyquist laws.

Combining with Round 20, in a finite hidden-dimension block at coordinate n ,

$$\varepsilon_n(t, b_n) = 1 - \sum_{j=0}^{b_n} [z^j] \prod_{d>t} G_{n,d}(z),$$

where $G_{n,d}$ is the one-dimension deficit polynomial. In the rare-defect regime,

$$\varepsilon_n(t, b_n) \approx 1 - e^{-\Lambda_{n,t}} \sum_{j=0}^{b_n} \frac{\Lambda_{n,t}^j}{j!}.$$

Thus the global design requirement becomes the explicit summability condition

$$\sum_{n<\omega} \left(1 - e^{-\Lambda_{n,t_n}} \sum_{j=0}^{b_n} \frac{\Lambda_{n,t_n}^j}{j!} \right) < \infty.$$

This is the first full-spine acquisition law: the local Poisson frontiers must be driven into an ℓ^1 tail.

21.3 Confidence schedules and budget inversion

The most useful operational form is to choose a summable failure budget sequence (η_n) , such as

$$\eta_n = \frac{c}{(n+1)^2} \quad \text{or} \quad \eta_n = 2^{-n-1},$$

and then choose the least acquisition/booster schedule satisfying

$$\varepsilon_n(t_n, b_n, N_n) \leq \eta_n.$$

Using the Round 18 confidence quantile and Round 20 suffix frontier, one obtains

$$N_n = \min \left\{ N : 1 - e^{-\Lambda_{n,t_n}(N)} \sum_{j=0}^{b_n} \frac{\Lambda_{n,t_n}(N)^j}{j!} \leq \eta_n \right\}.$$

For homogeneous critical bands of size m_n , observation probability p_n , and no boosters, this gives the approximate schedule

$$N_n \approx \frac{\log m_n + q \log \log m_n - \log(q!) + \log(1/\eta_n)}{p_n}.$$

If $\eta_n \asymp n^{-2}$, the cost of global almost-sure control adds roughly

$$\frac{2 \log n}{p_n}$$

random sensors at coordinate n beyond the single-coordinate constant-confidence threshold.

Principle 1 (Summable confidence principle). *Do not assign the same 99% local confidence to infinitely many spinal coordinates. A constant local failure probability produces infinitely many failures almost surely. Instead allocate a summable tail of local failure probabilities.*

21.4 Beyond Fin: density and summable recurrence ideals

The same framework gives useful laws for larger recurrence ideals. Let $F = \{n : F_n\}$.

Law 2 (Weighted summable-ideal sufficient law). *For the summable ideal*

$$J_w = \left\{ A \subseteq \omega : \sum_{n \in A} w_n < \infty \right\},$$

if

$$\sum_{n < \omega} w_n \varepsilon_n < \infty,$$

then

$$F \in J_w \quad \text{almost surely.}$$

Proof sketch. $\mathbb{E} \sum_n w_n 1_{F_n} = \sum_n w_n \varepsilon_n < \infty$, so the weighted failure sum is finite almost surely.

Law 3 (Density-ideal random Nyquist law). *For the asymptotic-density-zero ideal Z_0 , if the F_n are independent Bernoulli events, then*

$$\frac{1}{N} \sum_{n < N} 1_{F_n} - \frac{1}{N} \sum_{n < N} \varepsilon_n \rightarrow 0 \quad \text{almost surely.}$$

Hence

$$\frac{1}{N} \sum_{n < N} \varepsilon_n \rightarrow 0 \quad \implies \quad F \in Z_0 \quad \text{almost surely.}$$

If the Cesaro means of ε_n stay bounded away from 0, then $F \notin Z_0$ almost surely.

This separates recurrence standards sharply. A sequence such as $\varepsilon_n \asymp n^{-1/2}$ fails the Fin standard almost surely but succeeds for density-zero recurrence because its average tends to 0.

21.5 Finite experiments

The experiments below are finite shadows of the probabilistic spine. They do not prove the transfinite set-theoretic claims, but they test the numerical shape of the global failure laws.

Experiment A: prefix growth for summable versus divergent local failure probabilities. Each row used 200 independent simulations through 100,000 coordinates. The table reports expected and simulated mean failure counts at three prefixes.

Profile	E_{1k}	mean _{1k}	E_{10k}	mean _{10k}	E_{100k}	mean _{100k}
$0.8/(n+1)^2$	1.315	1.335	1.316	1.335	1.316	1.335
$2^{-(n+1)}$	1.000	0.960	1.000	0.960	1.000	0.960
$0.35/(n+1)$	2.620	2.620	3.426	3.425	4.232	4.310
$0.12/\sqrt{n+1}$	7.416	7.565	23.825	23.715	75.720	75.515

The summable profiles stabilize. The divergent profiles keep accumulating failures. This is exactly the global Borel–Nyquist distinction.

Experiment B: constructing a summable confidence schedule from local sensor laws. Parameters: critical band size $m = 20$, observation probability $p = 0.2$, failure tolerance $q = 1$, no boosters, and target $\eta_n = 0.6/(n+1)^2$. For each coordinate $n \leq 500$, the least N_n satisfying the exact binomial band-failure law was computed.

n	least N_n	exact ε_n	target $0.6/(n+1)^2$
1	32	0.133351	0.150000
2	36	0.062941	0.066667
5	43	0.015872	0.016667
10	49	0.004717	0.004959
25	58	0.000742	0.000888
50	64	0.000213	0.000231
100	71	0.000049	0.000059
250	79	0.000009	0.000010
500	86	0.000002	0.000002

Across the first 500 coordinates, $\sum \varepsilon_n = 0.3534$. In 5000 simulations of the derived independent failure field, the mean number of failed coordinates was 0.3518, the probability of no failures was 0.6938, and the probability of at most two failures was 0.9972. The schedule makes global finite failure plausible at finite prefixes and, by summability, almost sure in the infinite spine.

Experiment C: density ideal contrast. 100 simulations through 100,000 coordinates checked failure density rather than finite failure.

Profile	exp dens _{1k}	mean _{1k}	exp dens _{10k}	mean _{10k}	exp dens _{100k}	mean _{100k}
$0.7/\sqrt{n+1}$	0.0433	0.0438	0.0139	0.0139	0.0044	0.0044
constant 0.05	0.0500	0.0504	0.0500	0.0499	0.0500	0.0500
$1/\log(n+3)$	0.1763	0.1763	0.1245	0.1246	0.0963	0.0964

The $n^{-1/2}$ profile produces infinitely many failures almost surely, so it is not Fin-safe. But its density tends toward zero, so it is compatible with a density-zero recurrence ideal. Constant error is not.

21.6 Applications

Sensor-bank sizing. A local 95% or 99% reliability target is insufficient for an infinite or long-horizon spinal system. The correct engineering rule is to allocate local failure probabilities as a summable tail, then invert the local Poisson frontier to get N_n and b_n .

Verification and monitoring. For a countable family of subsystems, local monitors may each have small miss probability. The system-wide theorem says the miss probabilities must be summable if the specification allows only finitely many unresolved coordinates.

Privacy and re-identification. If each generation of auxiliary data has a small chance of clearing a critical hidden band, infinitely many generations can still breach privacy almost surely unless the breach probabilities are summably controlled. For density-style privacy standards, the average breach probability is the relevant variable instead.

Distributed identifiers and composite keys. The law separates local key robustness from global namespace safety. A namespace design with a constant per-layer collision risk creates infinitely many collision layers almost surely; a summable schedule creates only finitely many.

21.7 Summary and next target

Round 21 lifts the Poisson repair frontier from one coordinate to the entire spine. The new master equation is

$$\mathbb{P}(\text{global Fin-faithful repair}) = 1 \iff \sum_n \varepsilon_n < \infty.$$

For larger ideals the criterion changes: summable ideals weight the ε_n , and density ideals take Cesaro means. The practical wisdom is simple:

$$\boxed{\text{local confidence must be budgeted by the recurrence ideal.}}$$

The next natural target is an *ideal-calibrated random repair theorem*: for analytic P -ideals J_φ , characterize exactly which failure-probability fields (ε_n) make the random failure set belong to J_φ almost surely. That would turn recurrence ideals into a design language for stochastic sensor acquisition.

Round 22 — Recurrence-Ideal Risk Transform and the Block Chernoff-Nyquist Law

Status. Draft research continuation, not peer reviewed. This round extends the Round 21 spinal Borel-Nyquist summability law from Fin-style recurrence to block and density-style recurrence ideals. The goal is to find the right “confidence currency” for each recurrence ideal, rather than applying one local confidence level everywhere.

22.1 Question and blind spot

Round 21 established the Fin version: if F is the random set of coordinates where local repair still fails, then global almost-sure faithful observation requires summable local failure probabilities. The blind spot is that Fin is only one recurrence ideal. For block recurrence, density-zero recurrence, or weighted recurrence, the right design question is not “is each coordinate very likely to work?” but “does the random failure schedule fall inside the tolerated ideal?”

This suggests a new object: the ideal risk transform. Given local failure probabilities ϵ_n and an ideal J on the spine, define the random Bernoulli failure set

$$F_\epsilon = \{ n < \omega : X_n = 1 \}, \text{ with independent } X_n \sim \text{Bernoulli}(\epsilon_n).$$

The risk schedule is J -safe exactly when $F_\epsilon \in J$ almost surely. Different ideals impose different asymptotic currencies: Fin uses summability, summable ideals use weighted expectation, and block-density ideals use concentration of block averages.

22.2 Block ideals and the exact transform

Partition the spine into finite blocks B_k , with $b_k = |B_k|$. Let τ_k be a tolerance sequence, usually $\tau_k \rightarrow 0$. Define the block-threshold ideal

$$J_\tau = \{ A \subseteq \omega : |A \cap B_k| / b_k \leq \tau_k \text{ for all but finitely many } k \}.$$

For a Bernoulli failure field, define the k -th block violation event

$$V_k = \{ \sum_{n \in B_k} X_n > \tau_k b_k \}.$$

Because the blocks are disjoint, the V_k are independent. Therefore the exact Borel-Cantelli transform is:

$$R_{\{J_\tau\}}(\epsilon) = \sum_k \Pr(V_k).$$

Block ideal risk law.

$$R_{\{J_\tau\}}(\epsilon) < \infty \Rightarrow F_\epsilon \in J_\tau \text{ almost surely.}$$

$$R_{\{J_\tau\}}(\epsilon) = \infty \Rightarrow F_\epsilon \notin J_\tau \text{ almost surely, provided the block events are independent.}$$

This is a clean continuation of Round 21: Fin corresponds to singleton blocks with zero tolerance, where $\Pr(V_n) = \epsilon_n$ and the transform becomes $\sum_n \epsilon_n$.

22.3 Chernoff-Nyquist sufficient law

Let μ_k be the average local failure probability on block B_k :

$$\mu_k = (1/b_k) \sum_{n \in B_k} \epsilon_n.$$

For independent but not necessarily identical Bernoulli failures, Hoeffding gives

$$\Pr(V_k) \leq \exp(-2 b_k (\tau_k - \mu_k)_+^2).$$

Thus the first useful design law is:

$$\sum_k \exp(-2 b_k (\tau_k - \mu_k)_+^2) < \infty \Rightarrow F_\epsilon \in J_\tau \text{ almost surely.}$$

This is the Block Chernoff-Nyquist law. It says that a block is not safe merely because μ_k is below τ_k . The safety margin must beat the block fluctuation scale. Equivalently, one needs

$$\tau_k - \mu_k \gg \sqrt{\log k / b_k}$$

in a summable sense. This is the new operational correction: for density-like recurrence, local risk must clear both a mean barrier and a concentration barrier.

22.4 Coupling with the Poisson repair frontier

Round 20 gave the post-booster local repair failure probability at coordinate n . In the Poisson-frontier approximation, for threshold t_n , random acquisition N_n , q_n tolerated failures, and booster budget b_n , write

$$\varepsilon_n = 1 - \exp(-\Lambda_n) \sum_{j=0}^{b_n} \Lambda_n^j / j!,$$

where Λ_n is the suffix defect intensity above the local repair threshold. The Round 22 lift says: feed these ε_n into the ideal risk transform, not into a single coordinatewise confidence rule.

$$\text{Global } J_\tau\text{-safe design} \Leftarrow R_{\{J_\tau\}}(\varepsilon_n(N_n, b_n, t_n)) < \infty.$$

For the block ideal this becomes

$$\sum_k \Pr(\sum_{n \in B_k} \text{Bernoulli}(\varepsilon_n) > \tau_k b_k) < \infty.$$

This is the full-spine recurrence-aware acquisition equation. It merges the local Poisson repair frontier with the global recurrence standard.

Round 22 — Continued

22.5 New theory: confidence currencies by recurrence ideal

The theory now has a dictionary of confidence currencies:

Recurrence ideal	Almost-sure safety currency	Design meaning
Fin	$\sum \epsilon_n < \infty$	Only finitely many failed coordinates almost surely.
Weighted summable J_w	$\sum w_n \epsilon_n < \infty$ is a sufficient expectation weight	Weighted failure mass is finite almost surely.
Block threshold J_τ	$\sum_k \Pr(\text{block density} > \tau_k) < \infty$	Only finitely many blocks exceed tolerated density.
Density-zero shadow	Block or prefix averages plus concentration limit	Only finitely many failures allowed if asymptotic frequency vanishes.

The most useful conceptual result is that “99% local success” has no invariant meaning until a recurrence ideal is specified. Under Fin, even 99.999% constant local success fails almost surely over infinitely many coordinates. Under a density ideal, constant failure rate fails density-zero but a decaying average may be sufficient. Under block ideals, the relevant variable is not the coordinate probability alone but the block violation tail.

22.6 Phase law and inverse design

For homogeneous blocks with $\epsilon_n = p_k$ on B_k , the exact transform reduces to binomial tails:

$$R_{\{J_\tau\}}(p) = \sum_k \Pr(\text{Bin}(b_k, p_k) > \tau_k b_k).$$

This gives a direct inverse-design law. Given desired block transform budget α_k with $\sum_k \alpha_k < \infty$, choose local acquisition so that

$$\Pr(\text{Bin}(b_k, p_k) > \tau_k b_k) \leq \alpha_k.$$

A Chernoff-safe sufficient form is

$$p_k \leq \tau_k - \sqrt{\log(1/\alpha_k) / (2 b_k)}.$$

Then choose N_n , b_n , or booster budgets so that the local Poisson-frontier failure ϵ_n does not exceed this p_k budget. This creates a two-level Nyquist design: local repair must beat the hidden-dimension frontier, and the resulting coordinate failure field must beat the recurrence-ideal transform.

22.7 The exceptional finding

The exceptional discovery is that recurrence ideals turn probabilistic repair into a tail-transform problem. The local repair law does not finish the design; it merely produces an ϵ -field. The recurrence ideal then acts as a second Nyquist surface. In compact form:

$$\text{local sensor law} \rightarrow \epsilon\text{-field} \rightarrow \text{ideal risk transform} \rightarrow \text{global recurrence safety}.$$

So the real full-spine equation is not one Nyquist law but a composition of Nyquist laws:

$$\text{RepairSafe}_J = \{ \Gamma : F_{\{\epsilon(\Gamma)\}} \in J \text{ almost surely} \}.$$

For block ideals, this is exactly the summability of block violation probabilities. This appears useful because it tells an engineer how to allocate confidence nonuniformly across an infinite monitoring spine.

Round 22 — Experiments

22.8 Finite and computational sanity checks

The finite experiments tested three things: exact Poisson-binomial block tails, the block-risk transform, and simulated violation counts. These are finite shadows, not proofs of the transfinite ultrafilter statements, but they test the probability side of the new recurrence-aware design law.

Experiment A: heterogeneous block tail law

For heterogeneous Bernoulli probabilities inside each block, exact Poisson-binomial tails were computed by dynamic programming and compared with Monte Carlo. Hoeffding bounds were also recorded as conservative certificates.

Configurations	Max Monte Carlo error	Interpretation
27	0.0131	Exact DP and simulation agreed within sampling noise.

b	τ	mean μ	threshold	exact tail	MC tail	Hoeffding bound
16	0.10	0.0333	1	0.0975	0.0922	0.8673
16	0.15	0.0324	2	0.0137	0.0120	0.6422
64	0.20	0.0611	12	0.000108	0.0000	0.0847
64	0.20	0.0977	12	0.00798	0.00783	0.2617

Experiment B: summable versus divergent block-risk transforms

Three profiles were compared over the first 100 blocks. The summable-block profile uses large enough block sizes to force a rapidly summable tail. The thin-block profile keeps the same mean margin but uses too little block length, so violations accumulate. The above-threshold profile has mean failure above tolerance and fails strongly.

Profile	Σ first 100 block-violation probabilities	Simulated mean violations in first 40 blocks	Not-violation probability in first 40 blocks
summable block	1.57e-26	0.000	1.000
thin blocks	1.022	0.760	0.477
above threshold	85.41	31.91	0.000

The key finite lesson is that the same pointwise margin $\mu_k < \tau_k$ can either be safe or unsafe depending on block length. Concentration is part of the Nyquist budget.

22.9 Applications

Sensor-bank sizing. Infinite monitoring systems should allocate per-coordinate confidence by ideal-risk budgets, not by one uniform reliability target. Fin-style systems need summable failure probabilities; density-style systems need average decay plus concentration.

Verification and monitoring. A verifier can choose a recurrence ideal representing tolerated failure recurrence, then compute the required local repair schedule by inverting the ideal transform.

Privacy and re-identification. Re-identification failures can be allowed infinitely often if the privacy target is density-style, but not if the target is Fin-style. The same local leakage profile can be safe under one ideal and unsafe under another.

Distributed identifiers. Composite keys distributed across time or shards should be provisioned against block-violation tails. A shard block with many weak identifiers can be safe if concentration beats the recurrence threshold.

Round 22 — Summary and next targets

22.10 Summary of findings

Round 22 upgrades the full-spine acquisition theory from coordinatewise probability to recurrence-ideal probability. The main object is the ideal risk transform:

$$\varepsilon \mapsto \Pr(F_\varepsilon \notin J), \text{ or more usefully, } \varepsilon \mapsto R_J(\varepsilon).$$

For block-threshold ideals the transform is exact:

$$R_{\{J_\tau\}}(\varepsilon) = \sum_k \Pr(\sum_{n \in B_k} \text{Bernoulli}(\varepsilon_n) > \tau_k b_k).$$

The main sufficient law is the Chernoff-Nyquist condition:

$$\sum_k \exp(-2 b_k (\tau_k - \mu_k)^2) < \infty \Rightarrow F_\varepsilon \in J_\tau \text{ almost surely.}$$

The practical design equation is the composition

$$\text{local Poisson frontier } \varepsilon_n(N_n, b_n, t_n) + \text{ideal risk transform } R_J \Rightarrow \text{global recurrence safety.}$$

The exceptional conceptual result is that each recurrence ideal has its own confidence currency. Fin uses summability; block ideals use summability of block-violation tails; density-style ideals use vanishing averages plus concentration; weighted ideals use weighted failure mass.

22.11 Candidate law for the growing theory

A useful general target is the Spinal Risk-Transform Principle: for any recurrence ideal J with a computable submeasure or block presentation, robust random observation should be designed by making the induced Bernoulli failure set land in J almost surely.

$$\text{RobustFaithful}_J(\Gamma) \Leftrightarrow F_{\{\varepsilon(\Gamma)\}} \in J \text{ almost surely.}$$

In future work, the goal is to characterize this criterion for analytic P-ideals $\text{Exh}(\varphi)$, F_σ ideals $\text{Fin}(\varphi)$, density ideals, and Fubini sums of block ideals. The expected answer is a hierarchy of risk transforms, each translating local Nyquist failure into global recurrence failure.

22.12 Next targets

1. Analytic P-ideal Nyquist law. For $J = \text{Exh}(\varphi)$, find checkable conditions on ε ensuring $\varphi(F_\varepsilon \setminus \text{setminus } n) \rightarrow 0$ almost surely.
2. Fubini-risk composition. Combine block-risk transforms with earlier finite Fubini sensor kernels: inner blocks produce local defect probabilities, outer ideals decide recurrence safety.
3. Optimal confidence allocation. Given a cost function for reducing ε_n , minimize total acquisition cost subject to $R_J(\varepsilon) < \infty$ or $R_J(\varepsilon) \leq \delta$.
4. Adversarial dependence. Round 22 used independence across blocks. A stronger theorem should replace independence with martingale or negative-dependence hypotheses, giving robust concentration under adaptive acquisition.
5. Publication cleanup. Separate proven finite probability laws from conjectural transfinite sensor statements, then rebuild the theory as a chain of formal definitions, exact lemmas, and clearly marked speculative extensions.

One-line slogan. Local Nyquist repair buys an ε -field; the recurrence ideal decides whether that ε -field is globally meaningful.

Round 23 - Correlation-Robust Ideal Risk and the Dependence Tax

Rounds 21-22 converted local Nyquist failure probabilities into global recurrence-ideal safety. Round 22 used independent Bernoulli block tails. The blind spot is common-mode dependence: in real sensor banks, failures, blind spots, and adversarial suppressions are rarely independent. This round replaces the independent block transform by a sharp arbitrary-dependence transform based only on marginal local failure probabilities.

23.1 New object: the Frechet-Kwerel block tail transform

Let B_k be a finite block, $m_k = |B_k|$, and let X_n in $\{0,1\}$ mark local repair failure. Write $\epsilon_n = P(X_n=1)$. For a block-threshold ideal, violation occurs when

$$S_k = \sum_{n \in B_k} X_n \geq v_k, v_k = \text{floor}(\tau_k m_k) + 1.$$

Sort the block marginals in nonincreasing order: $p_{\{k,1\}} \geq \dots \geq p_{\{k,m_k\}}$. Define

$$K_k(\epsilon) = \min(1, \min_{0 \leq j < v_k} (\sum_{i=j+1}^{m_k} p_{\{k,i\}}) / (v_k - j)).$$

Interpretation: peel off the j largest marginal risks, then ask how much remaining marginal mass is available to force the remaining $v_k - j$ simultaneous failures. The $j=0$ term is the crude Markov bound; the later terms are the top-heavy correction.

Exceptional finding. K_k is the exact worst-case block violation probability over all joint laws with the prescribed marginals:

$$\sup \{ P(S_k \geq v_k) : P(X_n=1)=\epsilon_n \text{ for all } n \in B_k \} = K_k(\epsilon).$$

Upper-bound proof sketch. For each $j < v_k$, delete the j largest-risk indicators. On the event $S_k \geq v_k$, the undeleted indicators must contribute at least $v_k - j$ successes. Hence $P(S_k \geq v_k) p_{\{k,j\}} \leq \sum_{i>j} p_{\{k,i\}}$. Also $P \leq 1$. Taking the minimum gives K_k . Sharpness follows from the finite linear-program dual for atomic laws on $\{0,1\}^{m_k}$; the experiments below solve that primal LP directly and match the formula to numerical precision.

23.2 Correlation-robust recurrence law

For arbitrary dependence inside and across blocks, the first Borel-Cantelli lemma still gives a one-way safety theorem:

$$\sum_k K_k(\epsilon) < \infty \implies \{k : S_k \geq v_k\} \text{ is finite almost surely.}$$

Therefore, for the block-threshold ideal J_τ ,

$$\sum_k K_k(\epsilon) < \infty \implies F_\epsilon \text{ in } J_\tau \text{ almost surely, uniformly over all dependence couplings.}$$

This is the correlation-robust version of the Round 22 risk transform. Divergence is not by itself an impossibility theorem without additional independence or mixing assumptions; adversarial dependence can either concentrate or suppress infinitely many block violations. Thus the robust law is intentionally one-sided: it gives a dependable sufficient design certificate.

Practical law. Independent Chernoff-Nyquist safety is not correlation-safe. To survive unknown common-mode dependence, replace each independent block-tail probability by $K_k(\epsilon)$.

23.3 The dependence tax

In a homogeneous block, $p_{\{k,i\}} = p$ and v_k approximately $\tau_k m_k$. Then

$$K_k(\epsilon) = \min(1, m_k p / v_k) \text{ approximately } \min(1, p / \tau_k).$$

The block size no longer gives exponential concentration. Under independence, if $p < \tau_k$, the violation probability is roughly $\exp(-m_k D(\tau_k || p))$; under arbitrary dependence it is only of order p / τ_k . This is the dependence tax:

$$\text{Tax}_k = K_k(\epsilon) / P_{\text{ind}}(\text{Bin}(m_k, p) \geq v_k).$$

The tax can be astronomical, because dependence can align rare local failures into the same block-level catastrophe.

m	p	tau	v	independent tail	robust K	tax
50	0.02	0.10	6	4.782e-4	0.1667	3.49e2
100	0.02	0.10	11	5.646e-6	0.1818	3.22e4
200	0.02	0.10	21	9.332e-10	0.1905	2.04e8
500	0.02	0.08	41	6.680e-14	0.2439	3.65e12

Table: homogeneous blocks. Independence makes block risk exponentially small; arbitrary dependence leaves a linear-scale worst case.

23.4 Correlation firewall design

A sensor schedule that only makes the independent risk transform summable may still be globally unsafe under common-mode failures. A correlation firewall must enforce

$$\sum_k K_k(\epsilon | B_k) < \infty.$$

For homogeneous blocks this becomes the severe but transparent condition

$$\sum_k \min(1, p_k/\tau_k) < \infty.$$

Thus increasing block size is not a substitute for reducing marginal failure probability when dependence is uncontrolled. Large blocks help only after a credible independence, negative-dependence, martingale, or mixing certificate has been established.

Hybrid target. If failures are dependent only within clusters but clusters are sufficiently independent, collapse each cluster to a super-indicator with a cluster marginal, then run the ideal risk transform on the cluster field. The effective unit of Nyquist confidence is the common-mode cluster, not the raw coordinate.

23.5 Finite experiments

Experiment	Checks / configs	Failures	Max error / result
LP over atomic laws vs. K_k formula	324	0	8.05e-16
Homogeneous independence-vs-robust table	7 configs	--	tax up to 3.65e12
Summable profile, first 500 blocks	--	--	independent sum 0.0233; robust sum 2.190
Divergent $n^{-1/2}$ profile, first 500 blocks	--	--	independent sum 1.18e-5; robust sum 11.544
Constant $p=0.02$ below $\tau=0.10$, first 500 blocks	--	--	independent sum 0.00282; robust sum 90.909

The first row is the main verification: for random marginals on blocks of size 2 through 7 and all thresholds, a full linear program over all 2^m atoms matched the closed-form transform K_k . The remaining rows quantify the dependence tax. Independence can look safe while correlation-robust accounting sees a large recurrent-risk budget.

23.6 Applications and next target

Reliability engineering. A sensor bank sized by independent Chernoff tails can fail under shared power, shared model bias, weather, supply-chain defects, or coordinated attacks. The transform K_k gives a conservative audit number when only marginal failure rates are known.

Verification and monitoring. A monitoring theorem that assumes independent local alarms should expose that assumption as a resource. Without it, the correct certificate is summability of K_k , not summability of independent tails.

Privacy and re-identification. Rare leaks can be aligned by shared latent causes. The robust transform estimates the worst-case chance that a block contains enough simultaneous leaks to break anonymity even if each individual leak is marginally unlikely.

Composite keys and distributed identifiers. Repeated weak identifiers across a block are not exponentially protective unless their failures are independent or diversified by construction. Design should diversify common-mode clusters first, then count raw replicas.

Next target. Develop a cluster/mixing hierarchy between the independent Round 22 transform and the fully adversarial Round 23 transform. The expected law is a continuum: raw-coordinate Chernoff at one end, cluster-coordinate Kwerel risk at the other.

23.7 Summary

Round 23 identifies the missing dependence layer in the recurrence-risk program. The new law is:

global correlation-robust safety = summability of exact worst-case block tails K_k .

The exceptional conceptual point is that block size only buys exponential confidence when independence or sufficient mixing is real. In the blind spot where failures correlate, the Nyquist frontier must be paid in marginal risk reduction, diversification, or common-mode cluster breaking.

Round 24 - Latent Common-Mode Nyquist

Status. Draft theorem exploration; not peer reviewed. This round continues the risk-transform branch of the sensor/repair program. Round 22 treated independent block failures, and Round 23 showed that arbitrary dependence with fixed marginals can destroy Chernoff-style safety. Round 24 studies the intermediate and very common case: failures are conditionally independent given a hidden common-mode rate.

1. Motivation: the blind spot after correlation-robust risk

The Round 23 robust transform is deliberately pessimistic: it protects against any coupling of local failures with the same marginals. Engineering systems often have more structure. Failures may be independent after conditioning on a shared temperature, software release, vendor outage, network condition, or adversarial campaign intensity. This creates a latent variable Q_k for block B_k . Conditional on $Q_k=q$, the local failures behave independently with probability q . Unconditionally, they are positively correlated.

This model is a de Finetti/common-mode shadow. It interpolates between the independent Chernoff-Nyquist law and the arbitrary-dependence law. The new phenomenon is that increasing the block size no longer guarantees exponentially small block risk. The risk bottoms out at the tail of the latent common-mode distribution.

2. Latent block transform

For a block B_k of size m_k , let S_k be the number of failed coordinates and let the violation threshold be $v_k = \text{floor}(\tau_k m_k) + 1$. Assume

$$Q_k \sim L_k, \text{ and } X_{\{k,i\}} \mid Q_k \text{ are independent Bernoulli}(Q_k).$$

Define the binomial tail kernel

$$T_{\{m,\tau\}}(q) = \Pr[\text{Bin}(m,q) \geq \text{floor}(\tau m) + 1].$$

Then the exact block violation risk is the latent risk transform

$$r_k = R_k(L_k) = \int_0^1 T_{\{m_k,\tau_k\}}(q) dL_k(q).$$

This is the precise replacement for the independent binomial tail in Round 22. The global Borel-Nyquist rule becomes

$$\sum_k r_k < \infty \Rightarrow \text{the violation set belongs to } J_\tau \text{ almost surely.}$$

If the block violation events are independent across k , divergence of $\sum_k r_k$ gives infinitely many block violations almost surely. Thus, under block independence, the summability of the latent risk transform is the exact finite/infinite boundary.

3. Exceptional finding: the latent-tail barrier

Assume $\tau_k = \tau$ and Q_k has no atom at τ . By the conditional law of large numbers, S_k/m_k converges to Q_k given Q_k . Therefore, as m_k grows,

$$r_k = E[T_{\{m_k,\tau\}}(Q_k)] \rightarrow \Pr[Q_k > \tau].$$

This is the latent common-mode Nyquist barrier:

No amount of within-block replication makes block risk vanish below the probability that the hidden common-mode rate itself crosses the threshold.

In the independent case Q_k is deterministic, so this lower floor is zero whenever $p < \tau$. In the common-mode case, even with the same local mean $p < \tau$, a positive upper tail of Q_k produces a persistent block risk. This explains why the same observed marginal failure rates can be safe under independence but unsafe under latent dependence.

4. Two-layer confidence bounds

For any $\gamma > 0$, Hoeffding's inequality applied conditionally gives a useful sandwich:

$$\Pr[Q_k \geq \tau_k + \gamma] (1 - \exp(-2 m_k \gamma^2)) \leq r_k$$

$$r_k \leq \Pr[Q_k \geq \tau_k - \gamma] + \exp(-2 m_k \gamma^2).$$

Thus block risk splits into two currencies: latent-tail risk and sampling-noise risk. If $E[Q_k] = p_k \leq \tau_k - \delta$ and $\gamma = \delta/2$, then

$$r_k \leq \Pr[Q_k \geq p_k + \delta/2] + \exp(-m_k \delta^2/2).$$

A variance-based sufficient version follows from Cantelli's inequality:

$$r_k \leq \text{Var}(Q_k) / (\text{Var}(Q_k) + (\delta/2)^2) + \exp(-m_k \delta^2/2).$$

Consequently, if the means stay below threshold by a fixed margin, a practical global certificate is

$$\sum_k \text{Var}(Q_k) < \infty \text{ and } \sum_k \exp(-m_k \delta^2/2) < \infty.$$

For beta-binomial blocks with intraclass correlation ρ_k , $\text{Var}(Q_k) = \rho_k p_k(1-p_k)$. So persistent common-mode correlation is not harmless noise; to obtain Fin-style safety, the correlation tail itself must be summable, unless other structure gives a sharper latent-tail bound.

5. Beta-binomial specialization

A convenient finite model is $Q \sim \text{Beta}(\alpha, \beta)$, with mean p and intraclass correlation ρ :

$$\alpha = p(1/\rho - 1), \beta = (1-p)(1/\rho - 1).$$

Then S has a beta-binomial law:

$$\Pr[S=s] = C(m,s) B(s+\alpha, m-s+\beta) / B(\alpha, \beta).$$

The block risk is the beta-binomial upper tail, and as m grows it approaches $\Pr[\text{Beta}(\alpha, \beta) > \tau]$. This gives a computable, interpretable common-mode diagnostic: estimate p and ρ , then test the latent tail rather than trusting the marginal p alone.

6. Finite experiments

All experiments are finite shadows. They do not prove the transfinite sensor laws, but they test the new probabilistic transform and expose where independence-based confidence fails.

rho	risk at m=200	risk at m=1000	latent tail limit
0 (independent)	9.33e-10	2.28e-39	0
0.001	1.24e-7	1.04e-13	1.24e-24
0.005	9.82e-5	5.73e-6	1.69e-6
0.010	1.30e-3	5.14e-4	3.74e-4
0.050	3.43e-2	3.35e-2	3.33e-2

Table 1. Same marginal $p=0.02$ and threshold $\tau=0.10$. Small positive common-mode correlation can inflate the block risk by many orders of magnitude; large block size then converges to the latent tail rather than to zero.

Check	Rows/configurations	Failures / max error
Beta-binomial tail converges to beta latent-tail limit	100 parameter rows	m=2000 max abs error 0.00501
Two-layer Hoeffding sandwich	25 parameter rows	0 failures
Independent vs latent common-mode amplification	5 rho levels	risk floor appeared exactly as predicted
First-500 block global profiles	4 profiles	summable rho stabilized; constant rho accumulated

7. Global profile experiment

Profile, first 500 blocks	Total expected violations
independent deterministic $p=0.02$	4.67e-7
constant latent $\rho=0.01$	0.651
decaying $\rho=(k+20)^{-2}$	1.16e-5
slow decaying $\rho=0.05/\sqrt{k}$	0.201

This experiment used $m=200$, $p=0.02$, $\tau=0.10$. The independent profile is essentially safe. Constant common-mode correlation gives a persistent violation rate. Rapidly decaying correlation restores summability. Slow decay remains materially risky over the first 500 blocks and suggests a nontrivial tail audit is needed.

8. New law: common-mode summability

The practical global rule is now stronger than Round 21's local failure summability:

A spine is not globally confidence-safe merely because each coordinate or block has a small marginal error. It is safe when the recurrence-ideal risk of the whole latent failure field is finite.

For Fin-style safety with block thresholds, one sufficient certificate is

$$\sum_k [\Pr(Q_k \geq \tau_k - \gamma_k) + \exp(-2 m_k \gamma_k^2)] < \infty.$$

A variance-only, margin-based version is

$$p_k \leq \tau_k - \delta, \sum_k \text{Var}(Q_k) < \infty, \sum_k \exp(-m_k \delta^2 / 2) < \infty \Rightarrow \text{safety}.$$

For beta-binomial blocks this becomes approximately

$$\sum_k \rho_k p_k (1 - p_k) < \infty$$

under a fixed below-threshold margin. This is a clean engineering rule: common-mode correlation must be driven down across the spine, not just the local marginal failure probability.

9. Relationship to earlier rounds

Round 22: independent blocks. Risk currency is a sum of binomial or Chernoff tails.

Round 23: arbitrary dependence. Risk currency is the worst-case marginal coupling K_k , often much larger than the independent tail.

Round 24: latent common-mode dependence. Risk currency is an integral of binomial tails against a latent rate law. It is less pessimistic than arbitrary coupling but exposes a durable risk floor absent from independence.

This suggests a three-level diagnostic pipeline:

1. If independence is justified, use the Chernoff-Nyquist transform.
2. If a latent factor model is justified, use the latent-tail transform.
3. If no dependence model is justified, use the Round 23 correlation-robust transform.

10. Applications

Sensor-bank reliability. Shared vendor, firmware, weather, power, or network conditions create Q_k -like common modes. Redundancy is real only after common-mode tails are bounded.

Verification and monitoring. A test suite with many local checks may still fail in bursts after a release or configuration drift. The latent variable is release quality or environmental stress.

Privacy and re-identification. An anonymization system may pass many independent-looking local tests while a latent population or linkage variable increases many match probabilities at once.

Distributed identifiers and composite keys. Collision probabilities should be modeled with latent namespace stress, not only pairwise or marginal collision rates.

Active diagnostics. The highest-value experiment is to estimate or bound the common-mode tail $\Pr(Q_k > \tau_k)$, because this is the risk floor that block size alone cannot remove.

11. Summary of Round 24

$$r_k = \int T_{\{m_k, \tau_k\}}(q) dL_k(q)$$

$$r_k \rightarrow \Pr(Q_k > \tau_k) \text{ as } m_k \rightarrow \infty$$

$$\sum_k r_k < \infty \Rightarrow \text{recurrence-ideal safety}$$

$$\text{persistent common-mode tail} \Rightarrow \text{infinitely many block violations under block independence}$$

The exceptional discovery is the latent-tail barrier: block replication and local confidence do not defeat common-mode uncertainty. To make the spinal system safe, the common-mode tail itself must be made summable or pushed below the recurrence ideal's tolerance.

Round 25: Diversity-Nyquist and Common-Mode Repair

Status. Draft theorem exploration. This appendix continues the spinal-nullity sensor/repair program. The statements below are proof sketches and finite-shadow tests, not peer-reviewed theorems.

Motivation. Round 24 identified a latent common-mode barrier: if a whole block shares a hidden failure-rate Q , increasing the number of within-block replicas cannot push the block risk below the tail probability $P(Q > \tau)$. Round 25 asks what actually breaks that barrier. The answer is not more replicas inside the same latent source. It is independent latent-domain diversity.

Exceptional finding: Diversity-Nyquist law.

Split a block into s latent domains. Domain ℓ has hidden common-mode failure rate Q_{ℓ} . Conditional on Q_{ℓ} , observations inside that domain are independent. Let the domain weights be w_{ℓ} with $\sum w_{\ell} = 1$, and let the observed block failure fraction be R_{hat_m} . If every domain receives many replicas, then

$$R_{\text{hat}_m} \rightarrow R_s = \sum_{\ell \leq s} w_{\ell} Q_{\ell} \quad \text{almost surely.}$$

Therefore the block violation risk has the exact large-replica limit

$$\lim_{\{\text{within-domain replication} \rightarrow \infty\}} P(R_{\text{hat}_m} > \tau) = P(\sum_{\ell \leq s} w_{\ell} Q_{\ell} > \tau).$$

This is the new barrier and the new escape route. One common mode gives $P(Q > \tau)$. Many independent common modes give a mean-tail probability, which can decay exponentially in the number of truly independent domains.

Practical slogan: replication beats sampling noise; diversity beats common-mode noise.

Equal-domain iid form. If $Q_0, \dots, Q_{s-1}$ are iid copies of Q and all weights are $1/s$, then

$$r_s(\tau) = P((Q_0 + \dots + Q_{s-1})/s > \tau).$$

If $E[Q] < \tau$ and Q has a finite moment-generating function, Chernoff gives

$$r_s(\tau) \leq \exp(-s I_Q(\tau)),$$

$$I_Q(\tau) = \sup_{\{\lambda \geq 0\}} \{ \lambda \tau - \log E[\exp(\lambda Q)] \}.$$

Thus a target local block risk η is obtained by the domain-count rule

$$s_{\min}(\eta) = \text{ceil}(\log(1/\eta) / I_Q(\tau)).$$

This is a new Nyquist law, but the unit is no longer bits, sensors, or boosters. The unit is independent latent failure domains.

25.1 The Spinal Diversity Schedule

For a countable spine, each coordinate block k may have its own threshold τ_k , latent distribution L_k , and diversity count s_k . The Round 21 and Round 22 risk transforms say that global safety is decided by whether block violations belong to the recurrence ideal. The new diversity transform supplies the local epsilon-field:

$$\epsilon_k^{\text{div}} = P(\sum_{\ell < s_k} w_{\{k, \ell\}} Q_{\{k, \ell\}} > \tau_k) + \text{sampling_tail}_k.$$

For Fin-style global safety, choose the domain diversity schedule so that

$$\sum_k \epsilon_k^{\text{div}} < \text{infinity}.$$

For block-threshold ideals, replace this by the corresponding block-risk transform from Round 22. For density-style ideals, it is enough to force average diversity risk to zero.

Spinal Diversity-Nyquist schedule. In the equal iid case, a sufficient Fin-safe schedule is

$$s_k \geq \text{ceil} \left(\frac{a \log(k+2) + c_k}{I_{\{Q_k\}}(\tau_k)} \right), \quad \text{with } a > 1,$$

and with within-domain replica counts chosen so that $\sum_k \text{sampling_tail}_k < \text{infinity}$.

This turns a qualitative instruction - diversify your sensors - into a spine-level quantitative law. If the diversity rate function is small, the coordinate needs many independent domains. If the rate is zero because τ is not above the latent mean, no finite amount of diversity buys summable safety.

Domain distinctness warning. Merely labeling sensors as different is not enough. If all Q_{ℓ} share a higher supermode Q_0 , the effective rate function can collapse. In the extreme case $Q_{\ell} = Q_0$ for all ℓ , the diversity law reduces back to the Round 24 barrier:

$$P((Q_0 + \dots + Q_0)/s > \tau) = P(Q_0 > \tau).$$

So the relevant object is the number of independent or sufficiently mixing latent domains, not the raw number of vendors, replicas, probes, or measurements.

Connection to hidden-debt laws. Rounds 13-16 showed that the highest hidden dimension controls repair scale. Round 25 adds that a high dimension is not truly covered unless its coverage comes from enough independent domains. The cover count must be lifted from sensors to latent domains:

$$\text{coverage}(d) = \text{number of independent failure domains observing } d,$$

critical d is safe at risk η only when $\text{coverage}(d) \geq s_{\min_d}(\eta)$.

25.2 Finite Experiments

The finite tests below use beta-binomial common-mode blocks. They are sanity checks of the finite-shadow behavior, not proofs of the transfinite statements. The main model uses $Q \sim \text{Beta}(1,19)$, so $E[Q] = 0.05$, with threshold $\tau = 0.10$.

Experiment A: full beta-binomial risk converges to the latent mean-tail barrier. Each block is split into s equal latent domains. The full risk uses finite sampling inside each domain; the latent limit uses only $P(\text{mean}(Q_{\text{ell}}) > \tau)$.

latent domains s	block size m	full beta-binomial risk	latent mean-tail limit
1	160	0.13632	0.13627
1	1,600	0.13582	0.13627
1	16,000	0.13772	0.13627
2	160	0.09280	0.08628
2	1,600	0.08946	0.08628
2	16,000	0.08600	0.08628
4	160	0.05310	0.03574
4	1,600	0.03670	0.03574
4	16,000	0.03610	0.03574
8	160	0.02622	0.00655
8	1,600	0.00834	0.00655
8	16,000	0.00642	0.00655
16	160	0.01118	0.00034
16	1,600	0.00052	0.00034
16	16,000	0.00026	0.00034

Observation: for fixed s , increasing within-domain replication drives the full risk toward the latent mean-tail limit. But the limit itself only drops when s increases. This directly supports the Round 25 slogan: replication handles sampling noise; diversity handles common-mode noise.

Experiment B: independent latent-domain diversity creates exponential-looking decay. The table reports Monte Carlo estimates of $P(\text{mean}(Q_{\text{ell}}) > \tau)$ for several beta laws.

Beta latent law	τ	$s=1$	$s=2$	$s=4$	$s=8$	$s=16$
Beta(1,19), mean .05	.10	0.13607	0.08625	0.03559	0.00664	0.00029
Beta(2,38), mean .05	.10	0.08677	0.03720	0.00718	0.00028	0.00001
Beta(5,95), mean .05	.10	0.02578	0.00377	0.00010	<1e-5	<1e-5
Beta(1,9), mean .10	.20	0.13444	0.07875	0.02820	0.00407	0.00015

Observation: when the threshold exceeds the latent mean, risk falls rapidly with the number of independent domains. This decay is absent under the single-common-mode model of Round 24.

Experiment C: a summable diversity schedule. For $Q \sim \text{Beta}(1,19)$, $\tau = 0.10$, the Chernoff rate is approximately $I_Q(\tau) = 0.359$. Choosing $s_k = \text{ceil}(3 \log(k+2)/I_Q(\tau))$ gives a summable envelope.

k	required domains s_k	Chernoff envelope $\exp(-I s_k)$
1	10	2.76e-2
2	12	1.35e-2
5	17	2.24e-3
10	21	5.32e-4
50	34	5.00e-6
100	39	8.31e-7
500	52	7.81e-9
1000	58	9.06e-10

25.3 Applications

Sensor-bank engineering. If several sensors share the same firmware, vendor, power supply, site, dataset, training pipeline, or calibration process, they may represent one latent domain. A bank of twenty sensors from one latent domain can still behave like one common-mode observation for global spinal safety.

Reliability procurement. Round 16 gave the blocker view: an adversary or failure process erases coverage by hitting blockers. Round 25 says procurement should buy blocker diversity, not merely more units. A robust acquisition score should count independent latent domains per critical hidden dimension.

Verification and monitoring. Verification suites often replicate tests. If replicated tests share a generator, simulator, oracle, or benchmark construction process, they may have the same Q . The diversity-Nyquist law says a verification suite needs independent failure modes to lower common-mode risk.

Privacy and re-identification. Privacy attacks often combine many weak signals. If the signals are generated by one common latent source, extra samples may only reduce sampling noise. If the attacker obtains independent latent domains, the re-identification frontier can shift sharply. Defense should limit independent-domain leakage, not just per-sensor accuracy.

Composite keys and distributed identifiers. Adding more fields from the same authority may not diversify a key. Adding fields from independent authorities can reduce common-mode ambiguity. The law suggests a design audit: count independent domains contributing to each critical key dimension.

Active diagnostics. When a critical dimension remains high-risk, the best next test is not necessarily the most accurate test. It is often the test that comes from the least-correlated latent domain. Thus active learning should include a diversity value term:

value(test) = drop in highest critical-band risk after adding one independent domain,
not merely drop in marginal error.

Practical checklist.

1. Identify the critical hidden dimensions above the allowed repair budget.
2. For each critical dimension, cluster existing sensors by latent common-mode source.
3. Estimate or upper-bound the latent rate distribution Q for each domain.
4. Compute $I_Q(\tau)$, or use empirical tail estimates for the average of s domain rates.
5. Allocate new sensors to missing independent domains until the induced epsilon-field satisfies the recurrence-ideal risk transform.

25.4 Summary and Next Targets

Round 25 summary. The main new law is:

$$\lim P(\text{block violation}) = P(\text{weighted average of latent domain rates} > \tau).$$

If the domains are independent and the threshold exceeds the latent mean, then

$$P(\text{weighted average} > \tau) \leq \exp(-\text{effective_domain_count} * \text{rate}).$$

This turns common-mode mitigation into a Nyquist-style diversity law. The system must acquire enough independent latent domains for every critical hidden dimension, and the local risk generated by that diversity schedule must be summable or ideal-small at the spinal level.

Most useful equation.

$$s_{\min}(k, \eta_k) = \text{ceil}(\log(1/\eta_k) / I_{\{Q_k\}}(\tau_k)),$$

with $\sum_k \eta_k < \text{infinity}$ for Fin-style global safety.

Most important caution. Vendor count, sensor count, sample count, and test count are not the same as latent-domain count. A system can appear heavily replicated and still be common-mode fragile.

Next target A: dependency graph diversity. Replace independent domains by a dependency graph or mixing coefficient. Seek a risk law of the form

$$P(\text{weighted average } Q > \tau) \leq \exp(-s_{\text{eff}} * I_{\text{eff}}(\tau)),$$

where s_{eff} is an effective independent-domain count.

Next target B: adaptive diversity acquisition. Combine the Poisson booster frontier from Rounds 19-20 with domain diversity. After random acquisition, inspect which critical dimensions lack enough independent domains, then buy targeted domain boosters.

Next target C: ultrafilter version. Translate latent-domain diversity into a sensor-kernel statement: a critical coordinate is not robustly covered unless the observing kernels come from independently large failure filters whose blocker intersection is small in the effective ideal.

Compact final slogan.

In spinal sensor design, replication lowers variance, repair lowers aliasing, boosters clear defects, but only diversity breaks common-mode barriers.

Round 26 - Common-Mode Cut Nyquist Law

Status. Exploratory theorem proposal with exact finite model and asymptotic rare-shock checks. This round follows the Diversity-Nyquist law: replication suppresses sampling noise and independent diversity suppresses common-mode noise. The new blind spot is that nominally diverse domains may still share upstream latent causes. The correct object is therefore not the list of domains, but the hypergraph of common-mode shocks that can erase them together.

26.1 Discovery: diversity must survive quotienting by common-mode causes

Let D be a finite set of diversity domains with weights w_d summing to 1. Let A be a set of independent latent common-mode shocks. A shock a in A affects the domain subset S_a subset D with probability p_a . If the active shock set is T , the damaged weight is

$$L(T) = \sum_{d \in \bigcup_{a \in T} S_a} w_d.$$

A block violates the recurrence threshold τ when $L(T) > \tau$. The exact common-mode risk is therefore

$$r_H(\tau) = \sum_{T: L(T) > \tau} \prod_{a \in T} p_a \prod_{a \notin T} (1 - p_a).$$

This formula is the hypergraph analogue of the latent mean-tail from Round 25. It says that two systems with the same number of apparent domains can have radically different risk: the risk is controlled by shared-cause coverage geometry.

Practical slogan: domain labels are not diversity. Only independent common-mode cuts are diversity.

26.2 Single-source barrier and min-cut exponent

If one latent shock a covers weight $W_a = \sum_{d \in S_a} w_d$ greater than τ , then that shock alone violates the block. Hence

$$r_H(\tau) \geq 1 - \prod_{a: W_a > \tau} (1 - p_a).$$

This is the single-source common-mode barrier. It explains why adding more sensors within a large shared platform, data vendor, root service, power circuit, or software stack may fail to move the global repair frontier: the upstream source already exceeds the threshold.

There is also a rare-shock exponent. Suppose $p_a(c) = \exp(-c I_a)$ with independent shocks and $c \rightarrow \infty$. Then, provided some shock set can violate the threshold,

$$\lim_{c \rightarrow \infty} -(1/c) \log r_H(\tau) = C_H(\tau), \\ C_H(\tau) = \min \{ \sum_{a \in T} I_a : L(T) > \tau \}.$$

So robust diversity design is a weighted hypergraph min-cut problem. The dominant failure is the cheapest set of latent common-mode shocks whose union covers more than τ of the system.

This is the Common-Mode Cut Nyquist Law: to make the block globally safe, every violating common-mode cut must be made expensive enough, split below threshold, or moved into the tolerated ideal.

26.3 Spine/global version

For each spinal coordinate n , let H_n be the common-mode hypergraph of that coordinate and let $r_n(t_n)$ be its violation risk above the local repair threshold t_n . For the Fin recurrence standard, the global random failure set is finite almost surely whenever

$$\sum_{n < \omega} r_n(t_n) < \infty.$$

Under rare-shock scaling with exponents $C_n(t_n)$, a useful sufficient design criterion is

$$\sum_n \exp(-(1-o(1)) C_n(t_n)) < \infty.$$

For a general recurrence ideal J , replace the summability currency by the appropriate Round 22 risk transform. For block ideals, one applies the common-mode risk $r_{\{k,n\}}$ inside each block and then checks the ideal block-tail transform. In short: Round 25 gives the benefit of independent diversity; Round 26 says to first quotient by common-mode overlap, then apply the recurrence-ideal risk transform.

26.4 Engineering interpretation

A sensor portfolio can look diverse because it uses many sensors, vendors, models, regions, or identifiers. But if many of cause is not necessarily the most likely one; it is the one, or cheapest coalition, whose affected weight exceeds the threshold.

Design rule. Audit the common-mode hypergraph. For each source a , compute W_a . If $W_a > \tau$, it is a repair-rank blocker by itself. If no single source is enough, compute the cheapest violating union. Procurement should buy sensors that split large source weights or raise the cost/probability exponent of the cheapest violating cuts.

26.5 Finite experiments

The finite tests used binary common-mode shock hypergraphs. They are exact for the finite model and sanity checks for the transfinite design analogy.

Check	Configurations	Failures / max error	Interpretation
Exact hypergraph risk vs Monte Carlo	20 random hypergraphs	max MC error 0.00240	Enumeration formula matched simulation
Single-source barrier inequality	500 random hypergraphs	0 failures	Risk always exceeded barrier
Rare-shock exponent convergence	5 random hypergraphs	estimates approached $C_H(\tau)$	Dominant cut controls log-risk
Root common-mode illustration	5 domain counts	risk floor near p_{root}	Leaf diversity could not erase root risk

Sample exact-risk rows

m	shocks	τ	exact risk	MC risk	barrier	large single shocks
5	10	0.3	0.19363	0.19603	0.18556	4
5	6	0.4	0.03832	0.03673	0.0	0
6	6	0.35	0.06021	0.05883	0.03667	1
9	5	0.4	0.0912	0.09087	0.08431	1
8	8	0.35	0.25768	0.25765	0.24937	4

In the first row, most risk was already forced by four single common-mode shocks whose individual affected weight exceeded τ . More independent leaf replication would not remove this part of the risk; it must be split or hardened at the source.

26.6 Rare-shock min-cut data

For random hypergraphs with $p_a(c)=\exp(-c I_a)$, the table reports $C_H(\tau)$ and estimates of $-\log r_H(\tau)/c$ at $c=2,4,6,8,10$. The estimates move toward the min-cut exponent, as predicted.

$C_H(\tau)$	$c=2$	$c=4$	$c=6$	$c=8$	$c=10$	cut size
1.7	1.364	1.558	1.615	1.643	1.659	1
1.3	0.885	1.174	1.245	1.272	1.284	1
0.587	0.44	0.544	0.573	0.582	0.585	1
0.515	0.248	0.418	0.478	0.5	0.509	1
1.022	0.928	1.019	1.021	1.022	1.022	1

The cut size was one in these samples because random broad shocks often had enough affected weight to violate the threshold alone. That is itself a warning: a single upstream dependency can dominate an apparently diversified architecture. Future tests should target hypergraphs where no single edge violates τ , so the leading cut has size two or larger.

26.7 Root barrier demonstration

The next table shows a tree model with one root common-mode shock of probability 0.01 and independent leaf shocks of probability 0.02. Threshold $\tau=0.2$. As leaf count grows, independent leaf noise disappears, but the root floor remains.

leaf domains s	risk without root	risk with root
2	0.03960000	0.04920400
4	0.07763184	0.08685552
8	0.01033689	0.02023352
16	0.00024013	0.01023773
32	0.00000278	0.01000275

This is the clearest finite shadow of the new barrier: more independent leaves drive sampling risk toward zero, but they do not help against an upstream root that covers enough weight by itself.

26.8 Applications

Sensor-bank reliability. A bank of sensors is only as diverse as its common-mode dependency graph. Sensors sharing a vendor, firmware, power supply, dataset, model family, API, physical route, clock, or human procedure should be grouped under a common-mode source. Critical sources whose affected weight exceeds the safety threshold must be split or independently hardened.

Verification and monitoring. Test suites often repeat the same hidden assumption. The common-mode cut law says that additional tests do not improve the repair frontier if they share the same latent blind spot. Verification diversity should be bought by cutting assumption overlap, not merely by increasing test count.

Privacy and re-identification. Several anonymized identifiers can appear independent while sharing an upstream data broker, timestamping procedure, or demographic linkage source. If one source covers enough of the identifying key space, it creates a single-source re-identification barrier.

Composite keys and distributed identifiers. A composite key is robust only if its components do not collapse under the same common-mode outage or namespace collision. The useful metric is the cheapest latent cut that destroys more than the allowed key weight.

Active diagnostics. The best next diagnostic is often not another replicate sensor. It is a sensor that splits the largest common-mode edge or raises the exponent of the cheapest violating cut.

26.9 Summary of Round 26

1. Diversity must be measured after quotienting by common-mode dependencies.
2. Exact finite common-mode risk is a hypergraph union-tail probability.
3. A single source whose affected weight exceeds τ creates an irreducible risk floor.
4. In the rare-shock regime, log-risk is governed by the cheapest violating common-mode cut.
5. Global spinal safety requires the resulting common-mode risk field to satisfy the recurrence ideal risk transform.
6. The next target is a mixed model combining continuous latent rates from Rounds 24-25 with binary shock hypergraphs from this round.

Compact law: random acquisition creates coverage, diversity creates independent domains, but common-mode cuts decide whether that diversity is real.

Round 27 Addendum: Cut-Lift Hardening and the Exponent Budget Frontier

Status: exploratory theorem target with exact finite optimization laws, rare-shock asymptotics, and computational checks. This addendum continues the common-mode cut program from Round 26.

1. Blind spot after the common-mode cut law

Round 26 identified the dominant risk in a common-mode hypergraph: in the rare-shock regime, failure is governed by the cheapest violating common-mode cut, not by the number of apparent sensors. The next design question is inverse: how much deterministic hardening is needed to lift every dangerous cut above a required exponent?

This round turns the common-mode cut law into a hardening calculus. The result is useful because it says where money, redundancy, verification, or physical isolation should be spent: not evenly everywhere, but on the family of cuts that still violate the target threshold.

2. Setup: violation family and cut exponent

Let A be a finite set of latent common-mode shock sources and D a finite set of observed domains. A shock set $T \subset A$ activates a load $L(T)$, for example the total domain weight covered by shocks in T . A block violates threshold θ when $L(T) > \theta$. Define the violation family

$$V_\theta = \{ T \subset A : L(T) > \theta \}.$$

Each shock a has a rarity exponent I_a , so in the rare-shock scaling $p_a(c)$ approximately $\exp(-c I_a)$. The common-mode cut exponent is

$$C_H(\theta; I) = \min_{T \in V_\theta} \sum_{a \in T} I_a.$$

Round 26 showed that this exponent is the asymptotic decay rate of the block violation probability. If C_H is small, global spinal safety fails even if every sensor appears locally reliable.

3. New theorem: the cut-lift hardening law

Suppose hardening source a by $h_a \geq 0$ adds h_a exponent units and costs $r_a h_a$. To force the violation exponent above a target γ , we need

$$C_H(\theta; I+h) \geq \gamma.$$

This is equivalent to the following exact fractional blocker problem:

$$\begin{aligned} H_H(\gamma) = \min \sum_a r_a h_a \\ \text{subject to } \sum_{a \in T} h_a \geq (\gamma - \sum_{a \in T} I_a)_+ \text{ for every } T \in V_\theta, \\ h_a \geq 0. \end{aligned}$$

Interpretation: every still-too-cheap violating cut has a deficit. Hardening must hit each deficient cut by at least its deficit. This is not ordinary sensor counting; it is a weighted blocker of the dangerous common-mode cut family.

Dual form. The exact dual is a deficit-packing law:

$$\begin{aligned} H_H(\gamma) = \max \sum_{T \in V_\theta} \delta_T y_T \\ \text{subject to } \sum_{T \text{ contains } a} y_T \leq r_a \text{ for every } a, \\ y_T \geq 0, \delta_T = (\gamma - I(T))_+. \end{aligned}$$

This dual identifies the certificate of hardness: a packed family of deficient cuts whose combined demand cannot be blocked cheaply.

Discovery. The right design primitive is not marginal failure probability and not sensor count. It is the minimum cost of lifting all deficient common-mode cuts past the target exponent.

4. Uniform hardening gives a tropical cut-lift law

If every shock source receives the same exponent lift x , then the new cut exponent has a simple tropical form. For each cut size k define

$$C_k = \min \{ I(T) : T \text{ in } V_{\text{theta}} \text{ and } |T|=k \}.$$

Then

$$C_H(\text{theta}; I + x*1) = \min_k (C_k + k x).$$

This is a lower envelope of lines. The slope at a point is the size of the currently weakest violating cut. A root common-mode source that violates by itself creates a slope-1 bottleneck: uniform downstream hardening improves the exponent only one unit at a time until that root cut is no longer dominant.

This is the tropical analogue of the earlier hidden-debt frontier: risk does not improve smoothly because the identity of the weakest cut changes in staircase-like phases.

5. Global spine exponent budget

For a countable spine, let C_n be the post-hardening common-mode cut exponent at coordinate n . If the local failure probability is ϵ_n approximately $\exp(-C_n)$, the Round 21 Borel-Nyquist law says Fin-style global safety requires

$$\sum_n \exp(-C_n) < \text{infinity}.$$

A direct sufficient design is to choose a summable risk budget η_n and set

$$\gamma_n = \log(1/\eta_n), \quad C_n \geq \gamma_n.$$

The hardening cost at coordinate n is then exactly the cut-lift value

$$H_{\{H_n\}}(\gamma_n).$$

Thus the global design problem becomes an exponent-budget allocation problem:

$$\begin{aligned} \min \quad & \sum_n H_{\{H_n\}}(\log(1/\eta_n)) \\ \text{subject to} \quad & \sum_n \eta_n \leq \eta_{\text{total}}, \quad \eta_n > 0. \end{aligned}$$

This is a proposed next optimization target. It connects recurrence ideals, common-mode hypergraphs, and acquisition or hardening budgets into one calculable pipeline.

6. Finite experiments and checks

The following checks used random finite common-mode hypergraphs. A shock set violated when the total weight of domains hit by its active shocks exceeded θ . All violating shock sets were enumerated exactly.

Experiment	Checks	Failures	Max error / observation
LP primal-dual cut-lift law	645	0	max duality gap 3.55e-15; max certificate shortfall 4.44e-16
Uniform tropical cut-lift spectrum	1,592	0	max direct-vs-spectrum error 1.78e-15
Global summable exponent schedule	220 coords	-	base risk sum 106.435; hardened risk sum 0.03198

Sample global schedule. The target was $\gamma_n = 2 \log(n+2) + \log 20$, giving $\eta_n = 1/(20(n+2)^2)$. The first 220 simulated coordinates had baseline expected failure sum about 106.44. After cut-lift hardening to the summable schedule, the expected failure sum was about 0.03198.

n	baseline C_n	target γ_n	LP cost	base risk	hardened risk
0	0.544	4.382	13.936	0.5806	0.01250
1	1.692	5.193	10.334	0.1841	0.00556
9	0.402	7.792	40.704	0.6692	0.000413
49	0.160	10.859	58.695	0.8523	0.0000192
99	0.447	12.226	51.446	0.6393	0.00000490
199	0.735	13.602	37.824	0.4795	0.00000124

7. Applications

Reliability engineering. The law separates ordinary replication from hardening of common-mode sources. If the cheapest violating cut is a root cause, adding more leaf-level sensors may do almost nothing. The right intervention is to lift the root cause exponent or split it into genuinely independent sources.

Verification and monitoring. A verification suite should maintain a cut dashboard: list the currently cheapest violating cuts, their deficits against the global target γ_n , and the minimal hardening plan. This gives a proof-oriented safety certificate rather than a vague confidence score.

Privacy and re-identification. In a linkage system, common-mode identifiers are shocks. If one shared upstream identifier can connect too many domains, privacy risk is governed by that cheap cut. Protecting many downstream fields does not help until the upstream cut is hardened, fragmented, or removed.

Distributed identifiers and composite keys. Composite-key design should maximize the exponent of every violating linkage cut. The blocker dual says which weak key components are jointly responsible for unavoidable re-identification risk.

Sensor procurement. The procurement version is: buy the intervention with the best cut-deficit coverage per dollar, but verify against the LP frontier because greedy choices can fail when deficient cuts overlap in deceptive ways.

8. Summary of Round 27

Exceptional finding: common-mode risk admits an inverse hardening calculus. The risk exponent is a min-cut; the cost to raise it is a fractional blocker over the family of deficient violating cuts.

Main equations:

$$C_H(\theta; I) = \min_{T \in V_\theta} I(T).$$

$$H_H(\gamma) = \min \sum_a r_a h_a \text{ subject to } I(T) + h(T) \geq \gamma \text{ for every } T \in V_\theta.$$

$$C_H(\theta; I+x\mathbf{1}) = \min_k (C_k + kx).$$

$$\text{Global Fin-safe target: choose } C_n \geq \log(1/\eta_n) \text{ with } \sum \eta_n < \infty.$$

Practical slogan:

Do not merely add sensors. Lift the cheapest violating common-mode cuts until their exponents form a summable spine.

Next target. The natural continuation is adaptive exponent allocation: choose η_n and hardening vectors $h_{\{n,a\}}$ jointly so that a fixed global failure budget is met at minimum cost, and then generalize the same optimization to block, density, and other recurrence ideals.

Round 28: Log-Spine Cut-Lift Allocation and Exponent Water-Filling

Status. Draft theorem exploration, proof sketches, and finite sanity checks. This round continues the common-mode cut program: Round 26 identified cheapest violating cuts; Round 27 gave cut-lift hardening. The new blind spot is global recurrence across the countable spine. A local common-mode risk exponent may look excellent, but repeated across infinitely many coordinates it is still recurrent unless the residual risks are summable.

28.1 The new blind spot: a fixed local exponent is not spinal safety

Let coordinate n have a common-mode hypergraph of latent shock sources. Let V_n be its family of violating cuts and let $I_{\{n,a\}} + h_{\{n,a\}}$ be the hardened exponent of source a . The local cheapest violating cut after hardening is

$$C_n(h_n) = \min_{\{T \text{ in } V_n\}} \sum_{\{a \text{ in } T\}} (I_{\{n,a\}} + h_{\{n,a\}}).$$

In the rare-shock regime, the local block-violation probability is governed by $\exp(-C_n(h_n))$. The failure set along the spine is

$$F(h) = \{ n : \text{coordinate } n \text{ still violates its repair / reliability threshold} \}.$$

The exceptional discovery is that local cut hardening must cross a logarithmic spine barrier. Constant exponent hardening, no matter how impressive at one coordinate, leaves linearly accumulating exposure over an infinite Fin-spine.

Log-spine Borel-cut law

If $\sum_n \exp(-C_n(h_n)) < \text{infinity}$,
then $F(h)$ is finite almost surely.

If the coordinate failures are independent and
 $\sum_n \exp(-C_n(h_n)) = \text{infinity}$,
with comparable lower bounds on the true local risks,
then $F(h)$ is infinite almost surely.

Thus, for the Fin recurrence ideal, the critical exponent floor is C_n about $\log n$. More explicitly: $C_n \geq (1+\epsilon) \log(n+1)$ is summably safe, while $C_n = p \log(n+1)$ with $p \leq 1$ is not Fin-safe under independence. The classical Borel-Cantelli boundary reappears as a Nyquist boundary for hardened common-mode cuts.

A local exponent buys a coordinate. A logarithmic exponent schedule buys the spine.

28.2 Spinal cut-lift inverse law

Round 27 gave the local cost $H_n(\gamma)$ needed to lift every violating cut of coordinate n to exponent at least γ :

$$\begin{aligned} H_n(\gamma) &= \min \sum_a r_{\{n,a\}} h_{\{n,a\}} \\ \text{subject to} \\ \sum_{\{a \text{ in } T\}} h_{\{n,a\}} &\geq (\gamma - \sum_{\{a \text{ in } T\}} I_{\{n,a\}})_{+} \\ \text{for every violating cut } T \text{ in } V_n. \end{aligned}$$

Combining this with the log-spine law gives a clean global inverse. For a finite horizon N and target expected failure budget δ , the minimum hardening cost is

$$\begin{aligned} C_N(\delta) &= \min_{\{\gamma_0, \dots, \gamma_{N-1}\}} \sum_{\{n < N\}} H_n(\gamma_n) \\ \text{subject to} \\ \sum_{\{n < N\}} \exp(-\gamma_n) &\leq \delta. \end{aligned}$$

For the full Fin-spine, replace the finite constraint by $\sum_n \exp(-\gamma_n) < \text{infinity}$. A canonical safe schedule is

$$\gamma_n = \log A + p \log(n+1), \quad p > 1,$$

or, at the sharp harmonic boundary,

$$\gamma_n = \log A + \log(n+1) + (1+\epsilon) \log \log(n+2).$$

This is a useful bridge: local set-theoretic cut structure determines H_n , while the recurrence ideal determines which γ -profile is globally meaningful.

28.3 Exponent water-filling for weighted hardening

A particularly useful closed form appears when coordinate n has baseline risk $p_n^0 = \exp(-C_n^0)$, and spending x_n increases the exponent by x_n / r_n . The finite-horizon allocation problem is

$$\begin{aligned} & \text{minimize} && \sum_n x_n \\ & \text{subject to} && \sum_n p_n^0 \exp(-x_n/r_n) \leq \delta, \\ & && x_n \geq 0. \end{aligned}$$

The exact solution is a reverse water-filling law. There is a λ such that

$$\begin{aligned} q_n^* &= \min(p_n^0, \lambda r_n), \\ x_n^* &= r_n \log(p_n^0 / q_n^*), \\ \sum_n q_n^* &= \delta. \end{aligned}$$

Here q_n^* is the residual risk after hardening. Active coordinates have equal residual risk per hardening price:

$$q_n^* / r_n = \lambda.$$

Interpretation: hardening should not merely target the largest local risk. It should reduce risks until the next dollar buys the same marginal risk drop everywhere. This is the continuous counterpart of a dual cut-lift heat map: harden the coordinates and cut sources with the highest shadow price until all active shadow prices equilibrate.

This also identifies a common engineering error: hardening only the currently worst coordinate can be suboptimal when that coordinate has poor exponent-gain efficiency. The correct unit is not risk, but risk drop per cost.

28.4 Ideal-aware variants

For ideals weaker than Fin , the logarithmic floor changes. If J is a summable ideal with weights w_n , a sufficient transform is

$$\sum_n w_n \exp(-C_n(h_n)) < \infty \implies F(h) \text{ in } J \text{ almost surely.}$$

For density-zero style ideals, it is enough that the Cesaro mean of the local hardened risks vanish:

$$(1/N) \sum_{n \leq N} \exp(-C_n(h_n)) \rightarrow 0 \implies F(h) \text{ has density zero in expectation, with almost-sure conclusions under standard independence or concentration hypotheses.}$$

So the recurrence ideal selects the confidence currency: Fin asks for summability; summable ideals ask for weighted summability; density ideals ask for average decay. The same local hardening technology feeds different global risk transforms.

28.5 Finite tests and sanity checks

Finite experiments were run as sanity checks of the log-spine barrier and the water-filling equations. They do not prove the transfinite claims; they test the finite shadows of the new laws.

Exponent profile	N=100	N=1,000	N=10,000	N=100,000
constant $C=8$	0.033546	0.335463	3.354626	33.546263
$0.8 \log n$	7.159356	14.473788	26.111275	44.562612
$1.0 \log n$	4.197279	6.486470	8.787706	11.090156
$1.2 \log n$	2.606967	3.336016	3.799160	4.091584
$\log n + 2 \log \log n$	1.352611	1.424325	1.460486	1.482199

The table shows the phase boundary: $p=0.8$ and $p=1$ continue to grow; $p=1.2$ slows toward convergence; $\log n$ plus a log-log surcharge stabilizes fastest among the tested profiles.

Test	Checks / parameters	Result
Water-filling KKT equations	100 random weighted profiles	max constraint error 2.22e-16; max active-spread 5.55e-17
Constant exponent simulation	N=5,000, 100 repetitions	expected 1.677 failures; mean 1.860; std 1.371
Subcritical log simulation	$p_n=(n+1)^{-0.8}$	expected 22.028; mean 21.710; std 4.736
Supercritical log simulation	$p_n=(n+1)^{-1.2}$	expected 3.681; mean 3.780; std 2.138
Log-log summable profile	$1/((n+1) \log^2(n+2))$	expected 1.189; mean 1.130; std 0.934

28.6 Applications

Sensor-bank reliability. A procurement plan should not report only local success probability. It should report whether the residual failure exponent crosses the log-spine schedule required by the monitoring ideal.

Verification dashboards. The dashboard should show $C_n - \log(n+1)$, not only C_n . A coordinate with a large absolute exponent can still be below the spine floor if it appears very late in a long sequence.

Privacy and re-identification. Common-mode hardening maps directly onto attack-source hardening. If the adversary can exploit infinitely many coordinates, privacy protection must make the re-identification failure probabilities summable, not merely small at each coordinate.

Composite keys and distributed identifiers. Identifier hardening should equalize residual collision risk per cost across shards or dimensions. Over-hardening one shard while leaving many cheap residual shards above the log floor wastes budget.

Active diagnostics. Use the local cut-lift dual variables to find source-level hardening targets; use exponent water-filling to distribute the global budget across coordinates.

28.7 Summary of Round 28

New law: $\sum_n \exp(-C_n(h_n)) < \text{infinity}$ is the Fin-spine safety certificate.

New barrier: constant local exponent does not imply global faithful observation.

New inverse: $\min \sum H_n(\gamma_n)$ subject to $\sum \exp(-\gamma_n) \leq \delta$.

New optimizer: $q_n^* = \min(p_n^0, \lambda r_n)$, with q_n^*/r_n equalized on active coordinates.

New slogan: cut-lift hardens coordinates; log-spine allocation hardens the recurrence standard.

Most useful continuation target: combine the cut-lift LP dual from Round 27 with the water-filling multiplier from Round 28 into a single two-level shadow-price certificate. Locally, source shadows say which common-mode causes to harden. Globally, the spine multiplier says which coordinates deserve the next unit of budget.